Collins Aerospace

COMPONENT MAINTENANCE MANUAL

Coffee Maker Assembly

Component Maintenance Manual
(with Illustrated Parts List)

25-30-03

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SEPTEMBER 06, 2021

TO: HOLDERS OF COMPONENT MAINTENANCE MANUAL 25-30-03, COFFEE MAKER ASSEMBLY

DESCRIPTION OF REVISION 06, DATED SEPTEMBER 06, 2021 HIGHLIGHTS

This full revision replaces the manual that you currently have in its entirety.

All changed pages keep data necessary to do maintenance on all equipment models. Black bars on the side of the page identify the changes of this revision.

This revision does not include copies of service bulletins (SB) or service letters (SL) applicable to this equipment. This manual includes data from applicable SBs and SLs. For this revision, move the SBs and SLs from your manual to this manual.

As part of this full revision, inserted pages (e.g. 1001.1), inserted figures (e.g. Figure 4.1), inserted tables (e.g. Table 2004.1), etc., from previous revisions have been renumbered. Page numbers, figure numbers, table numbers, etc. have been allowed to increment normally. Also, identification of previous deletions have been removed. Pages where only figure references, table reference, etc. have changed or text has been removed by previous revisions are not listed below. Pages with relocated content are not listed below. The table below shows all other pages of the manual where content has been added, changed, or removed.

PAGE	**CHANGE**
LEP-1 and LEP-2	Revised to reflect current revision.
1001	Revised TABLE 25-30-03-99A-003-A01, Table 1001 to delete “CCA and Switch Module Test Box”.
1005	Revised TASK 25-30-03-700-802-A01, step 2.N. to update the para to “brew flow stops and will not resume”.
1007	Revised TASK 25-30-03-700-803-A01, step 3.D.(1) to update the para to “The manufacturer does not recommend the repair of the CCA”.
7001	Revised TABLE 25-30-03-99A-012-A01, Table 7001 to update PART/SPEC NUMBER to Never Seez NSWT-14.
7006	Revised TASK 25-30-03-400-801-A01, step 2.J.(2) to add “Install the four ring terminals (197) and heat shrink tube (199)”.
9001	Revised TABLE 25-30-03-99A-013-A01, Table 9001 to delete “CCA and Switch Module Test Box”.
9002	Revised TABLE 25-30-03-99A-013-A01, Table 9001 to update PART/SPEC NUMBER to Never Seez NSWT-14.
10000 Section	Revised Illustrated Parts List to reflect current revision.

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Coffee Maker Assembly

Component Maintenance Manual
(with Illustrated Parts List)

This manual includes coverage of the following equipment:

Unit	Collins Part Number
Coffee Maker Assembly	11225-1

B/E AEROSPACE, a part of Collins Aerospace
15701 W. 95th Street
Lenexa, KS 66219
USA
CAGE Code: 63367

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523-0833548

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REV. NO.	ISSUE DATE	INSERT DATE	BY	REV. NO.	ISSUE DATE	INSERT DATE	BY
NC	Sep 29/95						
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B	May 22/98						
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D	Mar 28/03						
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INTRODUCTION

TASK 25-30-03-990-807-A01

1. General

A. This manual has been prepared to fulfill the requirements of ATA iSpec 2200 for Component Maintenance Manuals.

B. All procedures established in this manual have been performed and verified acceptable by trained personnel at B/E Aerospace, Inc. A competent shop mechanic who is unfamiliar with these units should, with a minimum amount of effort and tools, be able to isolate the problems of a nonfunctional unit and restore it to service.

C. Procedures are provided for the full range of maintenance and include all the work functions that could or may be done to the unit throughout its entire service life.

D. A competent shop mechanic should use this manual first to test and isolate a faulty component. The mechanic should then go directly to the applicable manual subject to effect a repair in the specific problem area without disturbing other areas.

TASK 25-30-03-990-808-A01

2. Abbreviation

The following abbreviations may be used in this manual:

TABLE 25-30-03-99A-001-A01

ABBREVIATION	MEANING	ABBREVIATION	MEANING
AC	Alternating Current	° F	Degree Fahrenheit
AL	Aluminum	FIG	Figure
AR	As Required	FOD	Foreign Object Damage
ARP	Aerospace Recommended Practices	FT	Feet
ASSY	Assembly	GA	Gauge
BARG	Barometric Gauge	GPM	Gallons Per Minute
° C	Degree Celsius	HZ	Hertz
CAGE	Commercial and Government Entity	ID	Inner Diameter
CCA	Circuit Card Assembly	IN	Inch
CCW	Counter Clock Wise	IN-LBS	Inch-Pounds
CM	Centimeter	INOP	Inoperative
CW	Clock Wise	IPL	Illustrated Parts List
DIA	Diameter	IR	Infra Red
DIP	Dual In-line Package	IRED	Infra Red Emitting Diode
EFF	Effectivity	KPAG	Kilo Pascals Gauge
KVA	Kilo Volt-Ampere	NO	Number

ABBREVIATION	MEANING
LBS	Pound
LG	Long
LH	Left Hand
LKG	Locking
MAX	Maximum
MM	Millimeter
NHA	Next Higher Assembly

ABBREVIATION	MEANING
RF	Reference
RTD	Resistive Temperature Device
SB	Service Bulletin
SCR	Silicone Control Rectifier
TIL	Technical Information Letter
TPN	True Part Number
TTL REQ	Total Required

TASK 25-30-03-990-809-A01

3. Verification

The sections of this publication shown below have been verified on the date(s) specified. Where actual performance of the functions (tasks) cannot be done prior to issuing a manual, the function (tasks) must be verified by simulation. Any function (task) verified by simulation must be verified by actual performance at the earliest possible time.

VERIFIED SECTION	PART NO.	VERIFIED BY	VERIFIED DATE
Testing and Fault Isolation	11225-1	Performance	12/12/95
Disassembly	11225-1	Performance	12/12/95
Assembly	11225-1	Performance	12/12/95

TASK 25-30-03-990-810-A01

4. Maintenance Task Oriented Support System

A. The Maintenance Task Oriented Support System (MTOSS) permits the use of electronic data processing of maintenance data.

NOTE: For detailed information, refer to Component Maintenance Manual (CMM) of the Air Transport Association (ATA) Specification 2200.

(1) The MTOSS system uses standard and unique number combinations to identify maintenance tasks and sub tasks.

(2) The MTOSS structure is a logical approach to organizing maintenance tasks and sub tasks.

(3) The MTOSS numbering system includes ATA chapter-section-subject number as well as a function code and unique identifiers.

(4) The purpose of the MTOSS numbering system is to provide a means for automated sorting, retrieval, and management of digitized data.

B. The MTOSS numbering system is an expansion of the ATA three element numbering system. An MTOSS number has seven elements of which the first five elements are mandatory for each task and sub task. The sixth and seventh elements are applied when necessary. This CMM incorporates the use of six of the seven elements. Refer to Figure INTRO-1.

(1) Element numbers 1, 2, and 3. These elements are existing ATA numbers:

(a) Element number 1: Chapter

(b) Element number 2: Section

(c) Element number 3: Subject

(2) Element number 4. The fourth element numerically defines the maintenance function being performed.

(3) Element number 5. The fifth element is provided to create unique numbers for all tasks or sub tasks that are similarly numbered through the first four elements.

(a) Tasks are numbered from 801 - 999.

(b) Subtasks are numbered from 001 - 800.

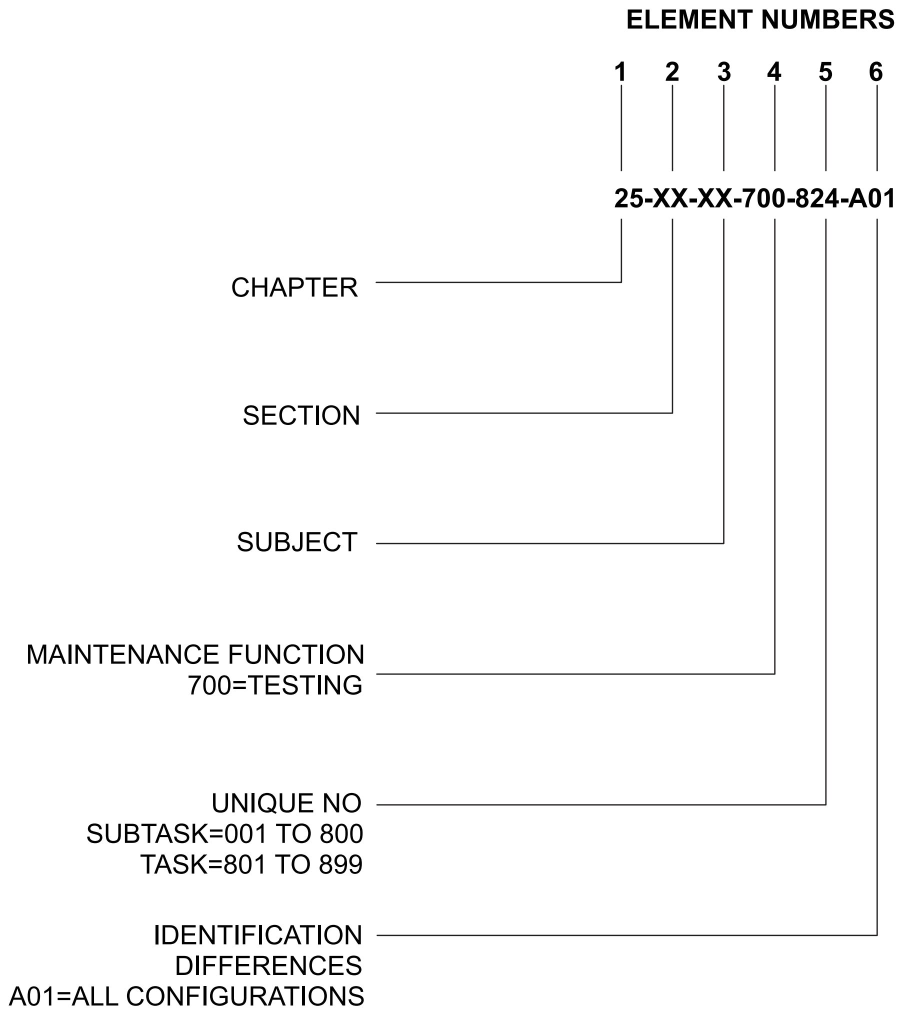
(4) Element number 6. The sixth element is a three position alphanumeric number that allows for identification of differences in configurations, methods, and techniques, variations of standard practice applications, etc.

(a) Most tasks in this CMM will use A01 as the sixth element. This indicates that the task applies to all configurations with no variation.

(b) If an S01 is used as the sixth element, it will indicate that a service bulletin applies to the task.

(5) Element number 7. The seventh element is a three position alphanumeric number reserved for customer use.

TASK 25-XX-XX-700-824-A01



NOTE: Element Number 7 is NOT USED.

GRAPHIC 25-30-03-99B-001-A01

Figure INTRO-1
Explanation of Maintenance Task Oriented Support System (MTOSS) Task Codes

DESCRIPTION AND OPERATION

TASK 25-30-03-870-801-A01

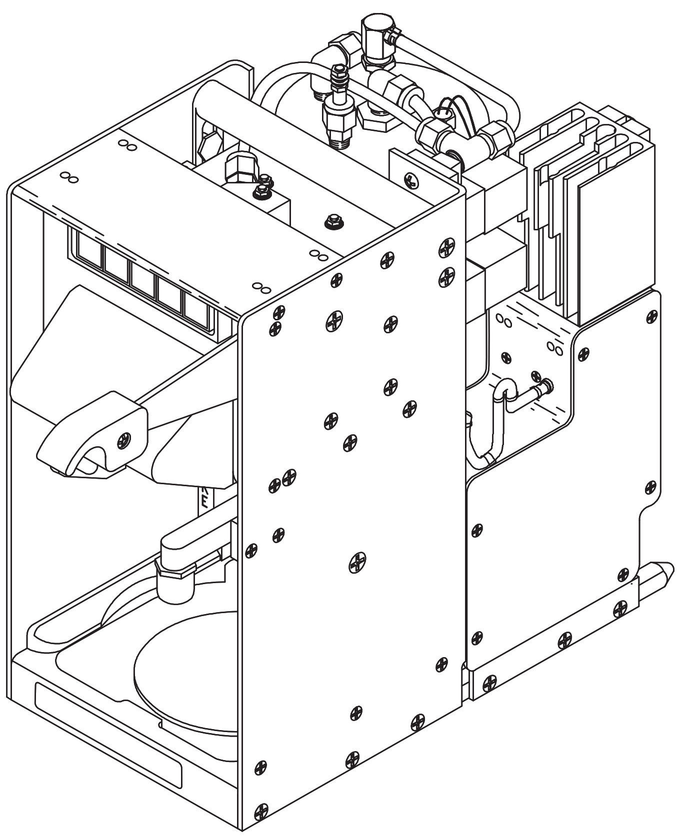
1. Description

A. Coffee Maker PN 11225-1 is manufactured at B/E Aerospace, Inc., Commercial Aircraft Products Group in Lenexa, Kansas.

B. Coffee Maker PN 11225-1 ia a two-solenoid Coffee Maker. It can be equipped with an optional solid state relay upgrade. This Coffee Maker is a galley insert on airline transport aircraft. The Coffee Maker fit in a galley Coffee Maker compartment. Power and potable water are obtained via a Coffee Maker rail such as B/E Aerospace, Inc. PN 11160-1.

C. The Coffee Maker consists of a black anodized aluminum chassis to which the water tank, solenoid manifold, electronics enclosure, brew assembly, and hot plate assembly are attached. Brief operating instructions are placarded on the face plate under the color coded operation switches.

D. Immersion heaters in the heavy duty stainless steel water tank provide hot water which is routed to “pillow pack” coffee in the brew cup. Coffee then drains into the server until it is full.



GRAPHIC 25-30-03-99B-002-A01

Figure 1
Coffee Maker Assembly

TASK 25-30-03-870-802-A01

2. General Operation Instruction

A. This section assumes an installed functional Coffee Maker (galley power and water turned on) and contains operating instructions for flight attendants.

B. Power

(1) Depress the red ON/OFF button if it is not already illuminated. Verify that TEST/LOW WATER button is not illuminated. (If it is, turn on the galley water supply.) Once power is on and the water tank is full, the water will automatically heat to a maximum nominal temperature of approximately 188° F ±3° F (86° C ±2° C) at which time the HOT WATER button will illuminate. You do not need to wait for hot water before initiating brew.

C. Brew

(1) Raise the brew handle and remove the brew cup. Verify that the brew cup is clean and free of coffee grounds. Place a pillow pack, seam down, on top of the screen in the brew cup. Replace the brew cup in the Coffee Maker.

(2) Insert an empty server into the server cavity and lower the brew handle. Verify that the server is in the server cavity as far as it will go by giving the server a gentle push.

(3) Press the green BREW button which will illuminate to indicate that brew is in progress. Brewing will occur only when the water is within the set maximum nominal operating temperature range of approximately 188° F ±3° F (86° C ±2° C). The Coffee Maker will cycle between heating water and brewing until the server is full. When the server is full the green brew light will turn off indicating that brew is complete.

(4) After the BREW light goes out raise the brew handle. Remove the brew cup and remove the used pillow pack. Tilting the brew cup slightly will limit dripping. Rinse the brew cup and replace it in the Coffee Maker. Coffee is ready to serve.

(5) The brew cycle can be interrupted at any time by raising the brew handle. To restart the brew, lower the brew handle and press the green BREW button.

D. Hot Plate

WARNING: DO NOT TOUCH HOT PLATE.

(1) Depress the orange HOT PLATE button to maintain server temperature until the coffee is served.

E. Hot Water

(1) Pushing the yellow HOT WATER button channels heated water from the tank to the faucet. This feature works irrespective of water temperature. To get fully heated water wait until the hot water light comes on.

(2) Hot water is also available during brew. The brew will be temporarily interrupted (indicated by the brew light going off), but will automatically resume (brew light will go back on) when the HOT WATER button is released. If enough water is drawn out to make the hot water light go out, the brew will resume only after the water has reheated.

(3) If a full server of hot water is desired, rinse the brew cup and server then perform a normal brew without a pillow pack.

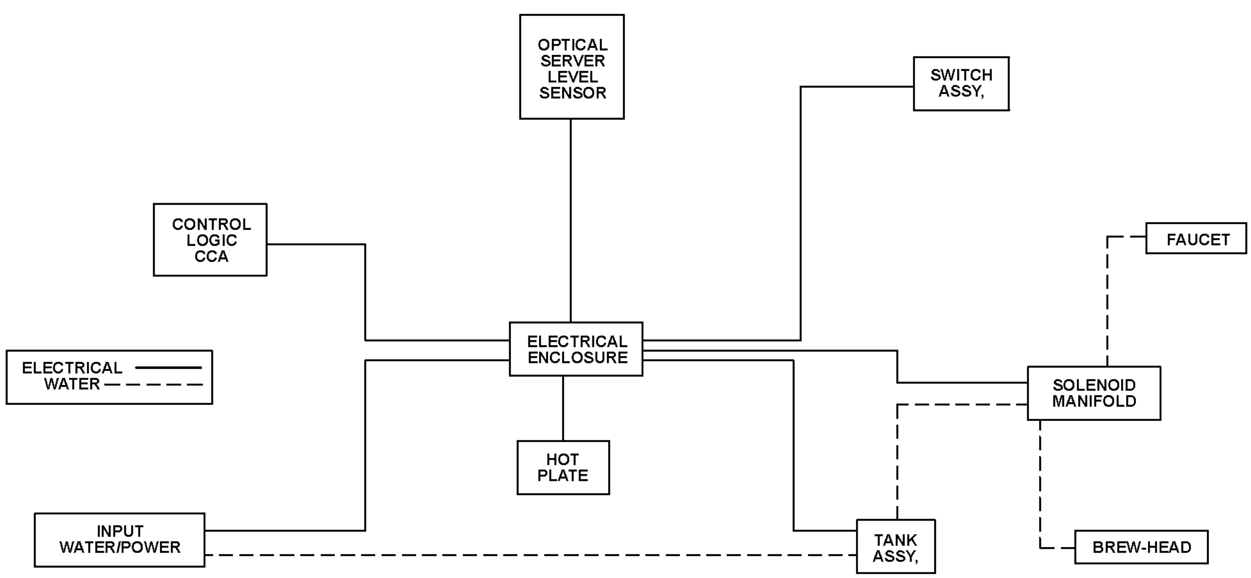
F. Lamp Test

Pushing the clear TEST/LOW WATER button while the ON/OFF button is depressed will illuminate all LEDs (except for ON/OFF, which is self-illuminating).

TASK 25-30-03-870-803-A01

3. <u>Detailed Description Of Operation</u>

A. This section will provide a detailed overview of Coffee Maker operation for qualified maintenance personnel. Circuit and system analysis, along with schematics can be found in SCHEMATIC AND WIRING DIAGRAMS section. The Coffee Maker block diagram in Figure 2 has been provided to help understand subsystem interactions.



GRAPHIC 25-30-03-99B-003-A01

Figure 2
Coffee Maker Block Diagram

B. Power

(1) Depress the ON/OFF switch which should illuminate (otherwise turn on galley power). The TEST/LOW WATER lamp should illuminate until the tank fills with water. The tank will finish filling because the electrical air vent was automatically activated when power was turned on.

(2) When the water reaches the tank full level sensor the low water lamp will go off and the three 900 watt immersion heaters in the tank will come on. Each heater is powered through a three phase power relay by a separate phase of 115 VAC line power. The heaters will remain on until the water reaches nominally 188° ±3° F (86° ±2° C) at which time the hot water light will come on and the heaters will turn off. The heaters will cycle on and off as required to maintain hot water temperature.

C. Temperature Control

(1) Hot water temperature is controlled by a negative feedback closed loop. A very low impedance RTD (Resistive Temperature Device) senses the water temperature above the immersion heaters. Comparator logic on the CCA (Circuit Card Assembly) then turns the three phase heater relay on and off as required and drives the hot water light to indicate when the water is hot. Additional protection circuitry on the CCA will open the circuit breaker if the water is above 235° F (112.77° C) or below 35° F (1.66° C).

(2) The nominal maximum hot water temperature of 188° F ±3° F (86° C ±2° C) is adjustable ±3° F on the CCA. This is a bench procedure and will be discussed in TESTING AND FAULT ISOLATION section.

(3) The control logic on the CCA contains heater control anticipator circuitry to optimize cycle time. This is only observable on a bench set up and will be described in TESTING AND FAULT ISOLATION section.

D. Brew

(1) Pushing the green BREW button attempts to arm the brew latch on the CCA. If the brew handle is not fully lowered, a micro switch closes a circuit that will allow the brew latch arm. Also, a full server in the Coffee Maker will allow the brew latch arm. Once the brew latch is armed the brew light will remain on. A brief power interruption, such as change over to aircraft power, will not disarm it.

(2) The brew latch can be armed without a brew cup or a server present, but this should be avoided because hot water will splash from the brew cup cavity.

(3) The control logic waits for a combination of brew latch and hot water, and then energizes the brew solenoid with 28 VDC. Cold water then enters the bottom of the tank pushing hot water out of the top. The hot water travels from the top of the tank through the brew solenoid to the brew head. The brew head directs the hot water into a “pillow pack” which is placed seam down in the brew cup. Coffee flows from the brew cup into the server.

(4) When the water tank temperature drops below the desired range the control logic closes the brew solenoid and turns on the heaters. When the water is hot the brew continues until the server is full.

E. Server Level

(1) Opto-electronic sensors are used to detect server level. An IRED (Infra Red Emitting Diode) transmits an IR (Infra Red) beam into the server. An IR pass filtered silicon photo transistor aimed into the server detects the IR beam only if it is reflected from coffee (or water) in a full server. Once the fluid detection level is reached, the brew solenoid and BREW light are turned off. If a pillow pack is in the brew cup it takes approximately five seconds for the coffee to drain from the brew cup. Therefore even though the server detection point is the same, server level with coffee will be approximately 0.25 in. (6.35 mm) higher than with water.

(2) Discarding the used pillow pack before serving the coffee is a good practice because it limits the residual coffee drips into the server cavity. (Drips and spillage in the server cavity drain to the galley sump.)

F. Hot Plate

WARNING: DO NOT TOUCH HOT PLATE.

(1) Whenever Coffee Maker power is on, the hot plate can be energized by pushing the orange HOT PLATE button. This lights the button and connects phase A line power to the 30 watt cal-rod heater or 35 watt platen heater. The heater assembly concentrates the heat into the aluminum hot plate.

G. Hot Water

(1) Hot water is available at the faucet by pushing the HOT WATER button. Hot water is drawn from the top of the water tank when the hot water solenoid is energized with 28 VDC through the control CCA.

TASK 25-30-03-870-804-A01

4. Leading Particulars

A. Refer to Table 1 for a list of leading particulars on the Coffee Maker assembly.

TABLE 25-30-03-99A-002-A01

Table 1
Leading Particulars

PARAMETER	LIMIT
FEATURES	
Opto-electronic sensors reliably maintain server level	
50 ounces of fresh brewed coffee in less than 4 minutes	
Hot water spigot	
Recovery time between brewing cycles less than 1 minute	
Power change over logic hold	
Electrical air venting	
ELECTRICAL	
Operating voltage	115 VAC, 400 Hz, 3 Phase

Table 1 (Continued)
Leading Particulars

PARAMETER	LIMIT
Wattage	Standard 3KVA Insert (2785 Watts)
DIMENSIONS	
Height	12.00 in (304.80 mm)
Width	6.30 in (160.02 mm)
Depth	16.07 in (408.18 mm)
Weight	23 lb (10.43 kg)
SAFETY FEATURES	
Pressure relief valve Over temperature protection Low water sensing Freeze protection IRED power monitor Secondary restraint	

TESTING AND FAULT ISOLATION

TASK 25-30-03-700-801-A01

1. General

   A. TESTING AND FAULT ISOLATION section covers the test equipment, bench setup, functional checkout procedure and fault isolation procedures required to return an inoperative coffee maker to service.

   B. Completely read and thoroughly understand the entire TESTING AND FAULT ISOLATION section. Become thoroughly familiar with the test sequences and their interactive correlations before performing any tests.

   C. The test equipment and materials required for testing of the Coffee Maker are listed in Table 1001.

   NOTE: Equivalent substitutes may be used for listed items.

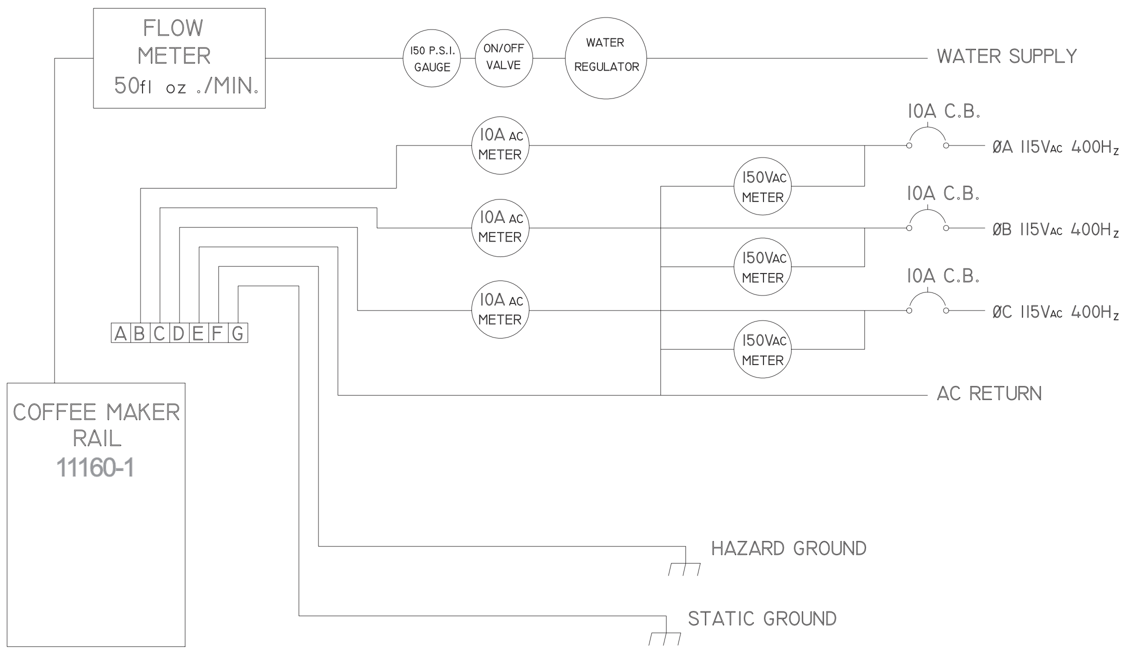
TABLE 25-30-03-99A-003-A01

Table 1001
Test Equipment and Materials

NOMENCLATURE	PARTS/SPEC NO	SOURCE (CAGE)
Deleted		
Coffee Maker Rail	PN 11160-1	B/E Aerospace, Inc. (V63367)
Meg-Ohm Tester	None	Commercially Available
Stop Watch	None	Commercially Available
Temperature Test Brew Cup	PN 11239-1	B/E Aerospace, Inc. (V63367)
Thermometers	None	Commercially Available

   D. Figure 1001 defines a standard Coffee Maker test bench configuration. This is the minimum test setup required to functionally check a Coffee Maker. All test and fault isolation procedures assume this setup and corresponding data availability.

   NOTE: The water should be regulated (while flowing) to 26 psig.



GRAPHIC 25-30-03-99B-004-A01

Figure 1001
Coffee Maker Test Bench

TASK 25-30-03-700-802-A01

2. Procedure

A. Functional Test

(1) This section contains a return to service test procedure intended to assure proper unit operation. This procedure is a logical starting point for an inoperative Coffee Maker. Refer to Table 1002 for Coffee Maker Functional Test.

B. Visual Inspection

(1) Visual inspection should precede and be a standard part of all testing. As a minimum, perform each check step in the INSPECTION/CHECK section. It should comply with the pre-operation check list.

NOTE: Make sure that the valve stem is closed.

C. Low Water

(1) Make sure that the water tank is empty. Attach Coffee Maker to test bench rail. (Do not turn on water supply.) Apply test bench line power. Close the Coffee Maker circuit breaker. Press the ON/OFF switch. ON/OFF and LOW WATER lamps should come on.

(2) Turn on the water and water tank should finish filling. The LOW WATER lamp should turn off when the tank is full.

NOTE: Phase A line current should be less than 0.1 amps. Phase B and C should be zero.

D. Water Leaks

(1) Turn off the ON/OFF switch. Turn on the water supply. Verify the water pressure is 26 psig while the water is flowing. Water flow rate should be greater than 32 fluid ounces per minute. The water tank should fill part way (until the air trapped in the tank is compressed to 26 psig) then water flow should stop completely. Residual water flow indicates a water leak. Visually check the Coffee Maker for leaks too small to cause a residual water flow.

E. Heater Current

(1) Press ON/OFF switch (this activates the electric vent valve). The water tank should finish filling. The LOW WATER lamp should turn off when the tank is full and each heater should draw approximately 8A ±1A on each of three power phases.

F. Hot Water Light

(1) Press HOT WATER button. Once the water is hot (approximately two minutes) the HOT WATER light should come on and the heaters should turn off (indicated by current drop).

G. Brew

(1) Starting with a hot tank, use a stop watch to time a brew cycle from when the BREW button is pushed until the BREW light goes out. The brew cycle time for Coffee Maker should be 3 minutes 15 seconds ±40 seconds.

NOTE: Brews started in mid cycle can be extended by a recovery time of up to 90 seconds.

(2) The server level measured in a 64 ounce server should be 4.1 ±0.3 in. (104.14 ±7.62 mm).

H. Temperature Discussion

(1) Several factors should be taken into account when testing Coffee Maker temperature.

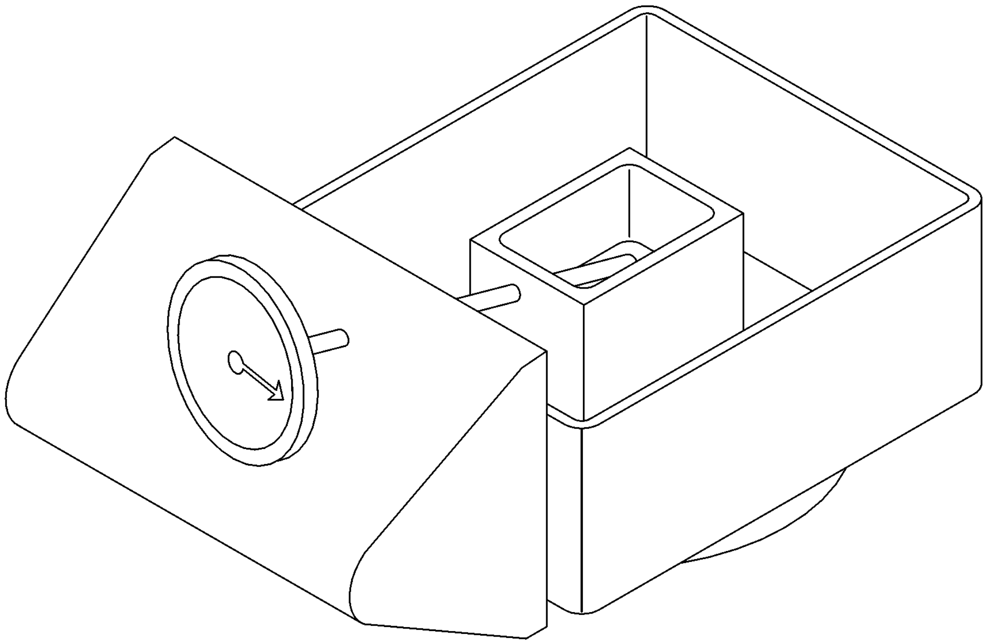
(2) A Coffee Maker does not warm up or heat soak as quickly on a bench setup as it does in an enclosed galley compartment. Partial compensation can be achieved by performing at least three brew cycles before taking temperature and timing measurements. However, ambient temperature and air flow differences will still introduce some error.

(3) Thermometers, even the digital lab grade models, have a set time delay and an accuracy limit.

(4) Water temperature drops very quickly as heat is withdrawn. The farther away from the source measurements are taken the less accurate they will be.

(5) The Coffee Maker is factory set to apply nominally 188° ±3° F (86° ±2° C) water to “pillow pack” in the brew cup. A ±3° F different nominal temperature may be set on the CCA. The best temperature measurement technique simulates the thermometer in a pillow pack. A special temperature test brew cup (refer to Table 1001 and Figure 1002) has been designed to perform this function.

(6) A second, much less accurate, technique to measure temperature is to hold the thermometer under the hot water faucet. Start the measurement right after the heaters cycle off. The thermometer will reach a peak temperature representative of the brew cup temperature, but it must be held steady in the water stream.



GRAPHIC 25-30-03-99B-005-A01

Figure 1002
Temperature Test Brew Cup

I. Temperature Test

(1) Measure the brew cup peak temperature using temperature test brew cup (refer to Table 1001). The temperature should be 188 ±3° F (86 ±2° C).

(2) Measure the server temperature after a brew cycle. The server temperature should be 175 ±10° F (79 ±6° C).

(3) The flight attendants have some control over server temperature by selective use of the hot plate during a brew cycle.

J. Heated Water

(1) Push the HOT WATER button and verify that heated water comes out of the faucet and that the flow rate is greater than 30 fluid ounces per minute (0.235 GPM).

K. Lamp Test

(1) Push the TEST/LOW WATER button and verify that all five lamps are illuminated.

L. Hot Plate

WARNING: THE HOT PLATE GETS VERY HOT.

(1) Press the HOT PLATE button and verify that the hot plate gets hot. Also, the HOT PLATE light should come on.

M. Power Interrupt

(1) During a brew cycle turn off the Coffee Maker then quickly turn it back on. The BREW light should remain on.

N. Brew Interrupt

(1) During a brew cycle raise the brew handle. The BREW light should go out, brew flow stops and will not resume.

O. IRED Monitor

(1) With the brew handle raised, insert a piece of 0.125 inch (3.175 mm) black heat shrink tubing in the left optical sensor cavity, blocking the infra-red beam. The circuit breaker should open.

P. Pressure Relief Valve

(1) With water tank full, increase source water pressure until the pressure relief valve opens. Make sure that the pressure relief valve opens at 95 ±10 psig, if not replace the pressure relief valve. Check for leaks in the Coffee Maker after 5 minutes.

TABLE 25-30-03-99A-004-A01

Table 1002
Coffee Maker Functional Test

Technician:	Date:	Ser. No.:
TEST	CRITERIA	RESULTS
Visual	Pass/Fail	
Low Water	Pass/Fail	
Water Leaks	Pass/Fail	
Phase A Current	8 ±1 amps	
Phase B Current	8 ±1 amps	
Phase C Current	8 ±1 amps	
Hot Water Light	Pass/Fail	
Brew Time	3'15" ±40"	
Brew Level	4.1 ±0.3 in. (104.14 ±7.62 mm)	
Peak Temperature	188° F ±3°F (86° C ±2° C)	
Server Temperature	175° F ±10°F (79° C ±6° C)	
Hot Water	>30 fl-oz/min	
Lamp Test	Pass/Fail	
Hot Plate	Pass/Fail	
Power Interrupt	Pass/Fail	
Brew Interrupt	Pass/Fail	
IRED Monitor	Pass/Fail	
Pressure Relief Valve	95 ±10 psig	

TASK 25-30-03-700-803-A01

3. Fault Isolation

A. This section contains the information, test procedures and techniques to identify and isolate Coffee Maker faults to a replacement part. Component reference designators will be used to help follow the System Wiring Diagram, refer to Figures 2001, 2002, or 2003, and the Circuit Card Assembly Schematic, Figure 2004.

B. Plumbing

Refer to Figure 2005 for Plumbing Schematic, included as an aid to isolating plumbing faults.

(1) Water Leaks

(a) Water can leak to the least desirable location. This could be the thermoswitches, power connector, electronics enclosure or switches etc. To all failure diagnosis procedures add a caveat to visually inspect for water near any components that might be related to the failure.

(b) The isolation of a water leak source requires visual detection. Look for water stains around all water fittings. During the tank fill test a mild solution of soapy water sprayed on a leak will bubble. This only tests the pressurized side of the water system, but that is where leaks are most likely to occur. Finally, carefully observe and inspect the Coffee Maker during operation until the leak is found.

(c) To fix a water leak, review the applicable disassembly and assembly procedures for the components involved. Generally, electrical parts (including the CCA) that do not visually appear damaged will still function once they are well dried.

(2) Slow Water Flow

(a) Slow water flow generally means that the Coffee Maker is due for periodic cleaning. To clear sluggish flow from a specific plumbing system component, isolate the component and blow air through it.

(3) Brew and Hot Water Solenoids

(a) The resistance across each of the two solenoids can be measured from TB1-3 (28 V) to TB1-4 (BREW) or to TB1-8 (HOT). This resistance should be approximately 85 ±10 ohms, if not remove and replace the solenoid.

(4) Leaky Water Faucet

(a) Water dripping from the faucet is usually caused by a small particle trapped in the valve seat. Often rapidly turning the valve on and off will dislodge the particle. Also, try blowing air through the valve with the valve energized. If the valve still leaks rework per CLEANING and REPAIR sections.

C. Wiring Errors

(1) It is important to inspect for and consider all possible wiring errors that could cause the fault. Touching ring terminals on terminal blocks and frayed wires etc. should be included in this category.

D. Circuit Card Assembly (CCA)

CAUTION: BEFORE USING THIS CCA PROCEDURE ASSURE THAT THE CCA WAS NOT DAMAGED BY AN EXISTING FAULT CONDITION. (I.E. 28 V SHORTED AT THE BREW BUTTON COULD DESTROY THE BREW LATCH ON THE CCA.)

(1) The manufacturer does not recommend the repair of the CCA.

(2) A general procedure for isolating a fault to the CCA requires using another CCA known to be functional. If use of the functional CCA corrects the fault, remove and replace the CCA.

E. Circuit Breaker CB1

(1) The circuit breaker is opened whenever the SCR (Q101) pulls current through the load resister (R101). This is designed to happen if the tank temperature as reported by the RTD is high (>235° F (112.78° C)) or low (<35° F (1.66° C)) or the IRED (U101) output power is low. Water leaking into the electronics enclosure usually trips one of these circuits and is the most common cause of circuit breaker faults. To isolate a circuit breaker fault open the electronics enclosure and remove the CCA. Inspect the CCA, especially the connector contacts, for water or contamination. If the CCA is wet, allow the CCA to dry. Close the circuit breaker and apply power with the CCA removed. If the circuit breaker opens, proceed to Paragraph 3.E.(2) If the circuit breaker remains closed proceed to Paragraph 3.E.(4).

(2) Circuit Breaker Opens Without CCA Installed

If the fault is not a water leak or a wiring error it must be the SCR (Q101), varistor (CR101), transformer (T1), hot plate (W1) or the circuit breaker (CB1).

(a) Hot Plate W1 (shorted to chassis)

Turn off the hot plate, close the circuit breaker and turn on power. If power stays on, depress the HOT PLATE switch. If the circuit breaker opens only when the HOT PLATE is turned on, remove and replace the hot plate assembly.

(b) Transformer T1

The transformer (T1) primary resistance (TB2-9 to TB3-4 with the Coffee Maker power switch off) should be 23.6 ohms and secondary resistance (TB1-1 to TB1-2) should be 0.5 ±0.3 ohm. The fault can be verified by removing either wire from TB2-9 and repeating the power on test. If the circuit breaker now remains closed, remove and replace the transformer.

(c) Circuit Breaker CB1

Disconnect all wires from the circuit breaker. The circuit breaker resistance should be 0.8 ohms, if not remove and replace the circuit breaker.

(d) Varistor CR101

Lift one leg of the varistor (CR101). The varistor resistance should be much greater than a standard ohm meter can measure (>>1 Gohms). If not remove and replace the varistor.

(e) SCR Q101 (shorted)

The SCR (Q101) anode (pin 2) to cathode (pin 1) resistance should be more than 1 Mohms. If not remove and replace the SCR.

(3) Circuit Breaker Opens Only With CCA Installed

Measure water tank temperature.

(a) Temperature under 35° F (1.66° C) - Coffee Maker is working correctly. Allow tank to warm before using Coffee Maker.

(b) Temperature between 35° F (1.66° C) and 235° F (112.78° C)

1 Server Level Optical Sensor U101

Remove CCA. Attach a jumper from TB2-4 to TB1-6. Reinstall CCA. Apply power. If the circuit breaker opens proceed to Paragraph 3.E.(3).(c).

Clean the optical sensor (U101) with an alcohol or water dampened cotton tipped swab, especially the IRED in the left hand cavity. Remove jumper, reinstall CCA and repeat power on test. If the circuit breaker opens, remove and replace the optical level sensor (U101).

If the circuit breaker opens when power is turned off, remove and replace the optical level sensor (U101).

2 RTD

An intermittent RTD is possible though unlikely. Disconnect the RTD at TB2-2 and TB2-4. The RTD resistance should be between 500 and 700 ohms depending on tank temperature per Table 1003. Also, the lead to case impedance should be greater than 1 Gohms. If not remove and replace the RTD.

TABLE 25-30-03-99A-005-A01

Table 1003
RTD Resistance Vs. Temperature

TEMPERATURE	RESISTANCE
32° F (0° C)	500 ohms
78° F (25.55° C)	549 ohms
212° F (100° C)	696 ohms

3 CCA

If water was found on the CCA while performing steps in Paragraph 3.E., verify that the CCA has dried. If replacing the CCA with a known good spare does not open the circuit breaker, replace the CCA (refer to Paragraph 3.D).

(c) Temperature over 235° F (112.78° C)

Allow the tank to cool before proceeding. Power the Coffee Maker and observe the current on each phase as the tank heats. When the hot water light turns on all three phase currents should drop to zero (phase A less than 0.1 amps). Note that if the brew latch has been armed the heaters will remain on for several seconds after the hot water light has turned on. Fully raise the brew handle to disarm the brew latch.

1 CCA

If all three phases remain at 8 ±1 amps, replace the CCA with a known good spare and repeat the test. If all three phases turn off when the hot water light turns on, replace the original CCA (refer to Paragraph 3.D.)

2 Relay

If only one or two of the phases turn off when the hot water light turns on see Paragraph 3.H. or 3.I. to troubleshoot the relay.

(4) Circuit Breaker Does Not Open per Paragraph 2.O.

Verify that test in Paragraph 2.O. was performed correctly. If light leaks around the piece of heat shrink, the circuit will not trigger.

(a) CCA

Remove CCA and measure resistance from CB1 (closed) to TB3-4 with the ON/OFF switch depressed. If the resistance is not approximately 24 ohms proceed to Paragraph 3.E.(4).(c) or 3.E.(4).(d). Remove wire number 57 from TB1-6 and reinstall the CCA. The circuit breaker should open on power on. If not remove and replace the CCA. (Verify with alternate CCA per 3.D.)

(b) Optical Sensor U101

Measure resistance from wire number 57 to TB2-4. If it is shorted, remove and replace the optical sensor (U101).

(c) Load Resistor R101

Measure the resistance across the load resistor (R101). If the resistance is not approximately 16 ohms, remove and replace the load resistor (R101).

(d) SCR Q101 (open)

Measure the SCR (Q101) gate (J1-1) to cathode (TB3-4) resistance. If the resistance is not approximately 70 ohms, remove and replace the SCR (Q101).

F. Switch Module SW1 - SW5

Although not required, the CCA and switch module test box (refer to Table 1001), provides a methodical approach to switch module fault isolation.

(1) LEDs

Move the suspect LED to a known functioning switch (i.e. either ON/OFF or HOT PLATE). If the LED still does not illuminate, remove and replace the LED.

(2) Switches SW1 - SW5

Using the wiring diagram verify proper continuity to P3 for each switch position. Nonconforming switches can be replaced individually or the module can be replaced and repaired separately.

(3) Connectors J3 and P3

Connector continuity can be verified point to point from the switches to the terminal blocks using the system wiring diagram. Defective connectors can be repaired per REPAIR section.

G. Water Tank Level Sensor Z1

(1) Remove the water tank level sensor Z1. Inspect for contamination and clean or replace as required.

H. Relay K1 (Mechanical)

(1) Remove the relay K1 and check continuity per Table 1004, replace if nonconforming per DISASSEMBLY and ASSEMBLY section.

TABLE 25-30-03-99A-006-A01

Table 1004
Relay K1 Continuity

PINS		CONTINUITY
2	10	Approximately 460 ohms
1	3	OPEN
1	4	SHORT
6	7	OPEN
6	5	SHORT
11	9	OPEN
11	8	SHORT
ALL	CASE	OPEN

I. Solid State Relay K1

(1) We expect the alternate and interchangeable solid state relay to be very reliable. To maintain an operational internal temperature the relay must be well attached to the heat sink. An overheated relay will not draw full current on all phases and must be replaced.

(2) To verify relay conformance remove the relay K1 per DISASSEMBLY section and check continuity per Table 1004A, replace if nonconforming, per ASSEMBLY section.

TABLE 25-30-03-99A-007-A01

Table 1004A
Relay K1 Continuity

PINS		CONTINUITY
+	-	OPEN
Pole 1-1	Pole 1-2	>2 MOhm
Pole 2-1	Pole 2-2	>2 MOhm
Pole 3-1	Pole 3-2	>2 MOhm
Pole 1-1	Pole 2-1	OPEN
Pole 1-1	Pole 2-2	OPEN
Pole 2-1	Pole 3-1	OPEN
Pole 2-1	Pole 3-2	OPEN
Pole 3-1	Pole 1-1	OPEN
Pole 3-2	Pole 1-2	OPEN

J. Heaters H1 - H3

(1) Verify the thermoswitches (S1, S2, and S3) per Paragraph 3.K. before testing the heaters. The heater resistance from the buss bar (BB1) to TB3-1, TB3-2 or TB3-3 should be approximately 15 ±2 ohms. If not remove and replace the corresponding heater.

(2) A commercially available, variable voltage Meg-ohm tester is required to test for potential heater failures due to increased insulation resistance. Set the tester to 500 VDC and measure the resistance between the water tank and AC RETURN (TB3-4). This tests all three heaters in parallel and essentially provides the value of the worst heater. A value under 1 Mohm indicates that at least one of the heaters needs to be replaced. Using the appropriate wiring diagram, remove the leads for each heater and measure the resistance between each lead and its heater case. Any heater which measures less than 1 Meg Ohm should be replaced. Any heater which has damage to the epoxy plug sealing the wires, or if the plug is pushed outward, should also be replaced. Lowered resistance or damage to the epoxy indicates a heater that is very likely to fail due to water intrusion.

K. Thermoswitches

(1) Verify that the thermoswitches are closed, by pushing on each red button. If a thermoswitch was open, but the Coffee Maker was not over temperature, verify all mechanical connections to the thermoswitch. If the fault repeats, remove and replace the snap disk.

(2) Thermoswitches (with the button in) should have <1 ohms resistance across their leads. If not remove and replace the thermoswitches.

L. Optical Sensor U101

(1) Clean the optical sensors with an alcohol dampened cotton tipped swab. Observe the water flow through the sensor. If the water flow is erratic or water is getting to the underside of the sensor, realign and reseal the sensor. If the average server level is still out of spec (i.e. short or overflow), remove and replace the optical sensor U101.

M. Microswitch SW6

(1) Disconnect the microswitch (SW6) leads from TB1-7 and TB2-4. Attach an Ohm meter to the leads and raise and lower the brew handle. The microswitch should be open when the brew handle is down and closed when the brew handle is up. If not verify that the actuator is not damaged. If the actuator is OK, then remove and replace the microswitch.

N. Temperature Adjust

(1) Minor temperature adjustments can be made by turning the trimpot (R27) on the CCA. CW increases the temperature. CCW decreases the temperature.

NOTE: Large temperature shifts indicate a fault mode. Check for wiring shorts on either RTD wire. Check the CCA per Paragraph 3.D. Check the RTD per Paragraph 3.E.(3).(b).2. Check the heater relay K1 per Paragraph 3.H. or 3.I.

O. Hot Plate W1

(1) If the hot plate does not get hot, check for unattached wires. If none, remove and replace the hot plate assembly.

SCHEMATICS AND WIRING DIAGRAMS

TASK 25-30-03-870-805-A01

1. General

   A. This section contains the schematics and wiring diagrams for the Coffee Maker Assembly.

TASK 25-30-03-870-806-A01

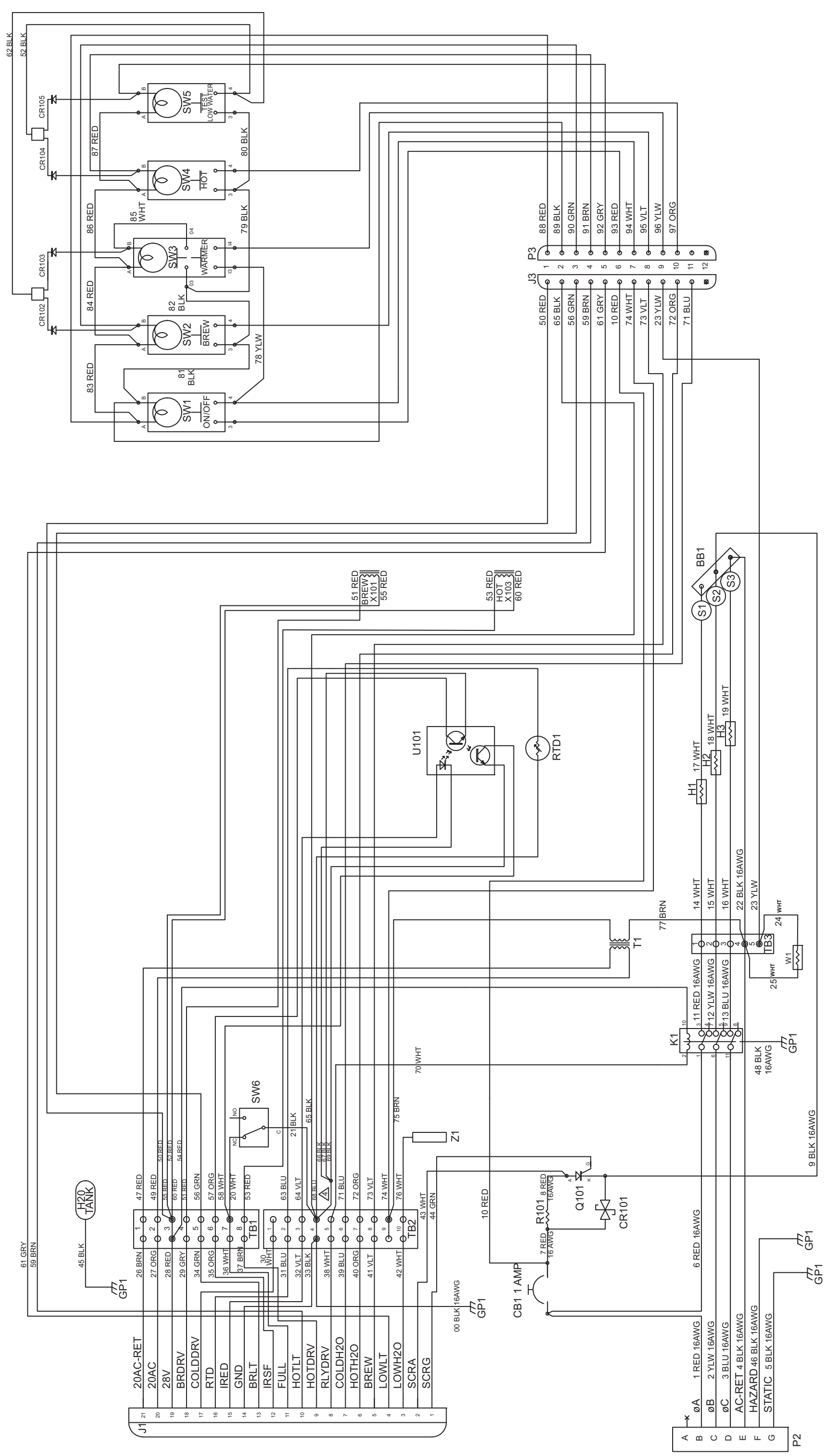
2. Wiring Diagrams

   A. The Coffee Maker receives electrical power from the galley through the Coffee Maker rail (PN 11160-1) via connector P2. Figure 2001 shows the System Wiring Diagram for a Coffee Maker equipped with a mechanical relay, minus the CCA. Figure 2002 shows the System Wiring Diagram for a Coffee Maker equipped with a solid state relay, minus the CCA. Figure 2003 shows the System Wiring Diagram for a Coffee Maker equipped with a solid state relay and “Brew Thru” base pan minus the CCA. Figure 2004 shows the Circuit Card Assembly Schematic, which interfaces with the Coffee Maker through connector J1.

TASK 25-30-03-870-807-A01

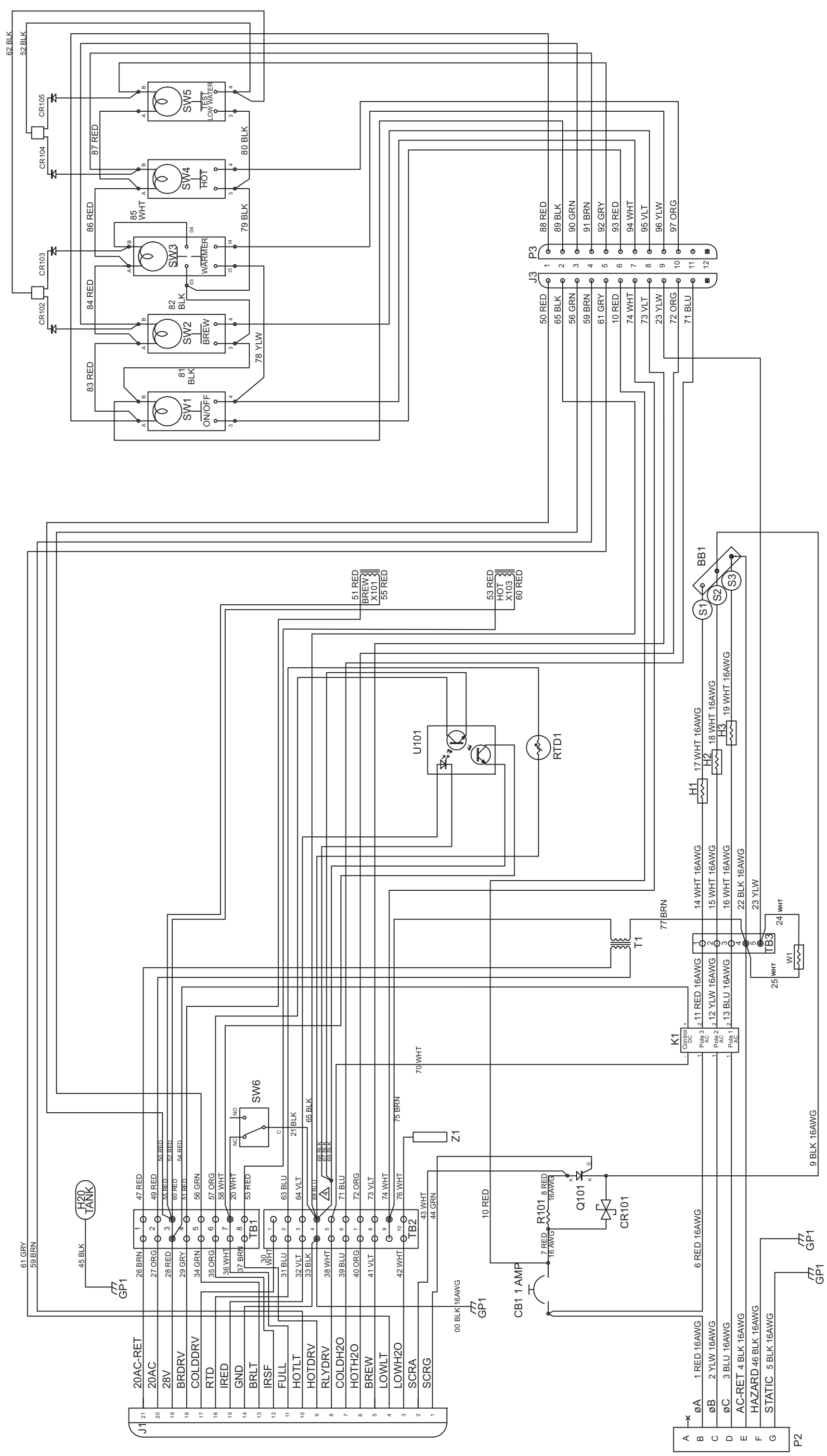
3. Plumbing Schematic

   A. The complete plumbing system is represented pictorially in Figure 2005. The plumbing components and their interconnections are shown with part number labels.



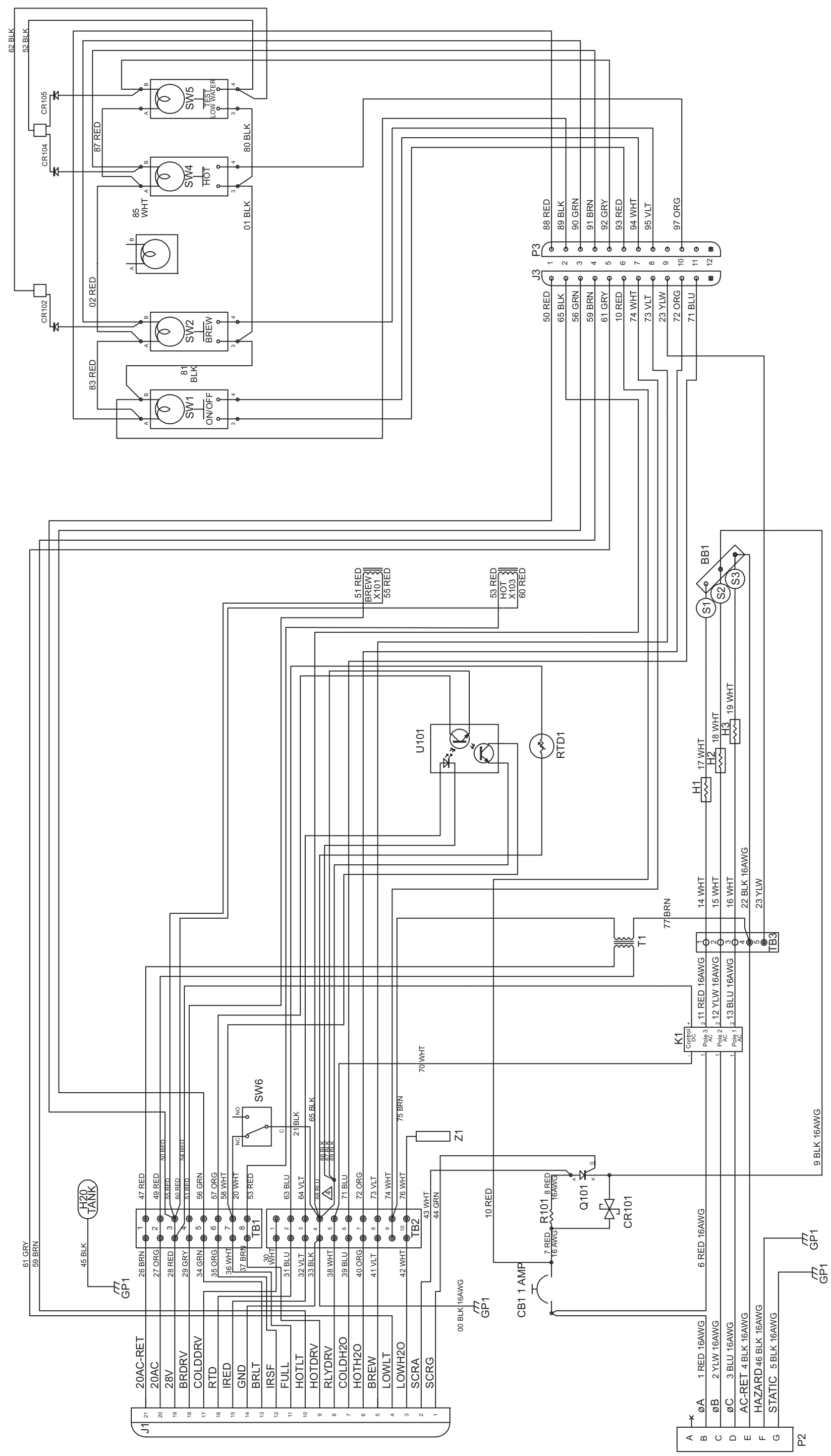
GRAPHIC 25-30-03-99B-006-A01

Figure 2001
System Wiring Diagram (Mechanical Relay)



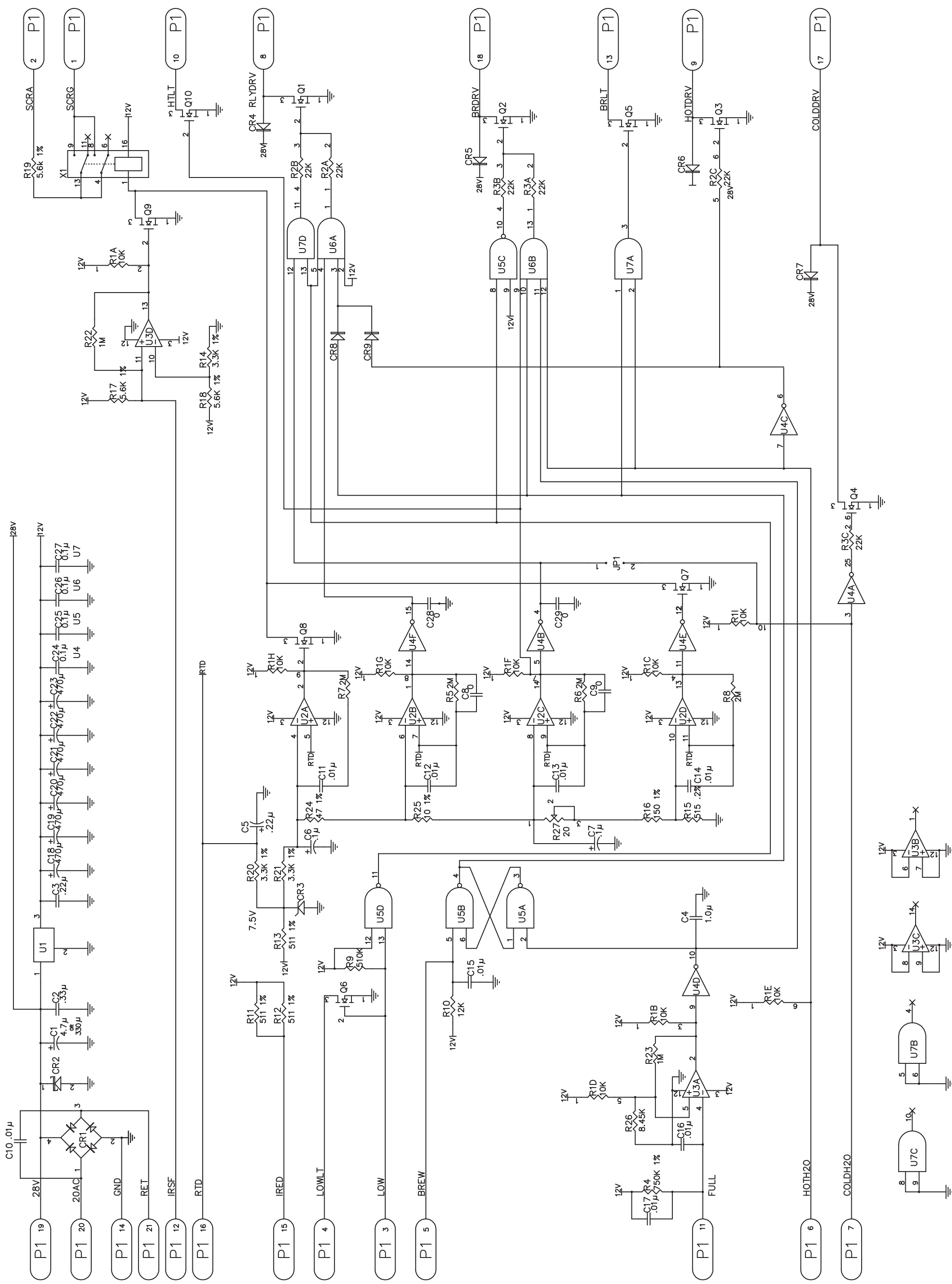
GRAPHIC 25-30-03-99B-007-A01

Figure 2002
System Wiring Diagram (Solid State Relay)



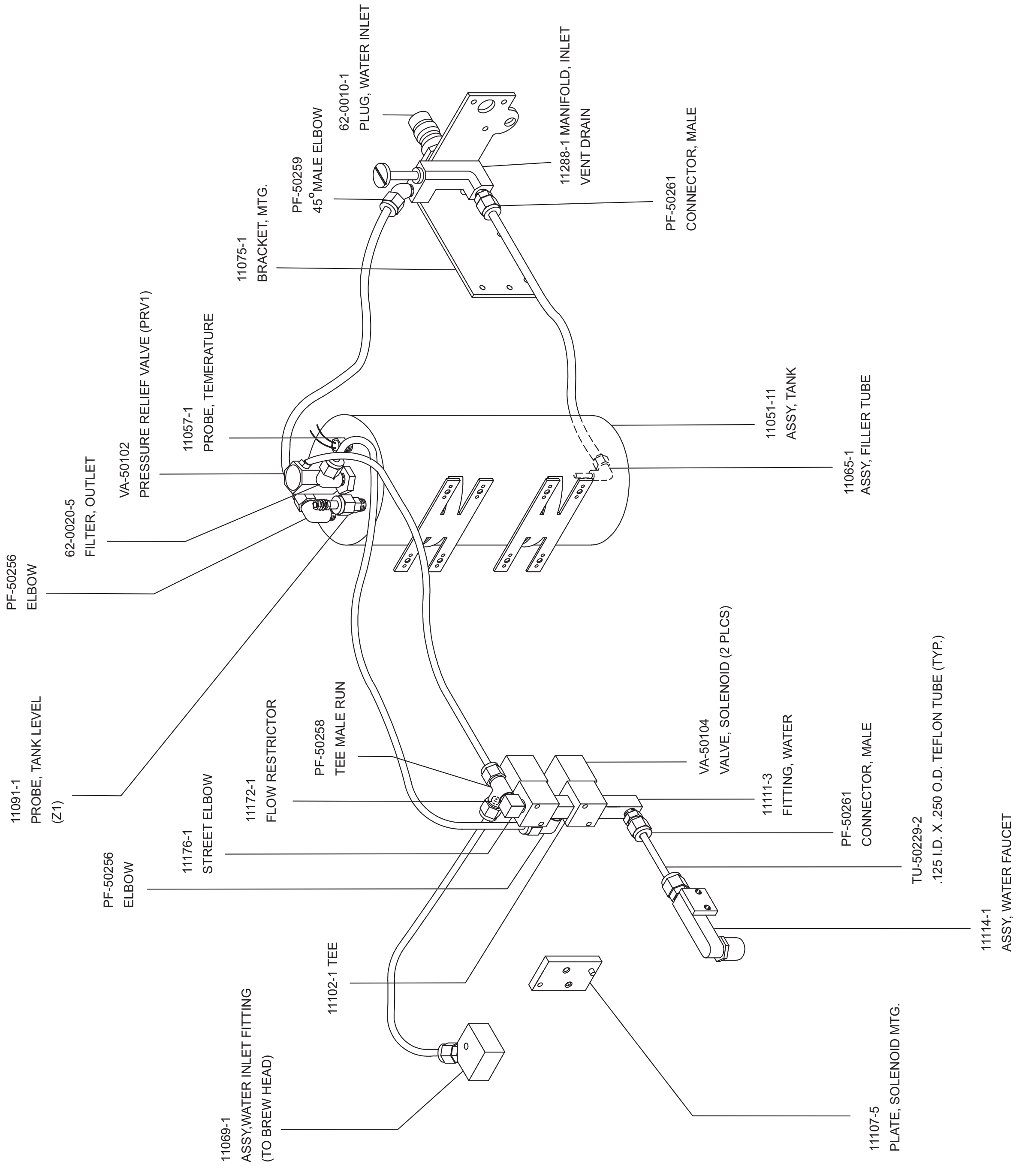
GRAPHIC 25-30-03-99B-008-A01

Figure 2003
System Wiring Diagram “Brew Thru”



GRAPHIC 25-30-03-99B-009-A01

Figure 2004
Circuit Card Assembly Schematic



GRAPHIC 25-30-03-99B-010-A01

Figure 2005
Plumbing Schematic

DISASSEMBLY

TASK 25-30-03-000-801-A01

1. General

A. Prior to disassembly carefully read all instructions applicable to the item of equipment undergoing maintenance.

B. Test each unit in accordance with the procedures in TESTING AND FAULT ISOLATION section. Refer to fault isolation section to determine the condition of the unit and the most probable cause of any malfunction. That will determine the extent of disassembly required to effect necessary repairs without complete teardown of the unit.

C. The list of disassembly tools required for the disassembly of the Coffee Maker are listed in the Table 3001.

NOTE: Equivalent substitutes may be used for listed items.

TABLE 25-30-03-99A-008-A01

Table 3001
List of Disassembly Tools

NOMENCLATURE	PARTS/SPEC NO	SOURCE (CAGE)
Angle Head Phillips Screw Driver	None	Commercially Available
Rachet with 5/32 in. socket	None	Commercially Available
Switch Lamp Remover	PN 02-906	EAO Switch Corporation (V61706)
Switch Lens Remover	PN 02-905	EAO Switch Corporation (V61706)

TASK 25-30-03-000-802-A01

2. Procedure

CAUTION: ALL DISASSEMBLY IS TO BE ACCOMPLISHED WITH THE COFFEE MAKER REMOVED FROM THE RAIL (I.E., POWER AND WATER SUPPLY ARE OFF) AND THE WATER TANK DRAINED.

A. Switch Module Assembly SW1 - SW5 (Refer to IPL Figure 1)

(1) Raise the brew handle and remove the brew cup assembly (5).

(2) Remove brew knob assembly (10) by removing screw (15).

(3) Remove the switch mount cover (20) by removing four screws (30) and one washer (25).

(4) Remove wire tie and disconnect the switch module connector (J3/P3).

(5) Remove the switch mount assembly (35) by removing four screws (40).

(6) Disassemble the switch mount assembly (35) as follows (refer to IPL Figure 6):

(a) Using switch lens remover (refer to Table 3001), grasp the switch lens cap (5, 10, 15, 20, or 25) to be removed along the vertical sides. Apply adequate pressure on the tool and pull the switch lens cap (5, 10, 15, 20, or 25) away from the switch housing.

(b) Use switch lamp remover (refer to Table 3001). Push the tool onto the LED (30) to be removed and pull the LED out.

B. Brew Head Assembly PN 11083-3 (Pre SB 25-30-03-9 and SB 25-30-03-11) (Refer to IPL Figure 1, unless specified)

(1) Remove the nut from the male run tee (95, IPL Figure 5) on the flow control assembly. The flow restrictor (75, IPL Figure 5) will fall free.

(2) Remove wire #20 from TB1-7 and wire #21 from TB2-4. Cut wire ties as required to loosen the brew head assembly (95). Remove cable wire clamp (60) by removing nut (75), washer (70), and screw (65).

(3) Remove two screws (180) from the handle assemblies (110 and 115, IPL Figure 3) and coffee maker housing (655).

(4) With the brew head assembly (95) raised, loosen set screws and remove pot retainer brackets (105 and 130). Do not remove screws (115 and 140), retainer bushings (120 and 145), or guide bushings (125 and 150) unless damaged or in need of replacement.

(5) Remove screws (160 and 170) and remove brew head guides (155 and 165) and brew head assembly (95) from the coffee maker housing (655).

(6) Disassemble the brew head assembly (95) as follows (refer to IPL Figure 3):

(a) Remove two shaft bushings (130) and retaining rings (125). Slide LH handle assembly (115) from RH handle assembly (110). Lift RH handle assembly (110) from brew head (5). Remove two roller bushings (120) from the handle assemblies (110 and 115).

(b) Remove the bolt (140), washer (145) and flat washers (150) to remove the ground wire assembly (135) from the RH handle assembly (110).

(c) Remove four nuts (30), flat washers (25), and screws (20) from the brew head (5). Remove retainer plate (10) and gasket (15) from the brew head (5).

(d) Remove nut (90), flat washer (85), and screw (80) from water inlet assembly (75). Lift water inlet assembly (75) out of the brew head (5). If necessary for replacement, remove male connector (95) from water inlet assembly (75).

(e) Remove screws (160) from the plunger assemblies (155) and pull plunger assemblies away from the brew head (5). Remove set screws (165) from the plunger assemblies (155). Invert plunger assemblies (155) to remove the plunger springs (170) and plungers (175) from the plunger housings (180).

(f) Remove the screws (65) and rod guide screws (70) to remove the slides (60) from the brew head (5).

C. Brew Head Assembly PN 12254-3 (Post SB 25-30-03-9 and SB 25-30-03-11) (Refer to IPL Figure 1)

(1) On the flow control assembly, remove the nut from the male run tee (95, IPL Figure 5). The flow restrictor (75, IPL Figure 5) will fall free.

(2) Remove wire #20 from TB1-7 and wire #21 from TB2-4. Cut wire ties as required to loosen the brew head assembly (95). Remove cable wire clamp (60) by removing nut (75), washer (70), and screw (65).

(3) Remove two brew handle screws (180) from the brew handle assemblies (105 and 110, IPL Figure 3A) and coffee maker housing (655).

(4) With the brew head assembly (95) raised, loosen set screws and remove pot retainer brackets (105 and 130). Do not remove screws (115 and 140), retainer bushings (120 and 145), or guide bushings (125 and 150) unless damaged or in need of replacement.

(5) Remove screws (160 and 170) and remove brew head guides (155 and 165) and brew head assembly (95A) from coffee maker housing (655).

(6) Disassemble the brew head assembly (95A) as follows (refer to IPL Figure 3A):

(a) Remove two shaft bushings (125) and retaining rings (120). Slide LH brew head handle assembly (110) from RH brew head handle assembly (105). Lift RH brew head handle assembly (105) from brew head (5). Remove two roller bushings (115) from the brew head handle assemblies (105 and 110).

(b) Remove the bolt (135), washer (140) and flat washers (145) to remove the ground wire assembly (130) from the RH brew head handle assembly (105).

(c) Remove four nuts (30), washers (25), and screws (20) from the brew head (5). Remove retainer plate (10) and brew head gasket (15) from the brew head (5).

(d) Remove nut (85), washer (80), and screw (75) from water inlet assembly (70). Lift water inlet assembly (70) out of the brew head (5). If necessary for replacement, remove male connector (90) from water inlet assembly (70).

(e) Remove two screws (40), star washers (45), and nuts (50) and remove the actuator angle (35).

(f) Remove the screws (60) and rod guide screws (65) to remove the slides (55) from the brew head (5).

D. Server Level Optical Sensor U101 (Refer to IPL Figure 1)

(1) Disconnect the following wires: #57 from TB1-6, #58 from TB1-7, #64 from TB2-3 and the black wire triplet (#66, #67, and #69) from TB2-4. Clip off the four ring terminals.

(2) Carefully clip the wire ties between the server cavity and the electrical enclosure terminal blocks. Carefully pull the six wires (#57, #58, #64, #66, #67, and #69) through the electrical enclosure grommet and wire clamp.

(3) Remove six screws (695) and remove the pillow pack plate assembly (690).

(4) Remove two screws (200) located at the top front edge of the pillow pack plate assembly (690). Break the seal between the server level sensor assembly (195) and the pillow pack plate assembly (690). Pull the server level sensor assembly (195) down, then forward when it clears the retainer guide rods (100). Carefully pull the six wires (#57, #58, #64, #66, #67, and #69) through the coffee maker housing (655) and remove the server level sensor assembly (195).

E. Limit Switch Angle Assembly (Refer to IPL Figure 1)

(1) Remove the screws (210) from the limit switch angle assembly (205). Cut ring terminals off both wires. Gently, pull wires through bundle to remove limit switch angle assembly (205).

(2) Remove the limit switch angle (215) and the actuator switch (235) by removing the two nuts (supplied with item 240), washers (230) and screws (supplied with item 240).

F. Water Faucet (Refer to IPL Figure 1)

(1) Remove water faucet assembly (245) by removing the two screws (250) and washers (255).

(2) Remove nut (40, IPL Figure 5) from male connector (30, IPL Figure 5) on flow control assembly (295) to disconnect hose of the water faucet assembly (245).

(3) Remove the faucet head (265) from the faucet subassembly (290). Remove the o-ring (270) and aerator (260) from the faucet head (265).

(4) Remove the connector (275) from the faucet subassembly (290).

G. Flow Control Assembly (Refer to IPL Figure 5)

(1) Remove screw (300, IPL Figure 1) and remove flow control assembly (295, IPL Figure 1) from coffee maker housing (655, IPL Figure 1).

(2) Remove two screws (65) and remove solenoid mounting plate assembly (5) from flow control assembly.

(3) Remove elbow (15) from flow control assembly. Disconnect hose assemblies from elbow (15) as required by removing nut (25) from elbow.

(4) Remove street elbow (45), male run tee (95) and male connector (30).

(5) Separate solenoid valves (10), water fitting (55) and tee (50).

(6) Remove ring terminals (70) from solenoid valves (10), only if damaged.

H. Locking Pin Assembly (Refer to IPL Figure 1)

(1) Remove screw (320), locking pin post (330), and locking pin bushing (325) and remove the locking pin assembly (310).

(2) Remove the E-ring (345) and groove pin (340) to remove the locking pin (350) and spring (315) from handle assembly (335).

(3) Remove screws (725) from angle (720) and remove the screws (735) to remove the locking pin block assembly (730).

I. Manifold Assembly (Refer to IPL Figure 1)

(1) Turn the inlet water plug (410) CCW and remove inlet water plug (410) and manifold (430) from the mounting bracket assembly (365).

(2) Remove and discard the o-ring (415) from the inlet water plug (410).

(3) If damaged or defective, remove and replace the valve stem assembly (445) and o-rings (450, 455, and 460) from manifold (430).

(4) Remove the male connector (435) and elbow (440) from the manifold (430).

(5) Remove screws (380), flat washers (375), lock nuts (370) and mounting bracket (420) from the mounting bracket assembly (365).

J. Electrical Enclosure Assembly (Refer to IPL Figure 1)

(1) Remove the screws (470 and 475) and washers (480) and remove the electrical enclosure assembly (465) from the coffee maker housing (655).

(2) Remove the electrical enclosure cover (485) by removing six screws (490).

(3) Remove screws (500 and 515) and washers (505 and 510) from the circuit card assembly (495).

(4) Disconnect wire #J101-1 from the circuit card assembly (495).

(5) Lift circuit card assembly (495) out of electrical enclosure assembly (465).

(6) Remove transformer (refer to IPL Figure 7):

(a) Remove the two screws (280) holding the transformer (275) in place. Carefully pull the transformer (275) away from the electrical enclosure (170) to expose the wires attached to it and beneath it.

(b) Disconnect the following wires: #47 from TB1-1, #49 from TB1-2, #75 from TB2-9 and #77 from TB3-4. Carefully cut wire ties as required and remove the transformer (275).

(7) If present, remove electromechanical relay K1 (refer to IPL Figure 1A):

(a) Remove the two screws (10) and washers (15) that hold the relay housing assembly (5) in place. Lift the housing (5) off the socket (50).

(b) Remove wire #54 from TB1-3. Remove wire #70 from TB2-5. Remove wire #6 from CB1-1. Remove wire #11 from TB3-1, wire #12 from TB3-2 and wire #13 from TB3-3. Extract pins C and D from P2 per REPAIRS section.

(c) Carefully lift relay socket (50) out of electrical enclosure assembly (465, IPL Figure 1).

(d) Remove two relay cover screws (30) and cover (35). Disconnect the ground wire #48 from the screw (245, IPL Figure 7).

(e) Carefully insert a screw driver through the relay housing assembly (5) slot and under the relay. Gently rotate the screw driver to dislodge the relay. Lift the relay out without snagging the ground wire.

(8) If present, remove solid state relay assembly K1 (refer to IPL Figure 1B):

CAUTION: ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) COMPONENT, USE PROPER HANDLING.

(a) Remove two screws (5 or 10), flat washers (20), hold down bar (15, if present), to remove the heat sink (55) and relay cover plate (65) from the electrical enclosure assembly (465, IPL Figure 1).

(b) Unsolder wires from solid state relay assembly (25), refer to system wiring diagram Figures 2002 or 2003 for wire numbers.

(c) To remove solid state relay assembly (25) from heat sink (55) remove two screws (30), lock washers (35) and flat washers (40).

(9) Remove circuit breaker CB1 (refer to IPL Figure 7):

(a) Remove wires #1 and #6 from CB1-1.

(b) Remove wires #7 and #10 from CB1-2.

(c) Remove retaining nut and washer from the circuit breaker CB1 (185).

(10) Remove SCR assembly (Refer to IPL Figure 7):

(a) Remove screws (210) and nuts (205) and remove the SCR assembly (200) from the electrical enclosure assembly (465, IPL Figure 1).

(b) It is recommended that SCR assembly (200) be removed prior to replacing the resistor (215), SCR (220), and varistor (225). The removal can be accomplished by using a low profile angle head Phillips screw driver (refer to Table 3001) and a small rachet with a 5/32 inch socket (refer to Table 3001).

(c) Remove the defective component by unsoldering the appropriate wires or replace entire assembly.

K. Tank Assembly (Refer to IPL Figure 4)

(1) Remove the screws (550, IPL Figure 1), standoffs (545B, IPL Figure 1), and thermostat cover (540, IPL Figure 1) from the tank assembly.

(2) Remove wire #68 from TB2-4 and wire #63 from TB2-2. Clip ring terminals and wire ties as required to remove wire from the electrical enclosure. Unscrew the temperature probe (5) CCW until it is free of the tank subassembly (215).

(3) Remove the self-locking nuts (25), washers (30), and screws (35) from thermoswitches (15).

(4) Disconnect wire #9 from S2-2 and wire #22 from S3-2 and remove the bus bar (20).

(5) Disconnect the wire (#17, #18, or #19) from the thermoswitches (15).

(6) Remove self-locking nuts (10) and pull the thermoswitches (15) from the mounting studs.

(7) Remove immersion heaters (45) from the tank assembly. Refer to Figure 3001 for immersion heater (45) orientation.

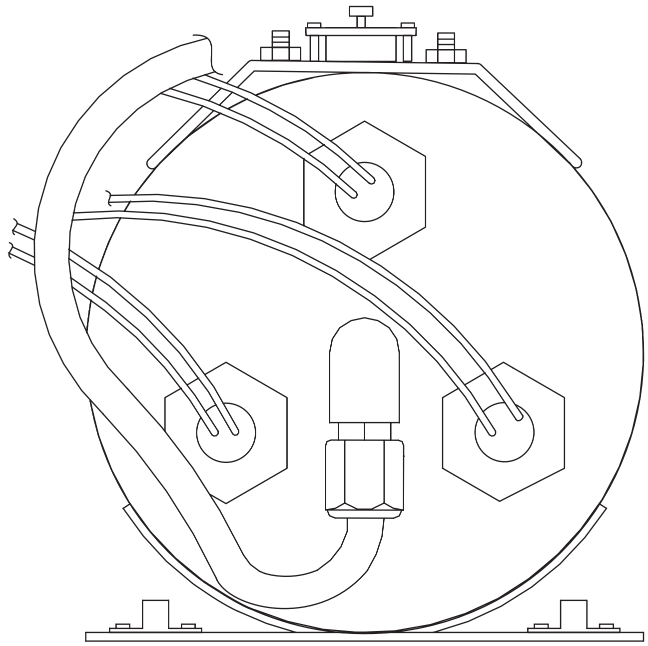
(a) Make sure that the water tank is drained.

(b) Remove terminal block cover (90, IPL Figure 7) from terminal block TB3 (95, IPL Figure 7).

(c) Disconnect wire #14 from TB3-1 and wire #17 from thermoswitch S1. Cut wire ties as required and pull heater wires through protective sleeving to the bottom of the tank subassembly (215). Turn the immersion heater H1 (45) CCW until it is free of the tank subassembly (215). Remove o-ring (50) from the immersion heater (45).

(d) Disconnect wire #15 from TB3-2 and wire #18 from thermoswitch S2. Cut wire ties as required and pull heater wires through protective sleeving to the bottom of the tank subassembly (215). Turn the immersion heater H2 (45) CCW until it is free of the tank subassembly (215). Remove o-ring (50) from the immersion heater (45).

(e) Disconnect wire #16 from TB3-3 and wire #19 from thermoswitch S3. Cut wire ties as required and pull heater wires through protective sleeving to the bottom of the tank subassembly (215). Turn the immersion heater H3 (45) CCW until it is free of the tank subassembly (215). Remove o-ring (50) from the immersion heater (45).



GRAPHIC 25-30-03-99B-011-A01

Figure 3001
Tank - Bottom View

(8) Remove nuts (65), washers (70) and wire #76 from the top of the tank level probe (60). Turn the tank level probe (60) CCW until the tank level probe (60) is free of the tank assembly.

(9) Remove the filler tube assembly (75) and elbow (80) from the tank assembly.

(10) Remove the pressure relief hose assembly (140) from the pressure relief valve (105B). Turn the pressure relief valve (105B) CCW to remove from the tank assembly.

(11) Remove elbow (175) from the tank assembly.

(12) Remove elbow (200) and filter assembly (190) from the tank assembly.

L. Hot Plate W1 (Pre SB 25-30-03-13) (Refer to IPL Figure 1)

(1) Remove screw (805), washer (810) and wire clamp (800) from the base pan assembly (815).

(2) Remove two screws (790), two washers (795) and ground wire assembly (785) from the base pan assembly (815).

(3) Remove the heating elements form the base pan assembly (815) as follows:

(a) Find and remove wires #24 and #25 from TB3-4 and TB3-5. Remove wire ties as required and carefully pull wires through protective sleeving to bottom of coffee maker base pan (890).

(b) Put the Coffee Maker on its side. Remove the screws (880) and washers (885).

(c) Put the Coffee Maker to the up right position and remove heater cover (845), o-ring (850), cal-rod heater (855) and heater plate (875) from base pan (890).

M. Hot Plate W1 (Post SB 25-30-03-13) (Refer to IPL Figure 1)

(1) Remove screw (805), washer (810) and wire clamp (800C) from the base pan assembly (815).

(2) Remove screws (790A), washers (795A) and ground wire assembly (785A) from the base pan assembly (815).

(3) Remove the heating elements from the base pan assembly (815) as follows:

(a) Find and remove wires #24 and #25 from TB3-4 and TB3-5. Clip off ring terminals (870), remove wire ties as required and carefully pull wires through protective sleeving to bottom of coffee maker base pan (890).

(b) Put the Coffee Maker on its side. Remove the screws (880A and 880B) and washers (885A).

(c) Put the Coffee Maker to the up right position and remove heater cover (845A), o-ring (850A), heater (855A) and heater retainer (875A) from base pan (890).

N. Base Pan Assembly (Refer to IPL Figure 1)

(1) Remove screws (820 and 830), washers (825 and 835) and remove the base pan assembly (815) from the base pan supports (745, 755, and 765).

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CLEANING

TASK 25-30-03-100-801-A01

1. General

CAUTION: TURN ALL POWER TO COFFEE MAKER AND COMPARTMENT OFF. EXCESSIVE WATER MAY CAUSE A SHORT CIRCUIT. USE A DAMP CLOTH ONLY.

CAUTION: DO NOT USE ANY SOLVENTS, DEGREASERS, DETERGENTS, AEROSOL SPRAY, ETC., AS THESE HARM SENSORS SWITCHES AND PLASTIC PARTS.

NOTE: The Coffee Maker should be completely cleaned and the brew head lubricated every 2000 operating hours.

A. The materials required for cleaning the Coffee Maker are listed in Table 4001.

NOTE: Equivalent substitutes may be used for listed items.

TABLE 25-30-03-99A-009-A01

Table 4001
Cleaning Materials

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
Cloths, Cleaning, Low-Lint Type 1	None	Commercially Available
Distilled Water	None	Commercially Available
Isopropyl Alcohol	TT-I-735	Commercially Available
Sulphamic Acid Based Cleaner	Sulphamic acid 5% solution, LimeAway, Vinegar, or other agent that removes calcium, lime, and rust deposits	Commercially Available

TASK 25-30-03-100-802-A01

2. Procedure

A. Coffee Maker Periodic Cleaning Instructions (Refer to IPL Figure 1)

WARNING: ALLOW WATER TANK TO COOL BEFORE DRAINING.

(1) Drain Coffee Maker by turning valve stem assembly (445) CCW until water flows from inlet vent drain manifold (430).

(2) Fill water tank with a sulphamic acid based cleaner that acts on calcium, lime, and rust deposits.

NOTE: Follow manufacturer's precaution for use of cleaner.

(3) Operate Coffee Maker for part of a cycle to allow cleaner to circulate.

(4) Draw half a cup of cleaner through the hot water solenoid and faucet.

(5) Turn off power and leave the cleaner in Coffee Maker for minimum amount of time required to remove the mineral deposits.

(6) Drain the cleaner from Coffee Maker as per Paragraph 2.A.(1).

(7) Flush all traces of the cleaner out of the Coffee Maker with distilled water. Also, flush hot water through the solenoid and faucet.

(8) Clean all surfaces in accordance with approved methods using warm water. Particular care should be taken to remove coffee residue that may have accumulated within the moving areas of the server retainer. If coffee leak is evident under the hot plate, replace o-ring (850). Refer to hot plate disassembly and assembly procedures in DISASSEMBLY AND ASSEMBLY sections of this manual.

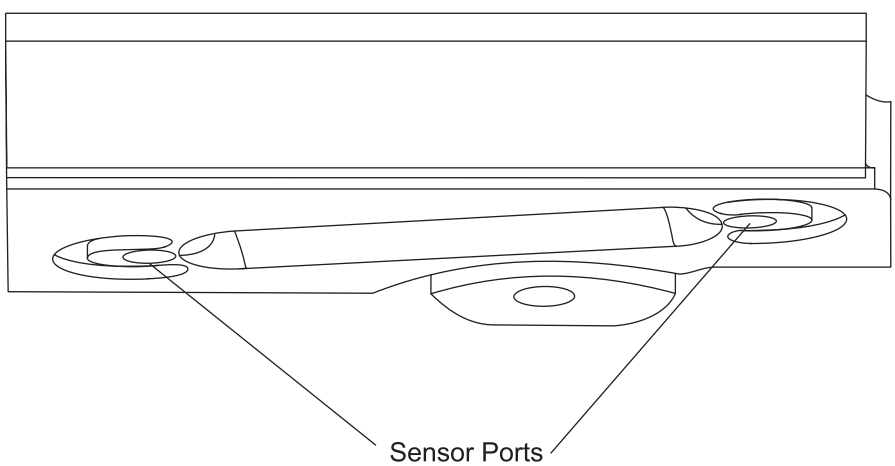
(9) Allow parts to dry thoroughly at room temperature, or blow dry parts with filtered, oil-free, compressed air.

B. Optical Server Level Sensor U101

CAUTION: ACETONE OR MEK WILL RUIN THE SENSOR. DO NOT USE UNAPPROVED SOLVENTS TO CLEAN THE SENSOR.

(1) Insert cotton tipped swab moistened with isopropyl alcohol into sensor ports and rotate to clean lens. Refer to Figure 4001 for location of sensor ports. Use the dry end of the cotton tipped swab to dry the lens. Check for lint left in ports.

(2) This cleaning should be done every week.



GRAPHIC 25-30-03-99B-012-A01

Figure 4001
Coffee Maker Assembly

C. Water Tank Level Sensor Z1

(1) The tank level probe should be inspected every time that a Coffee Maker has been removed for maintenance.

(2) Contamination and light surface rust can be removed from the probe tip with a non-metallic scouring pad, provided that either the pad is new or has only been used on stainless steel. Cleaning with metal tools or a scouring pad that has been used on different metals will embed iron particles on the probe and become a new source of rust.

(3) Buildup on the Teflon sleeving can be removed with alcohol.

(4) Polishing of the stainless steel probe tip is permitted provided that the polishing compound contains no metallic particles. Be sure to remove all traces of the polishing compound when finished.

(5) There are several types of corrosion which can attack the stainless steel of the tank level probe. Normally the passivation layer protects the stainless steel from corrosion. However, if the passivation layer is penetrated, lost or destroyed, corrosion can attack the stainless steel very rapidly. The most common types for this application are pitting, general corrosion, intergranular corrosion and stress corrosion cracking.

(6) The types of corrosion listed above have different characteristics but all will erode the stainless steel. More than one can be present at a time, and generally once started will continue to erode the stainless steel. Any probe that is attacked by any of these forms of erosion should be replaced.

(7) Remove water tank level sensor per DISASSEMBLY section, paragraph 2.K.

(8) Inspect per the above and determine if the probe is serviceable, if not obtain a replacement and install per ASSEMBLY section, paragraph 2.C.

(9) Determine and apply the appropriate method of cleaning.

(10) Install per ASSEMBLY section, paragraph 2.C.

D. Water Faucet Screen (Refer to IPL Figure 1)

(1) With soft grip pliers remove the faucet aerator (260B).

(2) Soak aerator screen in distilled vinegar and flush with water as required until the screen is free of debris. (Remove and replace the aerator (260B), if the screen is damaged.)

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INSPECTION/CHECK

TASK 25-30-03-200-801-A01

1. General

   A. The line operational check enables the Coffee Maker to be checked while installed on aircraft.

TASK 25-30-03-200-802-A01

2. Procedure

   A. Visual Inspection (Refer to IPL Figure 1)

      (1) Inspect the Coffee Maker for physical damage and verify that all parts are present and secure.

      (2) Inspect the power connector P2 for damage and verify that the pins are straight and not excessively worn. Refer to REPAIR section, paragraph 2.B.

      (3) Inspect the water inlet plug (410) for damage and contamination.

      (4) Inspect the bottom of the Coffee Maker to verify that all wires are properly tied and that the Coffee Maker is free of obstruction to installation.

   B. Check rail supports (775) for straightness.

      (1) Check brew handle for proper operation.

      (2) Inspect switch lens caps (refer to IPL Figure 6, Items 5, 10, 15, 20, and 25) for cracks, marks or illegible printing.

   C. Electrical Enclosure

      (1) Check that all parts are present and undamaged.

      (2) Check terminal block attachments for tightness.

      (3) Check that wires are undamaged, secured and routed out of the way as much as possible.

      (4) Check connector J1 for damaged contacts and good solder joints.

      (5) Inspect relay socket for corrosion or overheated wires.

      (6) Check the circuit breaker CB1 for proper operation.

   D. Pressure Relief Valve

      (1) Check the pressure relief valve (refer to IPL Figure 4, Item 105B) for proper operation. Refer to TESTING AND FAULT ISOLATION section, paragraph P.

E. Coffee Maker Line Operational Check (Refer to Table 5001)

(1) This section provides a short operational check that may be performed with the Coffee Maker installed on the aircraft. A wrist watch, small ruler and pocket thermometer is required to perform this check. Operational check presumes that the Coffee Maker is installed and that power and water are turned on.

(a) Clean the Coffee Maker. Refer to CLEANING section. The switches should not be sticky from coffee. The ON/OFF switch should illuminate when depressed. Install a server and lower the brew handle. The BREW light should illuminate when pressed.

(b) If the tank was not full the LOW WATER light will come on until it fills. If the tank was full skip this check.

(c) Throughout the entire check look for evidence of water or coffee leakage.

(d) The HOT WATER light should illuminate a couple of minutes after power is turned on and the tank is full. When the hot water light comes on the Coffee Maker will start brewing.

(e) Within five minutes the BREW light should go out indicating that brew cycle is complete.

(f) The server water or coffee level should be about one and a half inches from the top of the server.

(g) The temperature of the water or coffee in the server should be 175° ±10° F (79° ±6° C).

(h) Pushing the HOT WATER button should provide hot water at the faucet after the HOT WATER light comes on.

(i) Pushing the TEST/LOW water button should illuminate all five lamps.

WARNING: DO NOT TOUCH HOT PLATE.

(j) The HOT PLATE button should illuminate when depressed and the hot plate should get hot.

(k) Raising the brew handle should extinguish the BREW light and interrupt the brew cycle. Lower the brew handle and push BREW to continue operation.

TABLE 25-30-03-99A-010-A01

Table 5001
Coffee Maker Operational Check

Technician:	Date:	Ser. No.:
TEST	CRITERIA	RESULTS
Visual	Pass/Fail	
Low Water	Pass/Fail	
Water Leaks	Pass/Fail	
HOT WATER Light	~2 minutes	
Brew Time	<5 minutes	
Brew Level	~1.5 in (38.1 mm) from top	
Server Temperature	175° ±10° F (79° ±6° C)	
Hot Water	Pass/Fail	
Lamp Test	Pass/Fail	
Hot Plate	Pass/Fail	
Brew Interrupt	Pass/Fail	

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REPAIR

TASK 25-30-03-300-801-A01

1. General

   A. All parts found defective or inoperative in TESTING AND FAULT ISOLATION section and INSPECTION/CHECK section should be replaced or repaired.

   B. Standard parts such as screws, nuts, and washers should be replaced rather than repaired for economy purposes.

   C. The tools required for the repair of the Coffee Maker Assembly are listed in Table 6001.

   NOTE: Equivalent substitutes may be used for the listed items.

TABLE 25-30-03-99A-011-A01

Table 6001
Repair Materials

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
CCA and Switch Module Test Box	PN 11238-1	B/E Aerospace, Inc. (V63367)
J2/P2 Crimper	PN M22520/1-01	Daniels Manufacturing Corporation (V11851)
J2/P2 Crimper Turret	PN M22520/1-02	Daniels Manufacturing Corporation (V11851)
J2/P2 Pin Extraction Tool	PN M81969/19-08	Daniels Manufacturing Corporation (V11851)
J2/P2 Pin Insertion Tool	PN M81969/17-04	Daniels Manufacturing Corporation (V11851)
J3/P3 Crimper	PN 90325-1	Tyco Electronic Corporation (V00779)
J3/P3 Pin Extraction Tool	PN 455822-2	Tyco Electronic Corporation (V00779)
Switch Alignment Tool	PN 01-906	EAO Switch Corporation (V61706)
Switch Mounting Wrench	PN 01-907	EAO Switch Corporation (V61706)

TASK 25-30-03-300-802-A01

2. Procedure

   A. Circuit Card Assembly

   CAUTION: REPLACEMENT OF ELECTRONIC COMPONENTS ON THE CCA REQUIRES FUNCTIONAL TEST AND CCA CERTIFICATION. THE COFFEE MAKER COULD OVERHEAT AND/OR MALFUNCTION IF INSTRUCTIONS ARE NOT COMPLIED WITH.

   (1) Repair of the CCA requires the CCA and Switch Module Test Box (refer to Table 6001). An instructions and procedures manual comes with the test box.

B. Power Connector P2

(1) Pin Extraction

CAUTION: DO NOT ATTEMPT TO USE EXTRACTION TOOL (REFER TO TABLE 6001) IN ANY CONTACT HOLE UNLESS A CONTACT IS LOCKED IN PLACE.

CAUTION: DO NOT TWIST OR TURN TOOL WHILE EXTRACTING A CONTACT.

(a) Remove the strain from the pin to be extracted by disconnecting the opposite end of the wire (except pins C and D). Push extraction tool firmly onto pin such that the pin retention spring is unlocked. Push slider until it stops, this ejects the pin from the connector.

(2) Pin Replacement

(a) Extract damaged pins. If the pin is from position F or G the entire wire and ring terminal must be replaced. (Do not attempt to shorten these ground wires as this will strain the connector and inhibit alignment.) All other pins (positions B, C, D, and E) have sufficient wire length to be replaced at least once.

(3) Power Connector Removal

(a) Extract pins C and D. Disconnect wire #1 from CB1-1, wire #4 from TB3-4, and wires #5 and #46 from the ground stud. Slide the retaining ring of the wires.

(4) Pin Insertion

(a) Place crimped contact into the grommet hole at the rear of the connector. Using the insertion tool (refer to Table 6001) push the contact slowly into the connector until it snaps into position. Be careful not to cock the tool during insertion.

(5) Power Connector Installation

(a) Grind down the alignment pin on the connector to a maximum length of 0.08 inch (2.032 mm).

(b) Insert crimped contacts into positions B, E, F, and G. (Insert an uncrimped pin and filler into position A). Feed wires through hole in the electrical enclosure and align the connector. Slide the retaining ring over the wires onto the connector, but leave it loose. Insert pins C and D and tighten the retaining ring.

C. Switch Module

(1) Connectors J3 and P3

(a) Repair of the switch module connector requires both of J3/P3 related tools listed in Table 6001. Repair instructions are available free of charge from AMPFAX (1-800-522-6752) by requesting instruction sheet: IS 3231.

(2) Switches SW1 - SW4

(a) Carefully remove the wire ties from the switch module and spread the wires away from the defective switch. Unsolder all wires attached to the defective switch. Remove the retaining ring with switch mounting wrench (refer to Table 6001) and slide the switch out of the module. Insert new switch, align with switch alignment tool (refer to Table 6001) and tighten retaining ring with switch mounting wrench (refer to Table 6001). Solder wires back to the same position that they were removed from switch. (Refer to IPL Figure 6, as required.)

D. Relay Socket (Refer to IPL Figure 1A, Item 50)

(1) The relay socket contacts can be repaired by carefully bending the contact locking tabs and swapping the defective contact with an unused contact (socket positions 4, 5, or 8).

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ASSEMBLY

TASK 25-30-03-400-801-A01

1. General

CAUTION: DO NOT ALLOW LOCTITE THREAD SEALANT TO CONTACT THE BREW CUP OR BASE PAN.

When assembling the Coffee Maker refer to the figures in the Illustrated Parts List, along with the procedures in this section. The following table lists consumables that may be needed for assembly.

The list of equipment and materials required for assembly are listed in Table 7001.

NOTE: Equivalent substitutes may be used for listed items.

TABLE 25-30-03-99A-012-A01

Table 7001
List of Assembly Materials and Equipment

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
Adhesive, Silicone	736 Red AD-50005	Dow Corning Corporation (V71984) B/E Aerospace, Inc. (V63367)
Adhesive, Threadlock	Loctite® 242	Commercially Available
Coating, Alodine	1201 CM-50017	Henkel Corporation (V71410) B/E Aerospace, Inc. (V63367)
Heat Sink Compound	10-8109 CM-50019	Waldom, Inc. (V72653) B/E Aerospace, Inc. (V63367)
Nut Driver, 1/4 in.	None	Commercially Available
Nylon Wire Tie Downs	MS3367-4-9	Commercially Available
Open End Wrench 1/4 in., 7/8 in.	None	Commercially Available
Teflon Tape - 1/4 in	MS-STR-4 TT-7454	Swagelok Companies (V02570) B/E Aerospace, Inc. (V63367)
Thread Lubricant	Never Seez NSWT-14 CM-50016	Bostik, Inc. (V1Y0Z7) B/E Aerospace, Inc. (V63367)
Translucent Silicone Rubber Adhesive/Sealant	732 Clear RTV RTV 108 AD-50002	Dow Corning Corporation (V71984) Momentive Performance Materials, Inc. (V01139) B/E Aerospace, Inc. (V63367)

TASK 25-30-03-400-802-A01

2. Procedure

A. Base Pan Assembly (Refer to IPL Figure 1)

(1) Apply a thin coat of grease to the underside of base pan supports and install the base pan assembly (815) to the base pan supports (745, 755, and 765) and attach with screws (820 and 830) and washers (825 and 835).

B. Hot Plate W1 (Post SB 25-30-03-13) (Refer to IPL Figure 1)

(1) Install the heating elements to the base pan assembly (815) as follows:

(a) Place o-ring (850A) on the heater cover (845A) and attach heater (855A) to the heater cover.

NOTE: Use silicone rubber adhesive (refer to Table 7001) evenly between heater (855A) and heater cover (845A) and all around the perimeter.

(b) Add heater retainer (875A) on top of the heater (855A) and use silicone rubber adhesive (refer to Table 7001) to seal the perimeter to prevent moisture from getting to the wiring.

CAUTION: DO NOT USE THREAD SEALANT (I.E. LOCTITE) ON SCREWS.

(c) With the coffee maker in the upside down position, insert heater cover (845A), o-ring (850A), heater (855A) and heater retainer (875A) into the cavity of the base pan (890).

(d) Install screws (880A) and washers (885A) around the cavity and one screw (880B) and one washer (885A) in the center of the heater retainer (875A).

(e) Route wires #24 and #25 through protective sleeving (860) and attach ring terminals (870).

(f) Route and connect wire #24 to TB3- 4 and wire #25 to TB3-5. Use wire ties as required to be sure wiring does not touch the tank (530) or immersion heaters (45, IPL Figure 4).

(2) Install the ground wire assembly (785A) with screws (790A) and washers (795A).

NOTE: Use chemical coating to cover fastener heads and washers. The chemical coating must extend beyond the grounding terminal and be in contact with the assembly cover.

(3) Attach the heater wiring to the base pan assembly (815) using screw (805), washer (810) and wire clamp (800C).

C. Tank Assembly (Refer to IPL Figure 4)

(1) Apply teflon tape (refer to Table 7001) to the threads and install filter assembly (190) and elbow (200) on top of the tank assembly.

(2) Install the elbow (175) to the top of tank assembly.

(3) Apply teflon tape to the threads and install the pressure relief valve (105B) into the top of tank assembly and rotate CW. Connect the pressure relief hose assembly (140) to the pressure relief valve (105B).

(4) Apply teflon tape (refer to Table 7001) to the threads and install the filler tube assembly (75) and elbow (80) in the bottom of tank assembly and orient per Figure 3001 of DISASSEMBLY section.

(5) Apply thread lock compound to threads of tank level probe (60). Insert the tank level probe (60) into tank assembly and tighten.

CAUTION: DO NOT USE 0.56 INCH (14.29 MM) FITTING FOR REMOVAL OR INSTALLATION. OVERTIGHTENING THIS FITTING COULD GROUND THE TANK WATER LEVEL PROBE (60).

(6) Attach nut (65) onto the top of the tank level probe (60). Hand tighten the nut (65) against the Teflon shoulder. Place the washer (70) on top of the nut (65). Place the ring terminal of wire #76 next followed by the washer (70). Place the other nut (65) onto the 6-32 threads and make it finger tight. Using a quarter inch open end wrench (refer to Table 7001) to hold the bottom nut, tighten the top nut with a quarter inch nut driver (refer to Table 7001) until both of the lock washers are fully compressed.

NOTE: Some radial movement of the ring terminal may still be possible after tightening. As long as nuts, washers and the ring terminal rotate as a unit this is acceptable. Do not continue to tighten the nuts any further, as this will only deform the Teflon shoulder.

(7) Install the immersion heaters (45) to the tank assembly as follows:

(a) Install new o-ring (50) onto the immersion heater H1 (45) and insert into the bottom of the tank assembly. Rotate CW to advance threads. Continue until “finger” tight and then use a 7/8 wrench (refer to Table 7001) to tighten snugly into tank subassembly (215). Attach wire #14 to TB3-1 and wire #17 to thermoswitch S1.

NOTE: Immersion heater (45) should turn freely by hand to start threads properly.

(b) Install new o-ring (50) onto the immersion heater H2 (45) and insert into the bottom of the tank assembly. Rotate CW to advance threads. Continue until “finger” tight and then use a 7/8 wrench (refer to Table 7001) to tighten snugly into tank subassembly (215). Attach wire #15 to TB3-2 and wire #18 to thermoswitch S2.

NOTE: Immersion heater (45) should turn freely by hand to start threads properly.

(c) Install new o-ring (50) onto the immersion heater H3 (45) and insert into the bottom of the tank assembly. Rotate CW to advance threads. Continue until “finger” tight and then use a 7/8 wrench (refer to Table 7001) to tighten snugly into tank subassembly (215). Attach wire #16 to TB3-3 and wire #19 to thermoswitch S3.

NOTE: Immersion heater (45) should turn freely by hand to start threads properly.

(8) Apply teflon tape (refer to Table 7001) to the threads and insert the temperature probe (5) into tank assembly. Turn CW until “finger” tight.

(9) Route wires into electrical enclosure. Connect wire #63 to TB2-2 and wire #68 to TB2-4. Secure wires with wire tie as required.

(10) Install terminal block cover (90, IPL Figure 7) to terminal block TB3 (95, IPL Figure 7).

(11) Route wires through protective sleeving to prevent chafing and secure the wires with wire ties as required.

(12) Install thermoswitch S1 - S3 (15):

(a) Position the thermoswitches (15) onto the mounting stud and secure each thermoswitch with self-locking nuts (10).

(b) Attach wires #17, #18 and #19 to S1-1, S2-1, and S3-1 respectively. Then place buss bar (20) on S1-2, S2-2, and S3-2. Attach wire #9 to S2-2 and wire #22 to S3-2. Secure in place with screws (35), washers (30), and self-locking nuts (25).

(c) Attach thermostat cover (540, IPL Figure 1) with standoffs (545B, IPL Figure 1) and screws (550, IPL Figure 1) to the tank assembly.

D. Electrical Enclosure Assembly (Refer to IPL Figure 1):

(1) Install SCR assembly (Refer to IPL Figure 7):

NOTE: Per DISASSEMBLY section 2.J.(10) resistor (215), SCR (220), and varistor (225) should have been removed as an assembly or the defective part replaced.

(a) Attach the SCR assembly (200) to the electrical enclosure assembly (465, IPL Figure 1) with screws (210) and nuts (205).

(b) Route wires and attach per the System Wiring Diagram, Figures 2001, 2002, or 2003.

(2) Install circuit breaker CB1 (Refer to IPL Figure 7):

(a) Attach wires #1 and #6 to CB1-1.

(b) Attach wires #7 and #10 to CB1-2.

(c) Insert circuit breaker CB1 (185) with its alignment guide into hole on the back of the electrical enclosure (170) and attach with washer and retaining nut.

(3) If removed, install the solid state relay assembly K1 (Refer to IPL Figure 1B)

CAUTION: ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) COMPONENT, USE PROPER HANDLING.

(a) Begin by cleaning any remaining heat sink compound from solid state relay and heat sink. Apply heat sink compound (refer to Table 7001) to back side of solid state relay assembly (25) insuring no voids between heat sink (55) and white ceramic on back of solid state relay assembly (25).

(b) Securely attach solid state relay assembly (25) using flat washers (40), lock washers (35) and screws (30). Torque both screws to 64 in-oz.

(c) Solder wires to solid state relay assembly (25) leads per wiring diagram Figure 2002 or Figure 2003.

(d) Place the relay cover plate (65) on top of electrical enclosure (480, IPL Figure 7) and then attach heat sink (55) using screws (5) and washers (20) or heat sink (55A) using two screws (10) and hold down bar (15), if removed.

(4) If removed, install transformer (275) (refer to IPL Figure 7):

(a) Connect the following wires: #47 to TB1-1, #49 to TB1-2, #75 to TB2-9 and #77 to TB3-4. Secure with wire ties as required.

(b) Attach transformer (275) to electrical enclosure (170) with screws (280).

(5) Connect wire #J101-1 to the circuit card assembly (495).

(6) Install the circuit card assembly (495) with screws (500 and 515) and washers (505 and 510).

(7) Apply a bead of RTV (refer to Table 7001) around inside edge of electrical enclosure cover (485). Attach with six screws (490).

(8) Install the electrical enclosure assembly (465) to the coffee maker housing (655) with the screws (470 and 475) and washers (480).

E. Manifold Assembly (Refer to IPL Figure 1)

(1) Attach the mounting bracket (420) to the mounting bracket assembly (365) with screws (380), flat washers (375) and lock nuts (370).

(2) Install the male connector (435) and elbow (440) to the manifold (430).

(3) If removed, install the valve stem assembly (445) and o-rings (450, 455, and 460) to manifold (430).

(4) Install the o-ring (415) on the inlet water plug (410).

(5) Turn the inlet water plug (410) CW to install the inlet water plug (410) and manifold (430) to the mounting bracket assembly (365).

F. Locking Pin Assembly (Refer to IPL Figure 1)

(1) Attach the screws (725) to angle (720) and attach the screws (735) to install the locking pin block assembly (730).

(2) Install the locking pin (350) and spring (315) to the handle assembly (335) and install the groove pin (340) and E-ring (345).

(3) Install the locking pin assembly (310), locking pin post (330), and locking pin bushing (325) and attach with screw (320).

G. Flow Control Assembly (Refer to IPL Figure 5)

(1) If removed, install ring terminals (70) to the solenoid valves (10).

(2) Install solenoid valves (10), water fitting (55) and tee (50).

(3) Install street elbow (45), male run tee (95) and male connector (30).

(4) Install elbow (15) to the flow control assembly. Attach the hose assemblies to the elbow (15) as required by tightening the nuts (25) to elbows to complete the flow control assembly.

(5) Install two screws (65) and attach solenoid mounting plate assembly (5) to flow control assembly.

(6) Align the flow control assembly such that the male connector (30) enters the back of the coffee maker housing (655, IPL Figure 1). Install the screws (300, IPL Figure 1) through the coffee maker housing (655, IPL Figure 1) into the solenoid mounting plate assembly (5).

(7) Route wires through protective sleeving into electrical enclosure assembly (465, IPL Figure 1) and make any required connections as follows: Wire #51 to TB1-4, wire #62 to TB2-1, wire #53 to TB1-8 and wires #55, #52 and #60 to TB1-3. Wire tie as required.

H. Water Faucet (Refer to IPL Figure 1)

(1) Attach the connector (275) to the faucet subassembly (290).

(2) Install the o-ring (270) and aerator (260) to the faucet head (265). Install the faucet head (265) to the faucet subassembly (290).

(3) Connect hose of the water faucet assembly (245) to the male connector (30, IPL Figure 5) on flow control assembly (295) and tighten the nut (40, IPL Figure 5).

(4) Attach the water faucet assembly (245) to coffee maker housing (655) with the two screws (250) and washers (255).

I. Limit Switch Angle Assembly (Refer to IPL Figure 1)

(1) Attach the limit switch angle (215) and the actuator switch (235) with the two nuts (supplied with item 240), washers (230) and screws (supplied with item 240).

(2) Attach the limit switch angle assembly (205) with the screws (210).

J. Server Level Optical Sensor U101 (Refer to IPL Figure 1)

(1) Apply a thin coat of RTV (refer to Table 7001) to the top side of the server level sensor assembly (195) and attach to the pillow pack plate assembly (690) with two screws (200). Wipe off excess RTV from the server level sensor assembly (195) and pillow pack plate assembly (690).

(2) Carefully route sensor wire bundle through coffee maker housing (655) and route to the electrical enclosure assembly (465). Use RTV (refer to Table 7001) to seal openings through coffee maker housing (655) and electrical enclosure assembly (465). Install the four ring terminals (197) and heat shrink tube (199).

(3) Connect wire #57 to TB1-6, wire #58 to TB1-7, wire #64 to TB2-3 and the black wire triplet (#66, #67, and #69) to TB2-4. Wire tie as required.

(4) Attach the pillow pack plate assembly (690) with screws (695).

K. Brew Head Assembly, PN 11083-3 (Pre SB 25-30-03-9 and SB 25-30-03-11) (Refer to IPL Figure 1, unless specified)

(1) Assemble the brew head assembly (95) as follows (refer to IPL Figure 3):

(a) Install the water inlet assembly (75) to brew head (5) with screw (80), washer (85), and nut (90). If removed, install the male connector (95) to water inlet assembly (75).

(b) Insert gasket (15) and retainer plate (10) into brew head (5). Attach with screws (20), washers (25), and nuts (30).

(c) Install roller bushing (120) onto RH handle assembly (110). Place RH handle assembly (110) onto brew head (5) making sure roller bushing (120) is in the slot on the brew head (5). Install roller bushing (120) on LH handle assembly (115). Slide LH handle assembly (115) onto RH handle assembly (110) shaft so that roller bushing (120) goes into slot on the brew head (5). Install retaining rings (125) each side of handle onto hex shaft. Insert shaft bushing (130) into each end of the handle assemblies (110 and 115).

(d) Attach the ground wire assembly (135) to the RH handle assembly (110) with the bolt (140), washer (145) and flat washers (150).

(e) Insert plungers (175) and plunger springs (170) into plunger housings (180). Apply Loctite to set screws (165) and screw into plunger housings (180) until set screws (165) is in contact with the plunger springs (170). Attach plunger assemblies (155) to the brew head (5) with screws (160).

(f) Attach the slides (60) to the brew head (5) with screws (65) and rod guide screws (70).

(2) If removed, install the retainer bushings (120 and 145), or guide bushings (125 and 150). Install pot retainer brackets (105 and 130) to brew head with screws (115 and 140).

(3) Install brew head guides (155 and 165) with screws (160 and 170). Two brew head handle screws (180) go through the brew head guides (155 and 165) into the brew head handle assemblies (110 and 115, IPL Figure 3).

(4) Attach wire #20 from TB1-7 and wire #21 from TB2-4. Tie wire ties as required to tighten the brew head assembly (95). Attach cable wire clamp (60) by attaching nut (75), washer (70), and screw (65).

(5) On the flow control assembly, install the flow restrictor (75, IPL Figure 5) and attach the nut from the male run tee (95, IPL Figure 5).

L. Brew Head Assembly, PN 12254-3 (Post SB 25-30-03-9 and 25-30-03-11) (Refer to IPL Figure 1, unless specified)

(1) Assemble the brew head assembly (95A) as follows (refer to IPL Figure 3A):

(a) Install the water inlet assembly (70) to brew head (5) with screw (75), washer (80), and nut (85). If removed, install the male connector (90) to water inlet assembly (70).

(b) Insert brew head gasket (15) and retainer plate (10) into brew head (5). Attach with screws (20), washers (25), and nuts (30).

(c) Install roller bushing (115) onto RH handle assembly (105). Place RH handle assembly (105) onto brew head (5) making sure roller bushing (115) is in the slot on the brew head (5). Install roller bushing (115) on LH handle assembly (110). Slide LH handle assembly (110) onto RH handle assembly (105) shaft so that roller bushings (115) goes into slot on the brew head (5). Install retaining rings (120) on each side of handle onto hex shaft. Insert shaft bushings (125) into each end of the handle assemblies (105 and 110).

(d) Install two screws (40), star washers (45), and nuts (50) and attach the actuator angle (35).

(e) Attach the slides (55) to the brew head (5) with screws (60) and rod guide screws (65).

(f) Attach the ground wire assembly (130) to the RH brew head handle assembly (105) with the bolt (135), washer (140) and flat washers (145).

(2) If removed, install the retainer bushings (120 and 145), or guide bushings (125 and 150). Tighten set screws and install pot retainer brackets (105 and 130) to brew head with screws (115 and 140).

(3) Install brew head guides (155 and 165) with screws (160 and 170). Two brew handle screws (180) go through the guides into the brew head handle assemblies (105 and 110, IPL Figure 3A).

(4) Route wire bundle through protective sleeving. Attach wire #20 to TB1-7 and wire #21 to TB2-4. Wire tie as required. Attach cable wire clamp (60) with nut (75), washer (70), and screw (65).

(5) Attach wire #20 from TB1-7 and wire #21 from TB2-4. Tie wire ties as required to tighten the brew head assembly (95). Attach cable wire clamp (60) by attaching nut (75), washer (70), and screw (65).

(6) On the flow control assembly, install the flow restrictor (75, IPL Figure 5) and attach the nut from the male run tee (95, IPL Figure 5).

M. Switch Module Assembly SW1-SW5 (Refer to IPL Figure 1)

(1) Assemble switch mount assembly (35) as follows (refer to IPL Figure 6):

(a) Push in the LED (30) until it clicks into place.

(b) Position lens cap (5, 10, 15, 20, and 25) so that wording is properly oriented and push until it clicks into place.

(2) Install switch mount assembly (35) with four screws (40).

(3) Connect switch module connectors J3 and P3. Wire tie as required.

(4) Install switch mount cover (20) with four screws (30) and washer (25).

(5) Install brew knob assembly (10) with screw (15).

(6) Install the brew cup assembly (5).

SPECIAL TOOLS, FIXTURES, AND EQUIPMENT

TASK 25-30-03-940-801-A01

1. General

   A. All work described in this manual may be conducted in any normal work area with ambient environmental and physical conditions and whose environment has been cleaned for breathing oxygen equipment repair.

   B. Special tools, fixtures, and equipment required to perform the work function listed in all paragraphs of this manual are listed in Table 9001.

   NOTE: Equivalent substitutes may be used for listed items.

TABLE 25-30-03-99A-013-A01

Table 9001
Special Tools, Fixtures, and Equipment List

NOMENCLATURE	PARTS/SPEC NO	SOURCE (CAGE)
Adhesive, Silicone	736 Red AD-50005	Dow Corning Corporation (V71984) B/E Aerospace, Inc. (V63367)
Adhesive, Threadlock	Loctite® 242	Commercially Available
Angle Head Phillips Screw Driver	None	Commercially Available
Deleted		
Cloths, Cleaning, Low-Lint Type 1	None	Commercially Available
Coating, Alodine	1201 CM-50017	Henkel Corporation (V71410) B/E Aerospace, Inc. (V63367)
Coffee Maker Rail	PN 11160-1	B/E Aerospace, Inc. (V63367)
Distilled Water	None	Commercially Available
Heat Sink Compound	10-8109 CM-50019	Waldom, Inc. (V72653) B/E Aerospace, Inc. (V63367)
Isopropyl Alcohol	TT-I-735	Commercially Available
J2/P2 Crimper	PN M22520/1-01	Daniels Manufacturing Corporation (V11851)
J2/P2 Crimper Turret	PN M22520/1-02	Daniels Manufacturing Corporation (V11851)
J2/P2 Pin Extraction Tool	PN M81969/19-08	Daniels Manufacturing Corporation (V11851)
J2/P2 Pin Insertion Tool	PN M81969/17-04	Daniels Manufacturing Corporation (V11851)
J3/P3 Crimper	PN 90325-1	Tyco Electronic Corporation (V00779)
J3/P3 Pin Extraction Tool	PN 455822-2	Tyco Electronic Corporation (V00779)
Meg-Ohm Tester	None	Commercially Available
Nut Driver, 1/4 in.	None	Commercially Available

Table 9001 (Continued)
Special Tools, Fixtures, and Equipment List

NOMENCLATURE	PARTS/SPEC NO	SOURCE (CAGE)
Nylon Wire Tie Downs	MS3367-4-9	Commercially Available
Open End Wrench 1/4 in., 7/8 in.	None	Commercially Available
Rachet with 5/32 in. socket	None	Commercially Available
Stop Watch	None	Commercially Available
Sulphamic Acid Based Cleaner	Sulphamic acid 5% solution, LimeAway, Vinegar, or other agent that removes calcium, lime, and rust deposits	Commercially Available
Switch Alignment Tool	PN 01-906	EAO Switch Corporation (V61706)
Switch Lamp Remover	PN 02-906	EAO Switch Corporation (V61706)
Switch Lens Remover	PN 02-905	EAO Switch Corporation (V61706)
Switch Mounting Wrench	PN 01-907	EAO Switch Corporation (V61706)
Teflon Tape - 1/4 in	MS-STR-4 TT-7454	Swagelok Companies (V02570) B/E Aerospace, Inc. (V63367)
Temperature Test Brew Cup	PN 11239-1	B/E Aerospace, Inc. (V63367)
Thermometers	None	Commercially Available
Thread Lubricant	Never Seez NSWT-14 CM-50016	Bostik, Inc. (V1Y0Z7) B/E Aerospace, Inc. (V63367)
Translucent Silicone Rubber Adhesive/Sealant	732 Clear RTV RTV 108 AD-50002	Dow Corning Corporation (V71984) Momentive Performance Materials, Inc. (V01139) B/E Aerospace, Inc. (V63367)

ILLUSTRATED PARTS LIST

TASK 25-30-03-990-811-A01

1. Introduction

The ILLUSTRATED PARTS LIST is intended for use in provisioning, storing, and issuing replaceable parts for the component, and in identification of new and reclaimed parts.

TASK 25-30-03-990-812-A01

2. Vendor's Parts

A. Vendor's parts used in the unit and which are not altered by B/E Aerospace, Inc. are listed in the Detailed Parts List by the vendor's part number, vendor's description of the part, and vendor's code listed in parentheses following the description. Vendors' code symbols used in this publication are taken from the CAGE Cataloging Handbook, and consist of the applicable code symbol preceded by the Letter "V." Table 10001 shows a numerically arranged list of vendor codes used in this publication.

TABLE 25-30-03-99A-014-A01

Table 10001
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
VK8295	AVX Limited Admiral House Harlington Way Fleet, Hampshire GU51 4BB
V0JBM3	Hoke Inc Castle Controls Division 1901 Lynx Place Ontario, CA 91761
V00779	Tyco Electronics Corporation 2800 Fulling Mill Road Building-38 Middletown, PA 17057-3142
V01139	Momentive Performance Materials, Inc. 22 Corporate Woods Boulevard Albany, NY 12211-2374
V01295	Texas Instruments Incorporated DBA Texas Instruments Divison Semiconductor Group 12500 T I Boulevard Dallas, TX 75243
V02570	Swagelok Company 29500 Solon Road Solon, OH 44139-3449
V02697	Parker Hannifin Corporation 2360 Palumbo Drive Lexington, KY 40509-1048
V04713	Freescale Semiconductor Inc. 2100 East Elliot Road Tempe, AZ 85284

Table 10001 (Continued)
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V04845	Automatic Switch Company 50 Hanover Road Florham Park, NJ 07932-1591
V06090	Tyco Electronics Corporation Division Raychem 300 Constitution Drive Menlo Park, CA 94025-1140
V06383	Panduit Corp. DBA Panduit 17301 Ridgeland Avenue USA, IL 60477
V06915	Richco, Inc. 8145 River Drive Morton Grove, IL 60053
V06928	Teledyne Instruments Inc. DBA Teledyne 9855 Carroll Canyon Road San Diego, CA 92131
V1MPH7	Able Electronics Inc. 18 Sawgrass Drive Bellport, NY 11713
V1NUQ8	Teledyne Technologies Inc DBA Teledyne 1049 Camino Dos Rios Thousand Oaks, CA 91360
V1Y0Z7	Bostik, Inc. Construction and Distribution Division 205 Oliver Drive Calhoun, MI 49068-9543
V12060	Diodes, Inc. 3050 East Hillcrest Drive Westlake Village, CA 91362-3154
V13103	Thermalloy Company Inc. 2700 Research Drive Suite 200 Plano, TX 75074
V14726	Electro-Term, Inc. DBA Electro-Term/Hollingsworth-Plant 1 Crystal Park-Building. 4 1907 North West 40th Court Pompano Beach, FL 33064-8719

Table 10001 (Continued)
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V17547	Atco Technology, Inc. Bullock Magnetics Division 13560 Tonikan Road Apple Valley, CA 92307-0031
V18778	Thales Components Corporation 40 Commerce Way G Totowa, NJ 07512
V18612	Vishay Intertechnology Inc. 63 Lancaster Avenue Malvern, PA 19355
V2AA50	Aircraft Products Co. 10800 Pflumm Road Lenexa, KS 66215
V25281	Oetiker, Inc. 6317 Euclid Street Marlett, MI 48453-1426
V26405	Marathon Special Products Corporation 13300 Van Camp Road Bowling Green, OH 43402-9391
V27264	Molex Incorporated DBA Molex Automotive Division Molex Connector Corporation Use Cage Code 1UX99 For Cataloging. 2222 Wellington Court Lisle, IL 60532
V3A054	McMaster-Carr Supply Company 9630 Norwalk Boulevard Orlando, FL 32824-8133
V32ZE6	Neoperl Inc 171 Mattatuck Heights Waterbury, CT 06705
V32997	Bourns Inc. 1200 Columbia Avenue Riverside, CA 92507
V34371	Intersil Corporation 1001 Murphy Ranch Road Milpitas, CA 95035
V4M9W8	Xicon Passive Components 1000 North Main Street Mansfield, TX 76063

Table 10001 (Continued)
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V4Y667	ND Industries, Inc. 13929 Dinard Avenue Santa FE Springs, CA 90670-4920
V5V1P1	On Semiconductor Corporation 5005 East McDowell Road Maricopa, AZ 85008
V50088	ST Microelectronics Inc 1310 Electronics Drive Carrollton, TX 75006
V55104	Tri-Star Electronics International Inc. 2201 Rosecrans Avenue El Segundo, CA 90245
V59270	Selco Products, Inc. 7580 Stage Road Buena Park, CA 90621-1224
V6W912	MEG Technologies, Inc. 15381 Assembly Lane Huntington Beach, CA 92649-1327
V60495	Parker Hannifin Corporation Division Parflex Division Atlantic Tube Operations 200 Ram Ridge Road Chestnut Ridge NY 10977-6719
V61706	EAO Switch Corporation DBA Switches Plus 98 Washington Street Milford, CT 06460-3133
V63367	B/E Aerospace, Inc. 15701 W. 95th Street Lenexa, KS 66219
V66958	ST Microelectronics Inc. Lexington Corporation Center 10 McGuire Road 3rd Floor Lexington, MA 02421
V7X328	Replaced by 66958
V70485	Atlantic India Rubber Company, Inc 1437 Kentucky RT 1428 Hagerhill, KY 41222-8646
V71410	Henkel Corporation 32100 Stephenson Highway Oakland, MI-48071

Table 10001 (Continued)
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V71984	Dow Corning Corporation 2200 West Salzburg Road Midland, MI 48686-0001
V72653	Waldom, Inc. 1801 Morgan Street Winnebago, IL-61102
V73992	Tuthill Corp Hansem Coupling Div 1000 W Bagley Road Cuyahoga, OH-44017
V75915	Littelfuse Inc. 8755 W Higgins Road Suite 500 Chicago, IL 60631
V77342	Tyco Electronics Corporation 8010 Piedmont Triad Parkway Greensboro, NC 27409
V79074	Varflex Corporation DBA Manufacturing Electrical Insulating Sleeving 512 West Court Street Rome, NY 13440-4010
V82647	Sensata Technologies, Inc. 529 Pleasant Street Attleboro, MA 02703-2421
V83259	Parker Hannifin Corporation DBA Composite Sealing Systems Division Division Composite Sealing Systems Division 7664 Panasonic Way San Diego, CA 92154-8206
V83803	Peerless Electronics Inc. 700 Hicksville Road Suite 100 Bethpage, NY 11714
V85480	Brady United States 6555 West Good Hope Road Milwaukee, WI 53223
V91816	Circor Aerospace, Inc. DBA Circle Seal Controls 2301 Wardlow Road Corona, CA 92880-2894
V91833	Keystone Electronics Corp. 31-07 20th Road Astoria, NY 11105
V98978	DBA IERC Division CTS Electronic Components 413 North Moss Street Burbank, CA 91502

Table 10001 (Continued)
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V98893	Lavelle Industries, Inc. 665 Mchenry Street Burlington, WI 53105-2129
V99109	Chase Corporation DBA Royston Laboratories Division Humiseal Division 128 First Street Pittsburgh, PA 15238
V*10**	Replaced by 32ZE6
V**4**	Noble
V**1**	Replaced by 4M9W8
V**9**	The Rubber Group 3560 Lafayette Road Building. 2C Portsmouth, NH 03801

TASK 25-30-03-990-813-A01

3. Use of the Illustrated Parts List

A. When the part number is known:

Turn to the Numerical Index and locate the part number. Figure and item numbers on the illustration where the part appears are listed in the column to the right of the part number. Turn to the illustration and locate the item number. The corresponding item number in the accompanying Detailed Parts List will give part number, description, assembly relationship, quantity required for that particular application, and whether or not the part appears in the illustrations.

B. When the part number is not known:

Look through the figures and identify the part by appearance. Note the figure number and the item number. The corresponding Fig-Item number in the Detailed Parts List shows the part number, description, assembly relationship, effectivity code, and the number of units per assembly.

TASK 25-30-03-990-814-A01

4. Material Arrangement

A. Numerical Index

The Numerical Index is an alphanumeric listing of all the part numbers in the Detailed Parts List. Each part number is listed together with its corresponding item number, ILLUSTRATED PARTS LIST figure number, and the total required number of units per assembly. The figure-item number is composed of two parts. To the left of the dash is the figure number. It is the number of the figure in the ILLUSTRATED PARTS LIST that shows the part. To the right of the dash is the item number. It is the number that identifies a part in a figure.

B. Illustrations

The ILLUSTRATED PARTS LIST illustrations interrelate to the Numerical Index and the Detailed Parts List. The separate illustrations pictorially describe recommended disassembly/assembly sequence and support DISASSEMBLY and ASSEMBLY as well as the Numerical Index and the Detailed Parts List of this manual.

C. Detailed Parts List

(1) Fig-Item

The Detailed Parts List is arranged numerically by item numbers. These are the same item numbers listed in the Numerical Index. In the Detailed Parts List, the figure number portion of the figure-item number is listed only once per column. Immediately below a figure number are the item numbers that appear on that figure. A dash in front of an item number means the corresponding part is not shown in that or any other figure. Each item number is listed together with its corresponding part number, nomenclature (description, assembly relationship), effectivity code, and units per assembly.

(2) Part Number

The part number column contains the original manufacturer's part number. When standard parts are used, the standard part number is listed in this column. When B/E Aerospace, Inc. is not the original manufacturer the vendor's part number is shown in the part number column, and the CAGE vendor number and equivalent B/E Aerospace, Inc. part number are given in the nomenclature column.

(3) Airline Part Number

This column is left blank for airline internal use.

(4) Nomenclature

(a) Indenture System

The indenture system shows the relationship of parts and assembly to next higher assembly or installations, as follows:

1 2 3 4 5 6 7
Unit (Product which this manual describes)
. Detail parts for assembly into the unit
. Subassembly
. Attaching parts for subassembly
***
. . Detail parts for subassembly
. . Sub-subassembly
. . Attaching parts for sub-subassembly
***
. . . Detailed parts for sub-subassembly

(b) Attaching Parts

Attaching parts are captioned ATTACHING PARTS and are listed immediately following the parts attached. The *** symbol follows the last item of the attaching parts group.

(c) Vendor Code

Parts manufactured by companies other than B/E Aerospace, Inc. are identified by an appropriate vendor’s CAGE code following the nomenclature, and are preceded by the letter “V.” Standard parts such as AN, MS, etc., are not identified by a vendor code.

(5) Effectivity Code

When two or more units (main assembly) are listed in the same ILLUSTRATED PARTS LIST, a code letter (“A,” “B,” “C,” etc.) is assigned to each main unit. All subcomponents that are peculiar to a particular unit are identified by the same effectivity code letter as the unit. If parts are common to all units, the effectivity code column is left blank. The coding used in the ILLUSTRATED PARTS LIST in this manual is shown in Table 10002.

TABLE 25-30-03-99A-015-A01

Table 10002
Effectivity Guide

ASSEMBLY PN	IPL FIGURE NO.	EFF CODE
11225-1	1	A

(6) Units per assembly

Quantities specified in the “UNITS PER ASSY” column are the total number of each part required per assembly or subassembly and are not necessarily the total used per the complete unit.

TASK 25-30-03-990-815-A01

5. Parts Replacement Data

Interchangeability relationship between parts is identified in the “NOMENCLATURE” column. A list of terms used and their definition is as follows.

TERM	ABBREVIATION	DEFINITION
Alternate	ALT	Part fully meets functional and structural specifications, but differs either in overall external dimensions, connection installation, and/or mounting provisions, and may require additional rework or modifications to install in a specific application.
Optional	OPT	The part is fully interchangeable in form, fit, and function with other item numbers shown.
Preferred	PRFD	Part is preferred over the other optional parts shown.
Replaced By	REPLD BY	Part is replaced by and is interchangeable with item number listed in notation.
Replaces	REPLS	Part replaces and is interchangeable with number listed in notation.
Superseded By	SUPSD BY	Part is replaced by and is not interchangeable with item number listed in notation.
Supersedes	SUPSDS	Part replaces and is not interchangeable with item number listed in notation.
Service Bulletins	PRE SB POST SB SB RWK	Identifies when a service bulletin (SB) has an effect on a part. PRE shows the condition of the part before the work in the service bulletin is done; POST shows the condition after the work is done; RWK shows that the part is reworked by the service bulletin.

TASK 25-30-03-970-801-A01

6. Numerical Index

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
AN565D8H3		3	165	2
AN565D8H4		1	110	1
		1	135	1
AN960-08		1	70	1
		1	375	2
		1	395	2
		7	265	2
AN960-6		1B	20	2
		1B	40	2
AN960C4L		1	-565	6
		1	-585	6
AN960C6		7	140	1
AN960C6L		1	620	1
		1	810	1
		1	835	2
		1	-840	3
		1	885	5
		3	25	4
		3	85	1
		3A	25	4
		3A	80	1
AN960C8L		3	150	2
		3A	145	2
AN960D10L		7	-240	2
AN960D6L		1	25	1
		1	255	2
		1	795	2
B129831A001		1	-260A	1
C403B021R		7	15	1
		7	-15A	1
CCS25S8M0		1	-60A	1
		1	-610A	1
		1	-800A	1
		7	-255A	RF
CD4011BE		8	-45B	DEL
CD4011BF		8	45	DEL
		8	-45A	DEL
CD4049UBE		8	-40B	DEL
CD4049UBF		8	40	DEL
		8	-40A	DEL
CD4081BE		8	-55B	DEL
CD4081BF		8	55	DEL
		8	-55A	DEL
CD4082BE		8	-50B	DEL
CD4082BF		8	50	DEL
		8	-50A	DEL
CDC06E16S1P		7	-180	1

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
CO6505		7	-180A	1
CO6505-03		1B	-70A	2
E0540-2-008		1	-270A	1
E0540-802-016		4	50	3
		4	-195	DEL
E05402-005		1	-460A	1
E05402-012		1	-450A	1
E05402-016		4	-50A	3
EUS1512151W3D		6	-30B	5
F4NY250-0		1	-60B	1
		1	-610B	1
		7	-255B	RF
F4NY250NA		1	-800B	1
FN824C		1	550	4
GMT1056605		7	-160A	1
GMT105660S		7	-160B	1
GS14-010-03		3	135	1
		3A	130	1
HA1-3		1	-630	1
HA1-3-8		1	-625	1
HA1-4		1	-55	1
HA1-6		1	-305	3
HO#9BLK		1	-860	1
JE5		1	235	1
		1	-235A	1
KU63214		7	40	16
		7	65	20
		7	190	2
		8	-215	DEL
		8	-215A	DEL
L7812CV		8	30	DEL
		8	-30A	DEL
LM139AJ		8	-35A	DEL
LM239AJ		8	-35	DEL
LM239N		8	-35B	DEL
M23053-8-004C		1	-865	3
M230538-004C		1	-640	3
M83248-1-110		1	415	1
MCR264-008		7	-220	1
MPF960		8	-80A	DEL
MS15795-804		1	510	2
MS16562-189		1	-895	1
MS16624-4031		3	125	2
		3A	120	2
MS16633-4012		1	345	1
		1	-345B	1
MS18029-1S5		1	-520	1
		7	90	RF
MS204264D3-4		1	-715	4
MS21042-04		1	-560	3
		1	-580	3

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS21042-04		4	25	6
		7	205	2
MS21042-06		1	-605	6
		3	30	4
		3	90	1
		3A	30	4
		3A	85	1
		4	-10	6
		7	50	4
		7	75	4
		7	145	1
		8	-220	DEL
MS21042-08		1	75	1
		1	370	2
		1	390	2
		7	270	1
MS21042-3		7	235	1
MS21076-08		1	-710	2
MS24585C265		1	315	1
MS24693C24		1	490	6
		7	155	5
MS24693C25		1	735	3
		1	750	3
		1	760	2
		1	-760A	3
		1	770	3
		7	280	2
MS24693C26		1	30	4
		1	40	4
		1A	30	2
		7	135	2
MS24693C269		1	-115	1
		1	140	1
MS24693C270		1	685	4
MS24693C271		1	320	1
MS24693C28		3	20	4
		3A	20	4
		7	45	4
		7	70	4
MS24693C33		1B	10	2
		3	80	1
		3A	75	1
MS24693C4		1	500	2
		3	160	4
		7	-100A	2
		7	210	2
MS24693C49		1	200	2
		1	385	3
		1	695	6
		1	705	2
		1	-735A	3

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS24693C49		1	780	6
MS24693C50		1	65	1
		1	380	2
		1	400	2
		1	725	2
		5	65	2
		7	-260A	1
MS24693C51		1	-160	3
		1	170	3
MS24693C53		1	90	1
MS24693C8		7	25	2
MS26574-1		7	-185A	1
MS27212-1-5		7	95	1
MS3367-4-9		6	-80	4
MS35333-69		1	230	2
		3	50	2
		3A	45	2
MS35338-134		1	-230A	2
MS35338-135		1	505	2
MS35338-136		1	-555	4
		1B	35	2
		4	70	2
		7	30	16
		7	55	20
		7	115	5
		7	195	2
MS35338-42		3	145	1
		3A	140	1
MS35649-224		1	-225	2
		3	55	2
		3A	50	2
MS35649-244		7	105	2
MS51861-12C		1	-930	2
		1	-940	2
MS51957-13		1	210	2
		1	-570	3
		1	-590	3
		4	-35	6
MS51957-14		1	515	2
MS51957-16		1	-500A	2
MS51957-25		1	-880B	1
MS51957-26		1	615	1
		1	-790A	1
		1	820	6
		1	830	2
		1	-830A	3
MS51957-27		1	790	2
		1	805	1
		1	880	5
		1	-880A	7
MS51957-28		1	-805A	1

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS51957-28		1B	30	2
MS51957-30		1A	10	2
MS51957-32		1	-550A	4
MS51957-33		1	-550B	4
MS51957-43		1	300	1
		3	65	2
		3A	60	2
MS51957-45		1	475	4
		1	-535	8
		1	-535A	7
		7	260	1
MS51957-46		1	15	1
MS51958-62		7	245	1
MS51959-3		3A	40	2
MS51959-8		1	-220	2
		3	45	2
N2BK		1	-800C	1
NAS1149CN416R		4	30	12
NAS1149CN616R		1	-620A	1
		1	-835A	2
		1	-885A	8
		3A	-25A	4
		3A	-80A	1
		7	120	5
NAS1149CN632R		1B	-40A	2
		7	-140A	1
NAS1149CN816R		3A	-145A	2
NAS1149D0316H		7	240A	3
NAS1149DN616H		1	-25A	1
		1	-255A	2
		1	-795A	4
NAS1149DN816H		1	-480A	4
NAS1149FN832P		1	-70A	1
		1	-375A	2
		1	-395A	2
		7	-265A	2
NAS1635-06-48		1B	5	2
NAS1801-3-7		7	-245A	1
NAS1801-3-8		7	-245B	1
NAS1802-08-4		3	140	1
		3A	135	1
NAS671C6		4	65	2
		7	110	5
NDP24693C04		7	-85	2
NDP24693C25		1	250	2
NDP24693C26		2	15	2
NDP24693C4		7	100	2
OA250M		1	-595	3
OM250		4	15	3
OR067		1	405	1
R4105SBT		1	-645	3

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
R4105SBT		1B	-50A	3
		1B	-75A	3
		4	55	6
		5	-70	4
		7	-125	10
RFL1N08		8	80	DEL
RH25-16		7	-215	1
RV99-326		4	-105A	1
		4	105B	1
S0613A3303-044		1	-850A	1
SA105C103M		8	195	DEL
		8	-195A	DEL
SA105E104M		8	-205A	DEL
		8	-205B	DEL
		8	205	DEL
SA115E224M		8	175	DEL
		8	-175A	DEL
SA115E334M		8	170	DEL
		8	-170A	DEL
SA305E105M		8	180	DEL
		8	-180A	DEL
SC50269		7	-40B	16
		7	-65B	20
		7	-190B	2
SC50277		5	-65A	2
SS-400-2-2		4	85	1
SS400-1-2		1	275	1
		1	435	1
		3	95	1
		3A	90	1
		5	30	1
SS400-2-2		4	175	1
		4	-175A	1
		4	200	1
		4	-200A	1
		5	15	1
SS400-2-4		4	-120	1
SS400-3TMT		5	95	1
SS400-5-2		1	440	1
SS400SET		1	-285	1
		3	-96	1
		3A	-91	1
		4	90	1
		4	-115	1
		4	-150	1
		4	185	1
		4	210	1
		5	20	1
		5	35	1
		5	85	RF
		5	100	1

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
SS402-1		1	-280	1
		3	-97	1
		3A	-92	1
		4	95	1
		4	-110	1
		4	-145	1
		4	180	1
		4	205	1
		5	25	1
		5	40	1
		5	90	RF
		5	105	1
T85N11D114-12		8	25	DEL
TE097		1	-197	4
TFH		1	-635	1
TU310		1	-199	A/R
TYN640		7	-220A	1
U8225B6V		5	-10	2
UX8225B006V		5	10B	2
UX8225B6V		5	-10A	2
V23105A3003B201		8	-25A	DEL
V23105A50003A201		8	-25C	DEL
V250LA40A		7	-225	1
		7	-225A	1
W02F		8	-60B	DEL
		8	-60C	DEL
W02G		8	60	DEL
W02G		8	-60A	DEL
XAL50V330		8	10	DEL
		8	-10A	DEL
XR1903SNT		7	-230	2
XRL16V470		8	-200	DEL
		8	200A	DEL
ZVN4306A		8	-80B	DEL
AN960D8L		1	480	4
MS24693C49		1	470	3
NAS1149CN616R		1	-810A	1
.10M35		8	190	DEL
		8	-190A	DEL
.22M35		8	185	DEL
		8	-185A	DEL
03093113		1A	50	1
1-640512-0		6	50	1
1-640522-0		1	45	1
1.5KE39A		8	65	DEL
10 OHM		8	155	DEL
		8	-155A	DEL
10-23161089		6	-30A	5
		6	-30C	4
11051-11		1	530	1
		4	-1	RF

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
11051-13		4	-220	1
11051-5		4	-225	1
11051-7		4	-230	1
11051-9		4	215	1
11052-1		4	-235	1
11053-1		4	240	2
11055-1		1	-655	1
11056-1		4	45	3
11057-1		4	5	1
11058-1		6	75	1
11058-3		1	20	1
11059-1		7	170	1
11060-1		1	485	1
11061-1		1	745	1
11062-1		1	765	1
11063-1		1	755	1
11065-1		4	75	1
11065-3		4	100	1
11065-5		4	-80	1
11067-1		1	730	1
11067-3		1	-740	1
11068-1		3	105	1
		3A	100	1
11069-1		3	-75	1
		3A	-70	1
11069-3		3	-100	1
		3A	-95	1
11070-1		3	5	1
		3A	5	1
11071-1		1	690	1
11072-3		3	60	2
		3A	55	2
11073-1		3	15	1
11073-1		3A	15	1
11074-1		3	10	1
		3A	10	1
11075-1		1	420	1
11076-1		1	720	1
11080-1		1	540	1
11081-3		1	35	1
		6	-1	RF
11081-7		1	-35A	1
		6	-1A	RF
11082-3		1	-650	1
11082-7		1	-650A	1
11083-3		1	95	1
		3	-1	RF
11085-1		1	-310	1
11085-3		1	335	1
11085-5		1	350	1
11087-1		1	330	1
11088-1		1	675	1

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
11090-1		1	465	1
		7	-1	RF
11090-301		7	-5	1
11090-303		7	-165	1
11091-1		4	60	1
11093-1		7	-10	1
11094-1		4	-190	1
11095-301		8	-15	DEL
11095-503		1	495	1
11095-503		8	-1	RF
11095E		8	20	DEL
11098-1		7	150	1
11099-1		3	115	1
		3A	110	1
11099-3		3	110	1
		3A	105	1
11100-11		3	120	2
		3A	115	2
11100-13		1	-175	2
		3	130	RF
		3A	125	RF
11102-1		5	50	1
11106-1		1	180	2
11107-1		5	5	1
11107-5		5	5A	1
11110-1		1	775	2
11111-1		5	-55	1
11111-3		5	55A	1
11112-1		7	130	1
11114-1		1	-245	1
11114-7		1	265	1
11114-9		1	290	1
11117-1		1	195	1
11119-1		2	10	1
11120-1		1	-5	1
		2	-1	RF
11122-3		1	295	1
		5	-1	RF
11124-1		2	20	2
11125-1		1	355	1
11128-1		1	680	4
11132-1		1	360	1
11133-1		3	-180	2
11134-1		3	155	2
11135-1		3	-170	2
11136-1		3	-175	2
11154-1		1A	5	1
11154-3		1A	-20	1
11154-5		1A	-25	1
11154-7		1A	35	1
11170-1		1	340	1

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
11172-1		1	-307	1
		5	75	RF
11176-1		5	45	1
11201-1		6	70	1
11204-1		6	-85	RF
11205-1		4	-105	1
11205-3		4	-135	1
11205-5		4	-130	1
11206-1		8	5	DEL
11210-1		1	-925	1
11211-1		1	-905	1
11211-11		1	-920	1
11211-3		1	910	1
11211-5		1	-915	1
11222		8	-125A	DEL
11222-1		8	125	DEL
11224-1		7	20	1
11225-1		1	-1	RF
11225-5		1	-1A	RF
11230-1		1	-785A	1
11230-3		1	785	1
11240-1		6	5	1
11240-2		6	10	1
11240-3		6	15	1
11240-4		6	20	1
11240-6		6	25	1
11245-1		1	240	1
		3	35	1
11246-1		7	-80	2
1125		7	250	1
11258-1		1	-600	1
		4	40	1
1126		7	160	1
11261-1		1	-365	1
11263-1		1	325	1
11265-1		1	700	1
11267-1		1	525	RF
		1A	-1	RF
11268-1		1A	-45	1
11269-1		7	200	1
11274-1		6	-85A	RF
11279-1		1	130	1
11279-3		1	105	1
11280-1		1	120	1
		1	145	1
11281-1		1	-100	2
		3A	-160	RF
11284-1		3	70	2
		3A	65	2
11285-1		1	165	1
11285-3		1	155	1

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
11288-1		1	430	1
11289-1		1	445	1
11290-1		1	125	1
		1	150	1
11294-1		1	410	1
11296-1		1	-935	1
11298-1		1	-575	1
		4	20	1
11321-1		3	40	1
11327-1		7	175	1
11334-1		1	855	1
11335-1		1	875	1
11340-3		1	10	1
11376-1		1	845	1
11376-3		1	-845A	1
11377-1		1	-900	1
11378-1		1	-815	1
11389-1		1	890	1
1178-3		1A	40	1
11785-1		3	-135A	1
		3A	-130A	1
12252-1		3A	35	1
12254-3		1	95A	1
12254-3		3A	-1	RF
12255-1		1	215	1
12257-1		1	-205	1
12310-1		2	5	1
12321-1		1B	55	1
		1B	55A	1
12321-5		1B	15	1
12322-1		1B	-45	1
12349-1		1B	65	1
12355-3		1	525A	RF
12355-3		1B	-1	RF
12356-1		8	-235	DEL
12406-1		4	140	1
		5	-80	RF
12424-1		1B	25	1
12430-1		4	-160	1
12431-1		1	825	6
12432-1		1	670	1
12432-3		1	665	1
12618-1		1	-815A	1
		1	-130A	1
12628-3		1	-105A	1
12665-1		1	80	1
12678-1		6	-90	RF
12685		1	-425	1
12685-1		1	-425A	1
12688-1		1	85	1
12693-1		1B	-60	1

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
12K OHM		8	115	DEL
		8	-115A	DEL
150 OHM		8	130	DEL
		8	-130A	DEL
16700983		4	-155A	1
16700987		4	-155	1
1B31		8	-230	DEL
		8	-230A	DEL
1M OHM		8	145	DEL
1M OHM		8	-145A	DEL
1N4005		6	-60	4
		6	-60A	4
		6	-60B	3
1N755A		8	70	DEL
1N914		8	75	DEL
2-005E0540-80		1	460	1
		1	270	1
2-010E0540-80		1	455	1
2-010E0893-80		1	-455A	1
2-012E0540-80		1	450	1
2-044S0613-60		1	850	1
2-34105-2		1	-870	2
200298		7	275	1
205KB		1	260	1
2M OHM		8	105	DEL
		8	-105A	DEL
2N7000		8	85	DEL
3.3K OHM		8	140	DEL
		8	-140A	DEL
31-151-025		6	-40A	3
31-151025		6	40	3
		6	-40B	4
31-281-025		6	-35A	1
31-281025		6	35	1
31-282025		6	45	1
31-2820252		6	-45A	1
31-9310		6	-15A	1
317-1616-786		1B	-70	2
31818		6	-65	2
3201		1A	-15A	2
3386P1-200		8	-165A	DEL
		8	-165B	DEL
3386P200		8	165	DEL
34105		1B	-50	3
		1B	-75	3
350967-2		6	-55A	12
352-0981-1		1	-545B	4
3520-1038-01		5	-5B	1
3520-1054-03		1	-295A	1

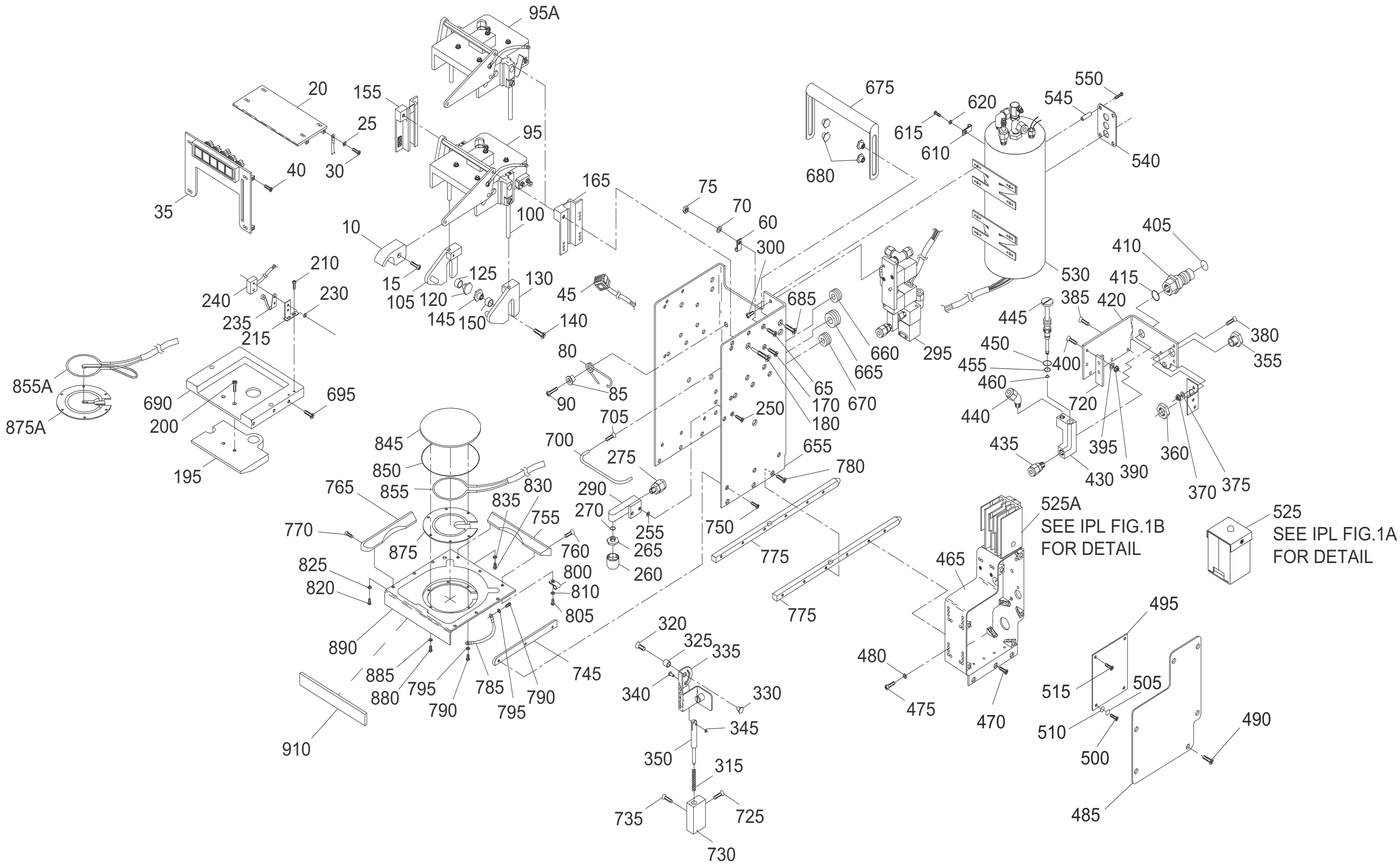
- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3520-1054-03		5	-1A	RF
3520-1071-01		5	-10C	2
3575-0549-01		1	875A	1
423030033		1	-260B	1
43-05-2		8	-210	DEL
		8	-210A	DEL
430531		3A	150	2
4443K721		5	-60	1
4606X102-223		8	95	DEL
		8	-95A	DEL
4610X101-103		8	-90	DEL
		8	-90A	DEL
47 OHM		8	150	DEL
47 OHM		8	-150A	DEL
5.6K OHM		8	135	DEL
		8	-135A	DEL
5100-12AS		1	-345A	1
510K OHM		8	110	DEL
		8	-110A	DEL
511 OHM		8	120	DEL
		8	-120A	DEL
532T12MP95		4	-125	1
534-3201		1A	15	2
55-0392-3		1	855A	1
600A10		7	60	1
600A8		7	35	1
600AGP10		7	-60A	1
600AGP8		7	-35A	1
62-0020-5		4	190A	1
640522		1	-45A	1
640545-2		6	-55	12
641300-2		1	-50	12
653		1	660	1
6MGMRG28N		6	30	5
7-179BA		8	225	DEL
7-179BA		8	-225A	DEL
7274-2-1		7	-185B	1
		7	-185C	1
7277-2-1		7	185	1
750K OHM		8	100	DEL
		8	-100A	DEL
8-1343792-8		8	-25B	DEL
8.45K OHM		8	160	DEL
		8	-160A	DEL
886		1	-545A	4
8876T13		1	60	1
		1	610	1
		1	800	1
		7	255	RF
910SBR		7	-250A	1
94639A106		1	545	4

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
9813504		7	-40A	16
		7	-65A	20
		7	-190A	2
		8	-215B	DEL

- ITEM NOT ILLUSTRATED



GRAPHIC 25-30-03-99B-013-A01

IPL Figure 1
Coffee Maker - Exploded View

TABLE 25-30-03-950-801-A01

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
-1	11225-1		COFFEE MAKER ASSEMBLY		RF
-1A	11225-5		DELETED		RF
-5	11120-1		• ASSEMBLY, BREW CUP (SEE IPL FIGURE 2 FOR DET BKDN)		1
10	11340-3		• ASSEMBLY, BREW KNOB, ONE PC		1
			ATTACHING PARTS		
15	MS51957-46		• SCREW, PAN HD (8-32 X 0.625)		1
			* * *		
20	11058-3		• COVER, SWITCH MOUNT		1
			ATTACHING PARTS		
25	AN960D6L		• WASHER, FLAT (SUPSD BY ITEM 25A)		1
-25A	NAS1149DN616H		• WASHER, ALUMINIUM (NO.6) (SUPSDS ITEM 25)		1
30	MS24693C26		• SCREW, FLAT HD (6-32 X 0.375)		4
			* * *		
35	11081-3		• ASSEMBLY, SWITCH MOUNT (SEE IPL FIGURE 6 FOR DET BKDN)		1
-35A	11081-7		DELETED		1
			ATTACHING PARTS		
40	MS24693C26		• SCREW, FLAT HD (6-32 X 0.375)		4
			* * *		
45	1-640522-0		• SOCKET HOUSING (V00779) (J3) (B/E AEROSPACE STOCK CODE TE-50448) (SUPSD BY ITEM 45A)		1
-45A	640522		• SOCKET HOUSING (V00779) (J3) (B/E AEROSPACE STOCK CODE TE-50448) (SUPSDS ITEM 45)		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
-50	641300-2		• SOCKET (V00779) (B/E AEROSPACE STOCK CODE TE-50452)		12
-55	HA1-4		• WHITE VARGLAS SILICONE RUBBER SLEEVING (6 IN) (V79074) (B/E AEROSPACE STOCK CODE TU-50034-4)		1
60	8876T13		• WIRE CLAMP 0.25 IN (6.35 MM) (V3A054) (SUPSD BY ITEM 60A)		1
-60A	CCS25S8M0		• WIRE CLAMP 0.25 IN (6.35 MM) (V06383) (SUPSDS ITEM 60) (SUPSD BY ITEM 60B)		1
-60B	F4NY250-0		• WIRE CLAMP 0.25 IN (6.35 MM) (V85480) (B/E AEROSPACE STOCK CODE CL-3201) (SUPSDS ITEM 60A)		1
			ATTACHING PARTS		
65	MS24693C50		• SCREW, FLAT HD (8-32 X 0.50)		1
70	AN960-08		• WASHER, FLAT (SUPSD BY ITEM 70A)		1
-70A	NAS1149FN832P		• WASHER, NO.8 (SUPSDS ITEM 70)		1
75	MS21042-08		• NUT, SELF-LOCKING (8-32)		1
			* * *		
80	12665-1		• SPRING, BREW HEAD		1
85	12688-1		• BUSHING, TORSION SPRING		1
			ATTACHING PARTS		
90	MS24693C53		• SCREW, FLAT HD (8-32 X 0.875)		1
			* * *		
95	11083-3		• ASSEMBLY, BREW HEAD (PRE SB 25-30-03-9 AND SB 25-30-03-11) (SEE IPL FIGURE 3 FOR DET BKDN)		1
95A	12254-3		• ASSEMBLY, BREW HEAD (POST SB 25-30-03-9 AND SB 25-30-03-11) (SEE IPL FIGURE 3A FOR DET BKDN)		1
-100	11281-1		• ROD GUIDE, RETAINER		2

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
105	11279-3		• BRACKET, POT RETAINER, LH		1
-105A	12628-3		DELETED		1
			ATTACHING PARTS		
110	AN565D8H4		DELETED		1
-115	MS24693C269		• SCREW, FLAT HD (10-32 X 0.3125)		1
120	11280-1		• BUSHING, RETAINER		1
125	11290-1		• BUSHING, GUIDE		1
			* * *		
130	11279-1		• BRACKET, POT RETAINER, RH		1
-130A	12628-1		DELETED		1
			ATTACHING PARTS		
135	AN565D8H4		DELETED		1
140	MS24693C269		• SCREW, FLAT HD (10-32 X 0.3125)		1
145	11280-1		• BUSHING, RETAINER		1
150	11290-1		• BUSHING, GUIDE		1
			* * *		
155	11285-3		• GUIDE, BREW HEAD SLOT, LH		1
			ATTACHING PARTS		
-160	MS24693C51		• SCREW, FLAT HD (8-32 X 0.625)		3
			* * *		
165	11285-1		• GUIDE, BREW HEAD SLOT, RH		1
			ATTACHING PARTS		
170	MS24693C51		• SCREW, FLAT HD (8-32 X 0.625)		3
			* * *		
-175	11100-13		• BUSHING, SHAFT		2

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
			ATTACHING PARTS		
180	11106-1		• SCREW, BREW HANDLE		2
			* * *		
195	11117-1		• ASSEMBLY, LEVEL SENSOR (U101)		1
-197	TE097		• • TERMINAL, RING (TPN TE-097)		4
-199	TU310		• • TUBING, HEAT SHRINK (TPN TU-310)		A/R
			ATTACHING PARTS		
200	MS24693C49		• SCREW, FLAT HD (8-32 X 0.4375)		2
			* * *		
-205	12257-1		• ASSEMBLY, LIMIT SWITCH ANGLE		1
			ATTACHING PARTS		
210	MS51957-13		• SCREW, PAN HD (4-40 X 0.25)		2
			* * *		
215	12255-1		• • LIMIT SWITCH ANGLE		1
			ATTACHING PARTS		
-220	MS51959-8		DELETED, SUPPLIED WITH ITEM 240		2
-225	MS35649-224		DELETED, SUPPLIED WITH ITEM 240		2
230	MS35333-69		• • WASHER (SUPSD BY ITEM 230A)		2
-230A	MS35338-134		• • WASHER, LOCK (SUPSDS ITEM 230)		2
			* * *		
235	JE5		• • ACTUATOR SWITCH (V91929) (SUPSD BY ITEM 235A)		1
-235A	JE5		• • ACTUATOR SWITCH (V83803) (TPN JE-5) (B/E AEROSPACE STOCK CODE SW-737) (SUPSDS ITEM 235)		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
240	11245-1		• • ASSEMBLY, LIMIT SWITCH		1
-245	11114-1		• ASSEMBLY, WATER FAUCET		1
			ATTACHING PARTS		
250	NDP24693C25		• SCREW, FLAT HD, NYLOK (6-32 X 0.3125) (V4Y667) (B/E AEROSPACE STOCK CODE SC-50268-25)		2
255	AN960D6L		• WASHER, FLAT (SUPSD BY ITEM 255A)		2
-255A	NAS1149DN616H		• WASHER, ALUMINIUM (NO.6) (SUPSDS ITEM 255)		2
			* * *		
260	205KB		• • AERATOR (V089X8) (SUPSD BY ITEM 260A)		1
-260A	B129831A001		• • AERATOR (V*10**) (SUPSDS ITEM 260) (SUPSD BY ITEM 260B)		1
-260B	423030033		• • AERATOR (V32ZE6) (TPN 42.3030.033) (B/E AEROSPACE STOCK CODE MISC-013) (SUPSDS ITEM 260A)		1
265	11114-7		• • HEAD, FAUCET		1
270	2-008E0540-80		• • O-RING (V83259) (SUPSD BY ITEM 270A)		1
-270A	E0540-2-008		• • O-RING (V02697) (B/E AEROSPACE STOCK CODE OR-50041-008) (SUPSDS ITEM 270)		1
275	SS400-1-2		• • MALE CONNECTOR (V02570) (TPN SS-400-1-2) (B/E AEROSPACE STOCK CODE PF-50261)		1
-280	SS402-1		• • • NUT (V02570) (TPN SS-402-1) (B/E AEROSPACE STOCK CODE PF-50262)		1
-285	SS400SET		• • • FERRULE SET (V02570) (TPN SS-400-SET) (B/E AEROSPACE STOCK CODE PF-50254)		1
290	11114-9		• • FAUCET SUBASSEMBLY		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
295	11122-3		• ASSEMBLY, FLOW CONTROL (SEE IPL FIGURE 5 FOR DET BKDN) (SUPSD BY ITEM 295A)		1
-295A	3520-1054-03		• ASSEMBLY, FLOW CONTROL (SEE IPL FIGURE 5 FOR DET BKDN) (SUPSDS ITEM 295)		1
			ATTACHING PARTS		
300	MS51957-43		• SCREW, PAN HD (8-32 X 0.375)		1
			* * *		
-305	HA1-6		• WHITE VARGLAS SILICONE RUBBER SLEEVING (6 IN) (457.2 MM) (V79074) (B/E AEROSPACE STOCK CODE TU-50034-6)		3
-307	11172-1		• FLOW, RESTRICTOR		1
-310	11085-1		• ASSEMBLY, LOCKING PIN		1
315	MS24585C265		• SPRING		1
			ATTACHING PARTS		
320	MS24693C271		• SCREW, FLAT HD (10-32 X 0.4375)		1
325	11263-1		• BUSHING, LOCKING PIN		1
330	11087-1		• POST, LOCKING PIN		1
			* * *		
335	11085-3		• • ASSEMBLY, HANDLE		1
340	11170-1		• • PIN, GROOVE		1
345	MS16633-4012		• • E-RING (SUPSD BY ITEM 345A)		1
-345A	5100-12AS		• • E-RING (V79136) (SUPSDS ITEM 345) (SUPSD BY ITEM 345B)		1
-345B	MS16633-4012		• • E-RING (SUPSDS ITEM 345A)		1
350	11085-5		• • PIN, LOCKING		1
355	11125-1		• BUSHING, GUIDE PIN		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
			ATTACHING PARTS		
360	11132-1		• NUT, ALIGNMENT BUSHING		1
			* * *		
-365	11261-1		• ASSEMBLY, MOUNTING BRACKET		1
			ATTACHING PARTS		
370	MS21042-08		• NUT, SELF-LOCKING (8-32)		2
375	AN960-08		• WASHER, FLAT (SUPSD BY ITEM 375A)		2
-375A	NAS1149FN832P		• WASHER, NO.8 (SUPSDS ITEM 375)		2
380	MS24693C50		• SCREW, FLAT HD (8-32 X 0.50)		2
385	MS24693C49		• SCREW, FLAT HD (8-32 X 0.4375)		3
390	MS21042-08		• NUT, SELF-LOCKING (8-32)		2
395	AN960-08		• WASHER, FLAT (SUPSD BY ITEM 395A)		2
-395A	NAS1149FN832P		• WASHER, NO.8 (SUPSDS ITEM 395)		2
400	MS24693C50		• SCREW, FLAT HD (8-32 X 0.5)		2
			* * *		
405	OR067		• • O-RING (TPN OR-067)		1
410	11294-1		• • PLUG, WATER INLET		1
415	M83248-1-110		• • O-RING (TPN M83248/1-110)		1
420	11075-1		• • BRACKET, MOUNTING		1
-425	12685		• • ASSEMBLY, VALVE (SUPSD BY ITEM 425A)		1
-425A	12685-1		• • ASSEMBLY, VALVE (SUPSDS ITEM 425)		1
430	11288-1		• • • MANIFOLD, INLET VENT DRAIN		1
435	SS400-1-2		• • • MALE CONNECTOR (V02570) (TPN SS-400-1-2) (B/E AEROSPACE STOCK CODE PF-50261)		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
440	SS400-5-2		• • • 45° ELBOW (V02570) (TPN SS-400-5-2) (B/E AEROSPACE STOCK CODE PF-50259)		1
445	11289-1		• • • ASSEMBLY, VALVE STEM		1
450	2-012E0540-80		• • • O-RING (V83259) (SUPSD BY ITEM 450A)		1
-450A	E05402-012		• • • O-RING (V02697) (B/E AEROSPACE STOCK CODE OR-50041-012) (SUPSDS ITEM 450)		1
455	2-010E0540-80		• • • O-RING (V83259) (PRE SB 25-30-03-10)		1
-455A	2-010E0893-80		• • • O-RING (V02697) (TPN 2-010-E0893-80) (B/E AEROSPACE STOCK CODE OR-50043-010) (POST SB 25-30-03-10)		1
460	2-005E0540-80		• • • O-RING (V83259) (SUPSD BY ITEM 460A)		1
-460A	E05402-005		• • • O-RING (V02697) (B/E AEROSPACE STOCK CODE OR-50041-005) (SUPSDS ITEM 460)		1
465	11090-1		• ASSEMBLY, ELECTRICAL ENCLOSURE (SEE IPL FIGURE 7 FOR DET BKDN)		1
			ATTACHING PARTS		
470	MS24693C49		• SCREW, FLAT HD (8-32 X 0.4375)		3
475	MS51957-45		• SCREW, PAN HD (8-32 X 0.50)		4
480	AN960D8L		• WASHER, FLAT (SUPSD BY ITEM 480A)		4
-480A	NAS1149DN816H		• WASHER, ALUMINIUM (NO.8) (SUPSDS ITEM 480)		4
			* * *		
485	11060-1		• COVER, ELECTRICAL ENCLOSURE		1
			ATTACHING PARTS		
490	MS24693C24		• SCREW, FLAT HD (6-32 X 0.25)		6
			* * *		
495	11095-503		• CIRCUIT CARD ASSEMBLY (SEE IPL FIGURE 8 FOR DET BKDN)		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
			ATTACHING PARTS		
500	MS24693C4		• SCREW (SUPSD BY ITEM 500A)		2
-500A	MS51957-16		• SCREW, PAN HD (4-40 X 0.4375) (SUPSDS ITEM 500)		2
505	MS35338-135		• SSP-WASHER, LOCK SPRING, REGULAR		2
510	MS15795-804		• WASHER, FLAT (0.125 X 0.03125)		2
515	MS51957-14		• SCREW, PAN HD (4-40 X 0.3125)		2
			* * *		
-520	MS18029-1S5		• COVER, TERMINAL BLOCK (TPN MS18029-1S-5)		1
525	11267-1		• RELAY ASSEMBLY (K1) (PRE SB 25-30-03-6)		RF
525A	12355-3		• KIT, SOLID STATE RELAY (POST SB 25-30-03-6)		RF
530	11051-11		• ASSEMBLY, TANK (SEE IPL FIGURE 4 FOR DET BKDN)		1
			ATTACHING PARTS		
-535	MS51957-45		• SCREW, PAN HD (8-32 X 0.50) (PRE SB 25-30-03-9)		8
-535A	MS51957-45		• SCREW, PAN HD (8-32 X 0.50) (POST SB 25-30-03-9)		7
			* * *		
540	11080-1		• COVER, THERMOSTAT		1
545	94639A106		• SPACER, (0.5 IN) (12.7 MM) (V3A054) (SUPSD BY ITEM 545A)		4
-545A	886		• SPACER, (0.5 IN) (12.7 MM) (V91833) (SUPSDS ITEM 545) (SUPSD BY ITEM 545B)		4
-545B	352-0981-1		• STANDOFF, THERMAL SWITCH (V2AA50) (SUPSDS ITEM 545A)		4
			ATTACHING PARTS		
550	FN824C		• SCREW, NYLON, 6-32 X 0.75 IN. (V6W912) (SUPSD BY ITEM 550A)		4

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
-550A	MS51957-32		• SCREW (SUPSDS ITEM 550) (SUPSD BY ITEM 550B)		4
-550B	MS51957-33		• SCREW (SUPSDS ITEM 550A)		4
-555	MS35338-136		• WASHER, LOCK		4
			* * *		
-560	MS21042-04		DELETED (SEE IPL FIGURE 4)		3
-565	AN960C4L		DELETED		6
-570	MS51957-13		DELETED (SEE IPL FIGURE 4)		3
-575	11298-1		DELETED (SEE IPL FIGURE 4)		1
-580	MS21042-04		DELETED (SEE IPL FIGURE 4)		3
-585	AN960C4L		DELETED		6
-590	MS51957-13		DELETED (SEE IPL FIGURE 4)		3
-595	OA250M		DELETED		3
-600	11258-1		DELETED (SEE IPL FIGURE 4)		1
-605	MS21042-06		DELETED (SEE IPL FIGURE 4)		6
610	8876T13		• WIRE CLAMP 0.25 IN (6.35 MM) (V3A054) (SUPSD BY ITEM 610A)		1
-610A	CCS25S8M0		• WIRE CLAMP 0.25 IN (6.35 MM) (V06383) (SUPSDS ITEM 610) (SUPSD BY ITEM 610B)		1
-610B	F4NY250-0		• WIRE CLAMP 0.25 IN (6.35 MM) (V85480) (B/E AEROSPACE STOCK CODE CL-3201) (SUPSDS ITEM 610A)		1
			ATTACHING PARTS		
615	MS51957-26		• SCREW, PAN HD (6-32 X 0.25)		1
620	AN960C6L		• WASHER, FLAT (SUPSD BY ITEM 620A)		1
-620A	NAS1149CN616R		• WASHER, NO. 6 (SUPSDS ITEM 620)		1
			* * *		

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
-625	HA1-3-8		• WHITE VARGLAS SILICONE RUBBER SLEEVING (4 IN) (101.6 MM) (TPN HA1-3/8)(V79074) (B/E AEROSPACE STOCK CODE TU-50034-1)		1
-630	HA1-3		• WHITE VARGLAS SILICONE RUBBER SLEEVING (12 IN) (304.8 MM) (V79074) (B/E AEROSPACE STOCK CODE TU-50034-3)		1
-635	TFH		• PTFE TUBE (0.25 X 0.125) (35.75 IN) (908 MM) (V60495) (B/E AEROSPACE STOCK CODE TU-50229-2)		1
-640	M230538-004C		• HEAT SHRINKING TUBING (0.62 DIA) (V06090) (TPN M23053/8-004-C) (B/E AEROSPACE STOCK CODE TU-310)		3
-645	R4105SBT		• TERMINAL, RING, (16-22 GA) (V14726) (B/E AEROSPACE STOCK CODE TE-097)		3
-650	11082-3		• ASSEMBLY, HOUSING		1
-650A	11082-7		DELETED		1
-655	11055-1		• • HOUSING, COFFEE MAKER		1
660	653		• • GROMMET (V70485) (B/E AEROSPACE STOCK CODE GR-50058)		1
665	12432-3		• • GROMMET		1
670	12432-1		• • GROMMET		1
675	11088-1		• • ASSEMBLY, HANDLE		1
			ATTACHING PARTS		
680	11128-1		• • ASSEMBLY, NUT STANDOFF		4
685	MS24693C270		• • SCREW, FLAT HD (10-32UNF X 0.375)		4
			* * *		
690	11071-1		• • ASSEMBLY, PILLOW PACK PLATE		1
			ATTACHING PARTS		
695	MS24693C49		• • SCREW, FLAT HD (8-32 X 0.738)		6
			* * *		

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
700	11265-1		• • REMOVAL HANDLE		1
			ATTACHING PARTS		
705	MS24693C49		• • SCREW, FLAT HD (8-32 X 0.738)		2
			* * *		
-710	MS21076-08		• • NUT, PLATE (8-32)		2
			ATTACHING PARTS		
-715	MS204264D3-4		• • RIVET SOLID		4
			* * *		
720	11076-1		• • ANGLE		1
			ATTACHING PARTS		
725	MS24693C50		• • SCREW, FLAT HD (8-32 X 0.5)		2
			* * *		
730	11067-1		• • BLOCK, LOCKING PIN ASSEMBLY		1
			ATTACHING PARTS		
735	MS24693C25		• • SCREW, FLAT HD (6-32UNC X 0.31) (SUPSD BY ITEM 735A)		3
-735A	MS24693C49		• • SCREW, FLAT HD (8-32 X 0.738) (SUPSDS ITEM 735)		3
			* * *		
-740	11067-3		DELETED		1
745	11061-1		• • SUPPORT, BASE PAN, LH		1
			ATTACHING PARTS		
750	MS24693C25		• • SCREW, FLAT HD (6-32UNC X 0.31)		3
			* * *		
755	11063-1		• • SUPPORT, BASE PAN END		1
			ATTACHING PARTS		

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
760	MS24693C25		• • SCREW, FLAT HD (6-32UNC X 0.31)		2
-760A	MS24693C25		DELETED		3
			* * *		
765	11062-1		• • SUPPORT, BASE PAN, RH		1
			ATTACHING PARTS		
770	MS24693C25		• • SCREW, FLAT HD (6-32UNC X 0.31)		3
			* * *		
775	11110-1		• • RAIL, SUPPORT		2
			ATTACHING PARTS		
780	MS24693C49		• • SCREW, FLAT HD (8-32 X 0.738)		6
			* * *		
785	11230-3		• • GROUND, WIRE ASSEMBLY (SUPSD BY ITEM 1380A) (PRE SB 25-30-03-13)		1
-785A	11230-1		• • GROUND, WIRE ASSEMBLY (SUPSDS ITEM 1380) (POST SB 25-30-03-13)		1
			ATTACHING PARTS		
790	MS51957-27		• • SCREW, PAN HD (6-32UNC X 0.312) (SUPSD BY ITEM 790A) (PRE SB 25-30-03-13)		2
-790A	MS51957-26		• • SCREW, PAN HD (6-32UNC X 0.25) (SUPSDS ITEM 790) (POST SB 25-30-03-13)		1
795	AN960D6L		• • WASHER, FLAT (SUPSD BY ITEM 795A) (PRE SB 25-30-03-13)		2
-795A	NAS1149DN616H		• • WASHER, FLAT (SUPSDS ITEM 795) (POST SB 25-30-03-13)		4
			* * *		

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
800	8876T13		• • WIRE CLAMP 0.25 IN (6.35 MM) (V3A054) (SUPSD BY ITEM 800A)		1
-800A	CCS25S8M0		• • WIRE CLAMP 0.25 IN (6.35 MM) (V06383) (SUPSDS ITEM 800) (SUPSD BY ITEM 800B)		1
-800B	F4NY250NA		• • WIRE CLAMP 0.25 IN (6.35 MM) (V85480) (TPN F4NY-250NA) (B/E AEROSPACE STOCK CODE CL-320) (SUPSDS ITEM 800A) (SUPSD BY ITEM 800C) (PRE SB 25-30-03-13)		1
-800C	N2BK		• • CLAMP, NYLON, BLACK 0.25 IN (6.35 MM) (V06915) (TPN N-2-BK) (B/E AEROSPACE STOCK CODE CL-315) (SUPSDS ITEM 800B) (POST SB 25-30-03-13)		1
			ATTACHING PARTS		
805	MS51957-27		• • SCREW, PAN HD (6-32UNC X 0.312) (SUPSD BY ITEM 805A)		1
-805A	MS51957-28		• • SCREW, PAN HD (6-32UNC X 0.375) (SUPSDS ITEM 805)		1
810	AN960C6L		• • WASHER, FLAT (SUPSD BY ITEM 810A)		1
-810A	NAS1149CN616R		• • WASHER, FLAT (NO.6) (SUPSDS ITEM 810)		1
			* * *		
-815	11378-1		• • ASSEMBLY, BASE PAN, HEATED, SUMP DRAIN		1
-815A	12618-1		DELETED		1
			ATTACHING PARTS		
820	MS51957-26		• • SCREW, PAN HD (6-32UNC X 0.25)		6
825	12431-1		• • WASHER, BASE PAN		6
830	MS51957-26		• • SCREW, PAN HD (6-32UNC X 0.25)		2
-830A	MS51957-26		DELETED		3
835	AN960C6L		• • WASHER, FLAT (SUPSD BY ITEM 835A)		2
-835A	NAS1149CN616R		• • WASHER, FLAT (SUPSDS ITEM 835)		2

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
-840	AN960C6L		DELETED		3
			* * *		
845	11376-1		• • • COVER, HEATER (PRE SB 25-30-03-13) (SUPSD BY ITEM 845A)		1
-845A	11376-3		• • • COVER, HEATER (POST SB 25-30-03-13) (SUPSDS ITEM 845)		1
850	2-044S0613-60		• • • SILICON O-RING, 60 SHORE (V02697) (B/E AEROSPACE STOCK CODE OR-50042-044) (SUPSD BY ITEM 850A)		1
-850A	S0613A3303-044		• • • SILICON O-RING, 60 SHORE (V02697) (B/E AEROSPACE STOCK CODE OR-50042-044) (SUPSDS ITEM 850)		1
855	11334-1		• • • HEATER, CAL-ROD (W1) (SUPSD BY ITEM 855A) (PRE SB 25-30-03-13)		1
855A	55-0392-3		• • • HEATER, 35 WATT (W1) (SUPSDS ITEM 855) (POST SB 25-30-03-13)		1
-860	HO#9BLK		• • • SLEEVING VARGLAS (V79074) (TPN HO#9-BLK) (B/E AEROSPACE STOCK CODE TU-50051)		1
-865	M23053-8-004C		• • • TUBING HEAT SHRINK (V06090) (TPN M23053/8-004-C) (B/E AEROSPACE STOCK CODE TU-310)		3
-870	2-34105-2		• • • TERMINAL, RING, 22-16 GAUGE, #6 LUG (V00779) (B/E AEROSPACE STOCK CODE TE-097) (POST SB 25-30-03-13)		2
875	11335-1		• • • HEATER PLATE, CAL-ROD (SUPSD BY ITEM 875A) (PRE SB 25-30-03-13)		1
875A	3575-0549-01		• • • HEATER, RETAINER (SUPSDS ITEM 875) (POST SB 25-30-03-13)		1
			ATTACHING PARTS		
880	MS51957-27		• • • SCREW, PAN HD (6-32UNC X 0.312) (PRE SB 25-30-03-13)		5
-880A	MS51957-27		• • • SCREW, PAN HD (6-32UNC X 0.312) (POST SB 25-30-03-13)		7

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
-880B	MS51957-25		• • • SCREW, PAN HD (POST SB 25-30-03-13)		1
885	AN960C6L		• • • WASHER, FLAT (PRE SB 25-30-03-13)		5
-885A	NAS1149CN616R		• • • WASHER, FLAT (NO.6) (POST SB 25-30-03-13)		8
			* * *		
890	11389-1		• • • BASE PAN, MACHINE		1
-895	MS16562-189		DELETED		1
-900	11377-1		• • • • BASE PAN, HEATED, MACHINE		1
-905	11211-1		DELETED		1
910	11211-3		• PLACARD		1
-915	11211-5		DELETED		1
-920	11211-11		DELETED		1
-925	11210-1		DELETED		1
-930	MS51861-12C		DELETED		2
-935	11296-1		DELETED		1
-940	MS51861-12C		DELETED		2

- ITEM NOT ILLUSTRATED

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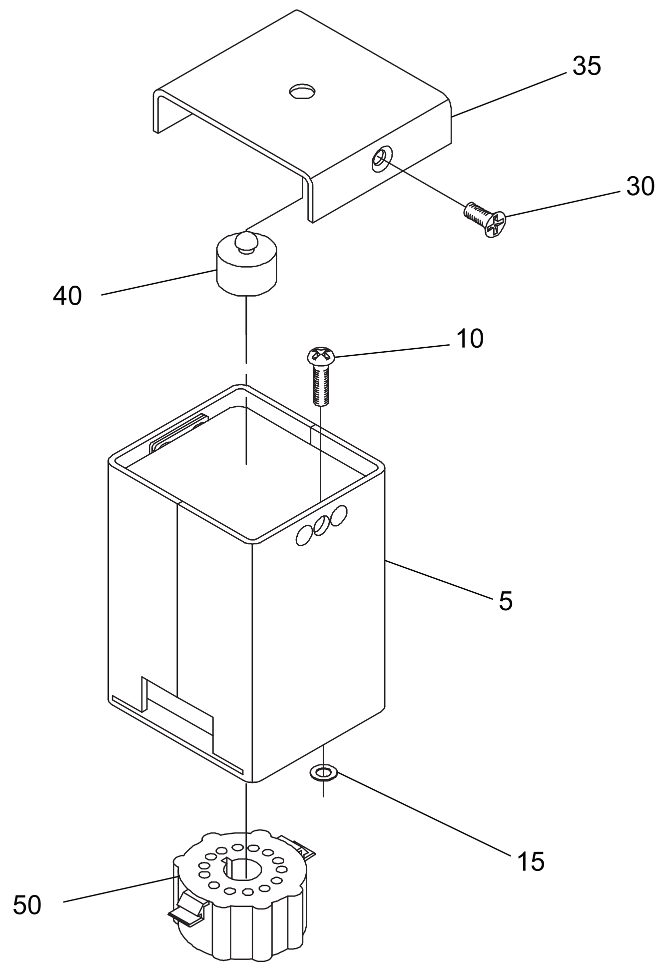


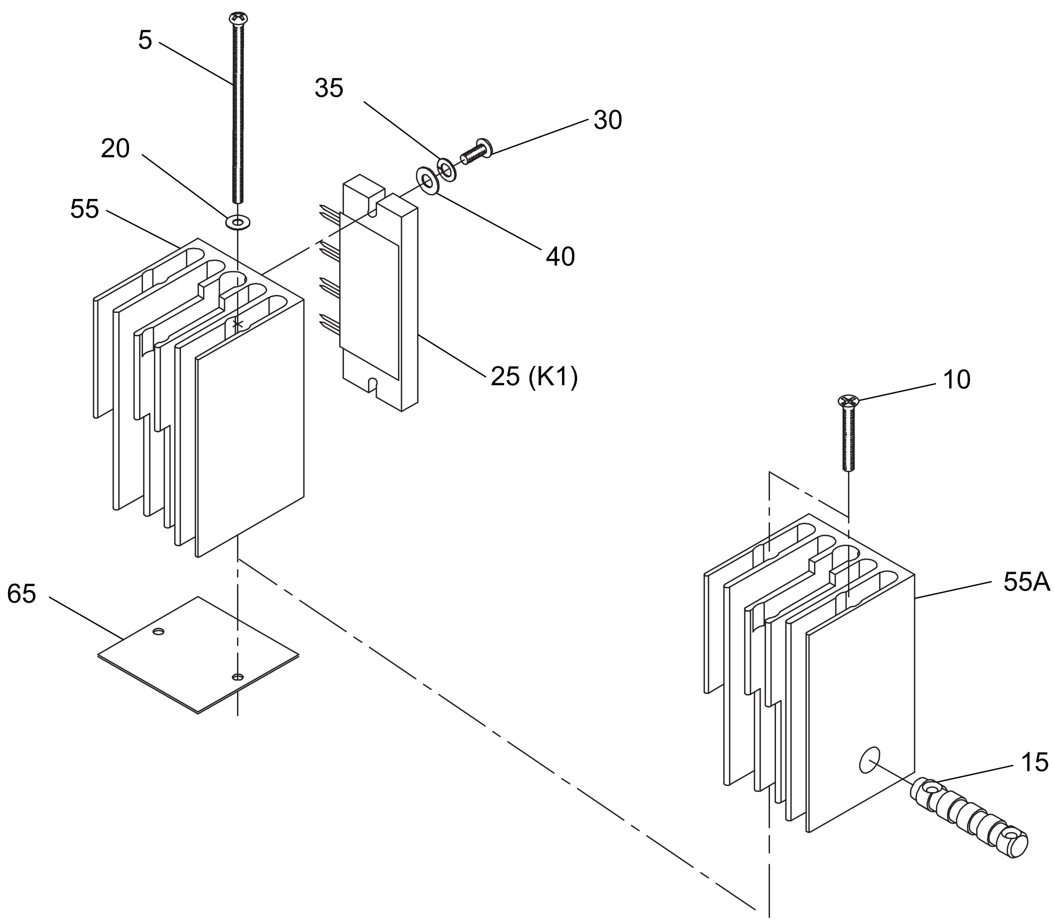
GRAPHIC 25-30-03-99B-014-A01

IPL Figure 1A
Relay Assembly
(SUPSD BY IPL FIGURE 1B)

TABLE 25-30-03-950-802-A01

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1A					
-1	11267-1		RELAY ASSEMBLY (K1) (SUPSD BY IPL FIGURE 1B) (SEE IPL FIGURE 1 FOR NHA)		RF
5	11154-1		• HOUSING, RELAY ASSEMBLY		1
			ATTACHING PARTS		
10	MS51957-30		• SCREW		2
15	534-3201		• WASHER, FLAT (SUPSD BY ITEM 15A)		2
-15A	3201		• WASHER, FLAT (V91833) (SUPSDS ITEM 15)		2
			* * *		
-20	11154-3		• • HOUSING		1
-25	11154-5		• • COVER HOUSING ASSEMBLY		1
			ATTACHING PARTS		
30	MS24693C26		• • SCREW		2
			* * *		
35	11154-7		• • COVER		1
40	1178-3		• • RUBBER BUMPER (V70485)		1
-45	11268-1		• SOCKET ASSEMBLY		1
50	03093113		• • SOCKET (V27264)		1

- ITEM NOT ILLUSTRATED



GRAPHIC 25-30-03-99B-015-A01

IPL Figure 1B
Solid State Relay Assembly
(SUPSDS IPL FIGURE 1A)

TABLE 25-30-03-950-803-A01

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1B					
-1	12355-3		KIT, SOLID STATE RELAY (SUPSDS IPL FIGURE 1A) (SEE IPL FIGURE 1 FOR NHA)		RF
			ATTACHING PARTS		
5	NAS1635-06-48		• SCREW (USED WITH ITEM 55A ONLY)		2
10	MS24693C33		• SCREW, PAN HD (6-32 X 0.75) (USED WITH ITEM 55 ONLY)		2
15	12321-5		• HOLD DOWN BAR (USED WITH ITEM 55A ONLY)		1
20	AN960-6		• WASHER, FLAT (USED WITH ITEM 55 ONLY)		2
			* * *		
25	12424-1		• • SOLID STATE RELAY ASSEMBLY (K1)		1
			ATTACHING PARTS		
30	MS51957-28		• • SCREW, PAN HD (6-32 X 0.375)		2
35	MS35338-136		• • WASHER, SELF LOCKING		2
40	AN960-6		• • WASHER, FLAT (SUPSD BY ITEM 40A)		2
-40A	NAS1149CN632R		• • WASHER, FLAT (SUPSDS ITEM 40)		2
			* * *		
-45	12322-1		• • • SOLID STATE RELAY		1
-50	34105		• • • TERMINAL, RING (V00779) (SUPSD BY ITEM 50A)		3
-50A	R4105SBT		• • • TERMINAL, RING (V14726) (B/E AEROSPACE STOCK CODE TE-097) (SUPSDS ITEM 50)		3
55	12321-1		• • HEAT SINK S.S. RELAY ASSEMBLY (SUPSD BY ITEM 55A) (BEFORE 08/29/03)		1
55A	12321-1		• • HEAT SINK S.S. RELAY ASSEMBLY (SUPSDS ITEM 55) (AFTER 08/29/03)		1
-60	12693-1		• • INSULATOR, S.S. RELAY		1
65	12349-1		• • COVER PLATE RELAY (USED WITH ITEM 55 ONLY)		1
-70	317-1616-786		• • PIN (V55104) (SUPSD BY ITEM 70A)		2

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1B					
-70A	CO6505-03		• • CONNECTOR - 7 PIN (TPN CO-6505-03) (SUPSDS ITEM 70)		2
-75	34105		• • RING TERMINAL (V00779) (SUPSD BY ITEM 75A)		3
-75A	R4105SBT		• • TERMINAL, RING (V14726) (B/E AEROSPACE STOCK CODE TE-097) (SUPSDS ITEM 75)		3

- ITEM NOT ILLUSTRATED

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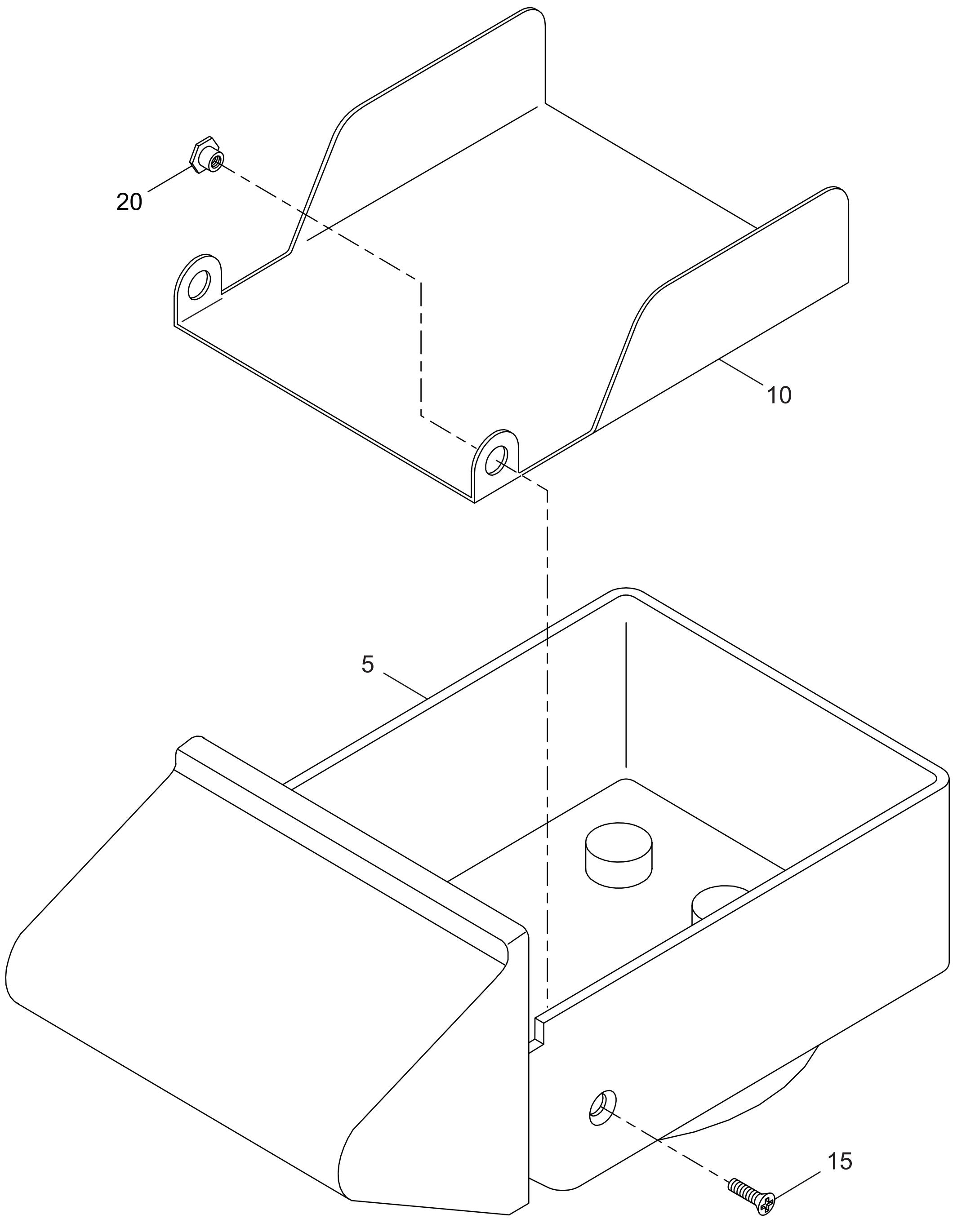


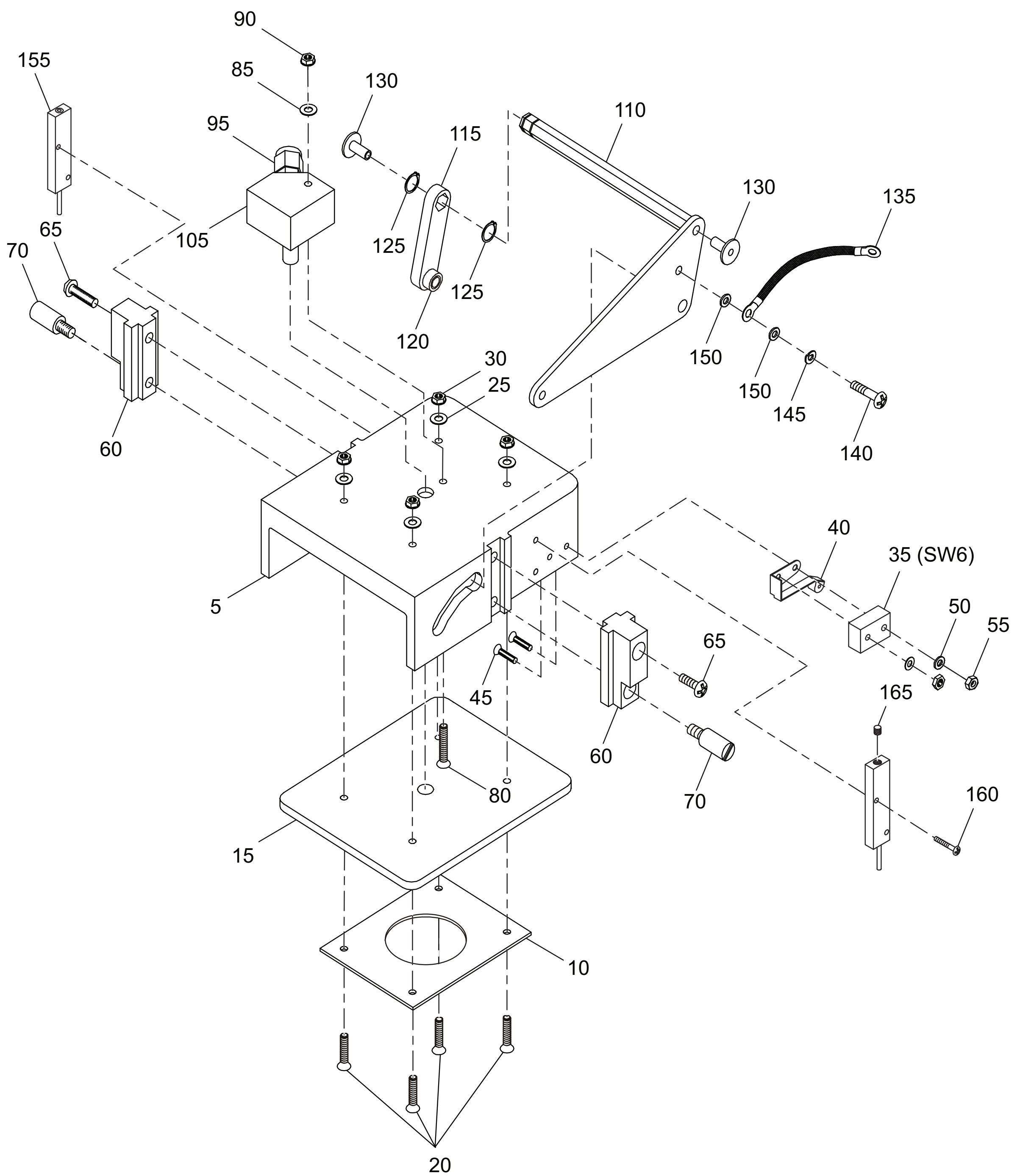
GRAPHIC 25-30-03-99B-016-A01

IPL Figure 2
Brew Cup Assembly

TABLE 25-30-03-950-804-A01

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
-1	11120-1		ASSEMBLY, BREW CUP (SEE IPL FIGURE 1 FOR NHA)		RF
5	12310-1		• BREW CUP (DEEP)		1
10	11119-1		• SCREEN		1
			ATTACHING PARTS		
15	NDP24693C26		• SCREW, NYLOCK (6-32 X 0.375) (V4Y667) (B/E AEROSPACE STOCK CODE SC-50268-26)		2
20	11124-1		• STANDOFF		2
			* * *		

- ITEM NOT ILLUSTRATED



GRAPHIC 25-30-03-99B-017-A01

IPL Figure 3
Brew Head Assembly

TABLE 25-30-03-950-805-A01

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3					
-1	11083-3		ASSEMBLY, BREW HEAD (PRE SB 25-30-03-9 & SB 25-30-03-11) (SEE IPL FIGURE 1 FOR NHA)		RF
5	11070-1		• BREW HEAD		1
10	11074-1		• PLATE, RETAINER		1
15	11073-1		• GASKET		1
			ATTACHING PARTS		
20	MS24693C28		• SCREW, FLAT HD (6-32 X 0.50)		4
25	AN960C6L		• WASHER, FLAT		4
30	MS21042-06		• NUT, SELF-LOCKING, HEX (6-32)		4
			* * *		
35	11245-1		• LIMIT SWITCH (SW6)		1
40	11321-1		• AUXILIARY ACTUATOR		1
			ATTACHING PARTS		
45	MS51959-8		• SCREW (2-56)		2
50	MS35333-69		• WASHER, STAR		2
55	MS35649-224		• NUT (2-56)		2
			* * *		
60	11072-3		• SLIDE		2
			ATTACHING PARTS		
65	MS51957-43		• SCREW, PAN HD (8-32 X 0.375)		2
70	11284-1		• SCREW ROD GUIDE		2
			* * *		
-75	11069-1		• ASSEMBLY, WATER INLET		1
			ATTACHING PARTS		
80	MS24693C33		• SCREW, FLAT HD (6-32 X 1.125)		1
85	AN960C6L		• WASHER, FLAT		1
90	MS21042-06		• NUT, SELF-LOCKING, HEX (6-32)		1
			* * *		

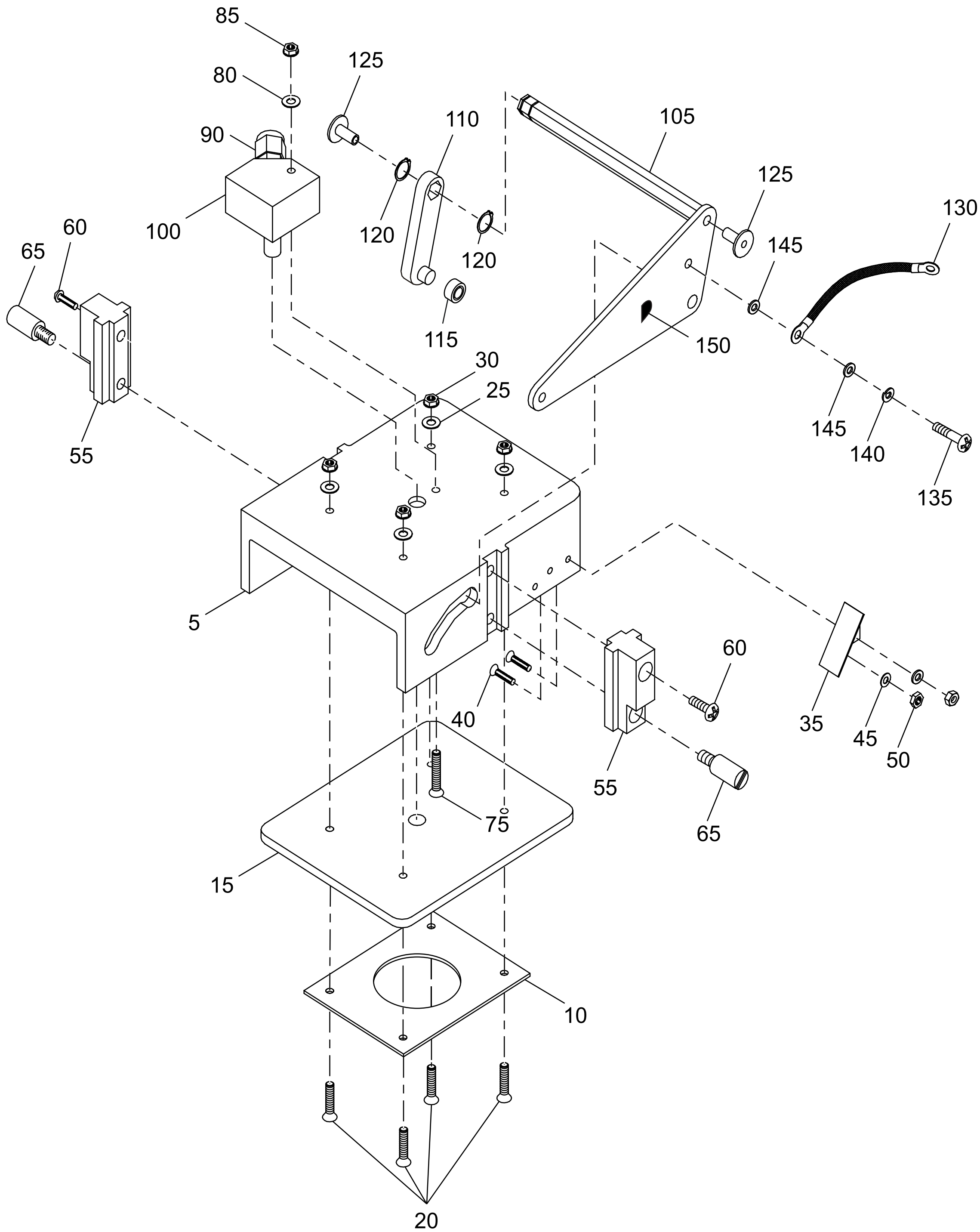
- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3					
95	SS400-1-2		• • CONNECTOR, MALE (V02570) (TPN SS-400-1-2) (B/E AEROSPACE STOCK CODE PF-50261)		1
-96	SS400SET		• • • FERRULE SET (V02570) (TPN SS-400-SET) (B/E AEROSPACE STOCK CODE PF-50254)		1
-97	SS402-1		• • • NUT (V02570) (TPN SS-402-1) (B/E AEROSPACE STOCK CODE PF-50262)		1
-100	11069-3		• • STEM		1
105	11068-1		• • FITTING WATER INLET		1
110	11099-3		• ASSEMBLY, HANDLE, RH		1
115	11099-1		• ASSEMBLY, HANDLE, LH		1
			ATTACHING PARTS		
120	11100-11		• BUSHING, ROLLER		2
125	MS16624-4031		• RING, RETAINING - EXTERNAL		2
130	11100-13		• BUSHING, SHAFT (SEE IPL FIGURE 1, ITEM 175)		RF
			* * *		
135	GS14-010-03		• GROUND WIRE ASSEMBLY (V1MPH7) (SUPSD BY ITEM 135A)		1
-135A	11785-1		• GROUND STRAP (SUPSDS ITEM 135)		1
			ATTACHING PARTS		
140	NAS1802-08-4		• BOLT (8-32 X 0.1875)		1
145	MS35338-42		• WASHER, SELF-LOCKING (NO. 8)		1
150	AN960C8L		• WASHER (NO. 8)		2
			* * *		
155	11134-1		• PLUNGER ASSEMBLY		2
			ATTACHING PARTS		
160	MS24693C4		• SCREW, FLAT HD COUNTER SUNK (4-40 X 0.375)		4
			* * *		

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3					
165	AN565D8H3		• • SCREW, SET (8-32)		2
-170	11135-1		• • SPRING, PLUNGER		2
-175	11136-1		• • PLUNGER		2
-180	11133-1		• • HOUSING, PLUNGER		2

- ITEM NOT ILLUSTRATED



GRAPHIC 25-30-03-99B-018-A01

IPL Figure 3A
Brew Head Assembly

TABLE 25-30-03-950-806-A01

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3A					
-1	12254-3		ASSEMBLY, BREW HEAD, TALL (POST SB 25-30-03-9 & SB 25-30-03-11) (SEE IPL FIGURE 1 FOR NHA)		RF
5	11070-1		• BREW HEAD		1
10	11074-1		• PLATE, RETAINER		1
15	11073-1		• GASKET		1
			ATTACHING PARTS		
20	MS24693C28		• SCREW, FLAT HD (6-32 X 0.50)		4
25	AN960C6L		• WASHER, FLAT (SUPSD BY ITEM 25A)		4
-25A	NAS1149CN616R		• WASHER, FLAT (NO. 6) (SUPSDS ITEM 25)		4
30	MS21042-06		• NUT, SELF-LOCKING, HEX (6-32)		4
			* * *		
35	12252-1		• ACTUATOR ANGLE		1
			ATTACHING PARTS		
40	MS51959-3		• SCREW, FLAT HD (2-56 X 0.25)		2
45	MS35333-69		• WASHER, STAR		2
50	MS35649-224		• NUT (2-56)		2
			* * *		
55	11072-3		• SLIDE		2
			ATTACHING PARTS		
60	MS51957-43		• SCREW, PAN HD (8-32 X 0.375)		2
65	11284-1		• SCREW ROD GUIDE		2
			* * *		
-70	11069-1		• ASSEMBLY, WATER INLET		1
			ATTACHING PARTS		
75	MS24693C33		• SCREW, FLAT HD (6-32 X 1.125)		1

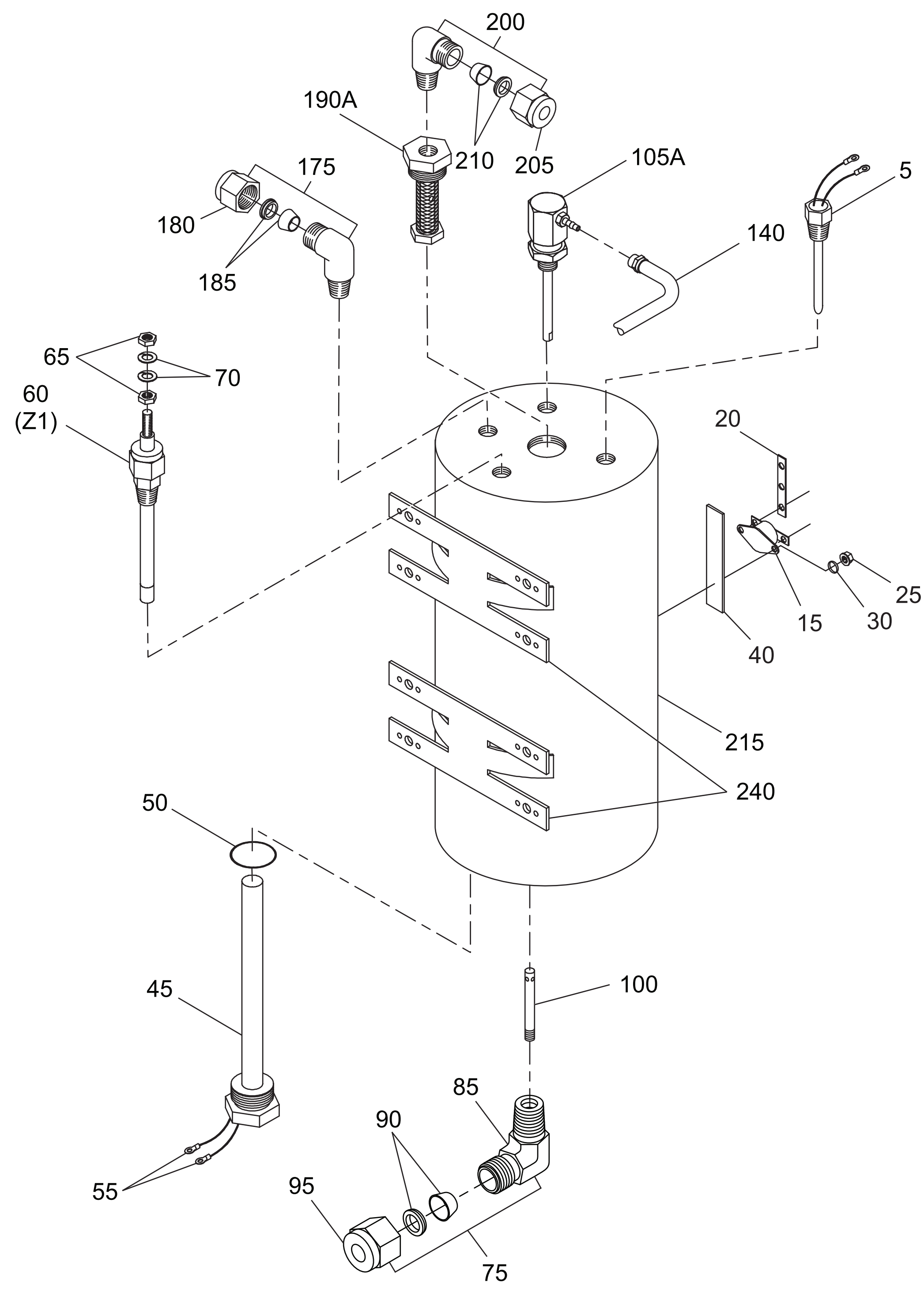
- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3A					
80	AN960C6L		• WASHER, FLAT (NO. 6) (SUPSD BY ITEM 80A)		1
-80A	NAS1149CN616R		• WASHER, FLAT (NO. 6) (SUPSDS ITEM 80)		1
85	MS21042-06		• NUT, SELF-LOCKING, HEX (6-32)		1
			* * *		
90	SS400-1-2		• • CONNECTOR, MALE (V02570) (TPN SS-400-1-2) (B/E AEROSPACE STOCK CODE PF-50261)		1
-91	SS400SET		• • • FERRULE SET (V02570) (TPN SS-400-SET) (B/E AEROSPACE STOCK CODE PF-50254)		1
-92	SS402-1		• • • NUT (V02570) (TPN SS-402-1) (B/E AEROSPACE STOCK CODE PF-50262)		1
-95	11069-3		• • STEM		1
100	11068-1		• • FITTING WATER INLET		1
105	11099-3		• ASSEMBLY, BREW HEAD HANDLE, RH		1
110	11099-1		• ASSEMBLY, BREW HEAD HANDLE, LH		1
			ATTACHING PARTS		
115	11100-11		• BUSHING, ROLLER		2
120	MS16624-4031		• RING, RETAINING - EXTERNAL		2
125	11100-13		• BUSHING, SHAFT (SEE IPL FIGURE 1, ITEM 175)		RF
			* * *		
130	GS14-010-03		• ASSEMBLY, GROUND WIRE (V1MPH7) (SUPSD BY ITEM 130A)		1
-130A	11785-1		• GROUND STRAP (SUPSDS ITEM 130)		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3A					
			ATTACHING PARTS		
135	NAS1802-08-4		• BOLT (8-32 X 0.18)		1
140	MS35338-42		• WASHER, SELF-LOCKING (NO. 8)		1
145	AN960C8L		• WASHER, FLAT (SUPSD BY ITEM 145A)		2
-145A	NAS1149CN816R		• WASHER, FLAT (NO. 8) (SUPSDS ITEM 145)		2
			* * *		
150	430531		• DECAL, DIE-CUT		2
-160	11281-1		• ROD, GUIDE RETAINER (SEE IPL FIGURE 1, ITEM 100)		RF

- ITEM NOT ILLUSTRATED



GRAPHIC 25-30-03-99B-019-A01

IPL Figure 4
Tank Assembly

TABLE 25-30-03-950-807-A01

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
4					
-1	11051-11		TANK ASSEMBLY (SEE IPL FIGURE 1 FOR NHA)		RF
5	11057-1		• PROBE, TEMPERATURE (RTD1)		1
-10	MS21042-06		• NUT, SELF-LOCKING (6-32)		6
15	OM250		• THERMOSWITCH (S1 TO S3) (V59270) (TPN OM-250) (B/E AEROSPACE STOCK CODE TS-50075)		3
20	11298-1		• BUSS BAR (BB1)		1
			ATTACHING PARTS		
25	MS21042-04		• NUT, SELF-LOCKING (4-40)		6
30	NAS1149CN416R		• WASHER NO.4		12
-35	MS51957-13		• SCREW, PAN HD (4-40 X 0.25)		6
			* * *		
40	11258-1		• HEAT SINK		1
45	11056-1		• HEATER, IMMERSION (H1 TO H3)		3
50	E0540-802-016		• O-RING (TPN E0540-80 2-016) (SUPSD BY ITEM 50A)		3
-50A	E05402-016		• O-RING (V02697) (B/E AEROSPACE STOCK CODE OR-50041-016) (SUPSDS ITEM 50)		3
55	R4105SBT		• TERMINAL, RING (V14726) (B/E AEROSPACE STOCK CODE TE-097)		6
60	11091-1		• PROBE, TANK LEVEL (Z1)		1
			ATTACHING PARTS		
65	NAS671C6		• NUT		2
70	MS35338-136		• WASHER, LOCKING		2
			* * *		
75	11065-1		• ASSEMBLY, FILLER TUBE		1
-80	11065-5		• • ELBOW		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
4					
85	SS-400-2-2		• • • ELBOW MALE (V02570) 90°, 0.25 OD X 0.625 NPT, SST (B/E AEROSPACE STOCK CODE PF-50256)		1
90	SS400SET		• • • • FERRULE SET (V02570) (TPN SS-400-SET) (B/E AEROSPACE STOCK CODE PF-50254)		1
95	SS402-1		• • • • NUT (V02570) (TPN SS-402-1) (B/E AEROSPACE STOCK CODE PF-50262)		1
100	11065-3		• • FILLER TUBE		1
-105	11205-1		• PRESSURE RELIEF ASSEMBLY (SUPSD BY ITEM 105A)		1
-105A	RV99-326		• PRESSURE RELIEF VALVE (95 PSIG) (V91816) (SUPSDS ITEM 105) (SUPSD BY ITEM 105B)		1
105B	RV99-326		• VALVE, PRESSURE RELIEF, 95 PSIG CRACK (V0JBM3) (B/E AEROSPACE STOCK CODE VA-50102) (SUPSDS ITEM 105A)		1
-110	SS402-1		• • NUT (V02570) (USE WITH ITEM 105)		1
-115	SS400SET		• • FERRULE SET (V02570) (USE WITH ITEM 105)		1
-120	SS400-2-4		• • ELBOW (V02570) (USE WITH ITEM 105)		1
-125	532T12MP95		• • PRESSURE RELIEF VALVE, 95 PSIG (V91816) (TPN 532T1-2MP-95) (USE WITH ITEM 105)		1
-130	11205-5		• • STREET ELBOW (USE WITH ITEM 105)		1
-135	11205-3		• • STEM (USE WITH ITEM 105)		1
140	12406-1		• ASSEMBLY, PRESSURE RELIEF HOSE		1
-145	SS402-1		• • NUT (V02570) (TPN SS-402-1) (B/E AEROSPACE STOCK CODE PF-50262)		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
4					
-150	SS400SET		• • FERRULE SET (V02570) (TPN SS-400-SET) (B/E AEROSPACE STOCK CODE PF-50254)		1
-155	16700987		• • CLAMP (V25281) (SUPSD BY ITEM 155A)		1
-155A	16700983		• • CLAMP (V25281) (B/E AEROSPACE STOCK CODE CL-50010) (SUPSDS ITEM 155)		1
-160	12430-1		• • FITTING, BARBED		1
175	SS400-2-2		• ELBOW (V02570) (TPN SS-400-2-2) (SUPSD BY ITEM 175A)		1
-175A	SS400-2-2		• ELBOW, MALE (V02570) (TPN SS-400-2-2) (B/E AEROSPACE STOCK CODE PF-50256) (SUPSDS ITEM 175)		1
180	SS402-1		• • NUT (V02570) (TPN SS-402-1) (B/E AEROSPACE STOCK CODE PF-50262)		1
185	SS400SET		• • FERRULE SET (V02570) (TPN SS-400-SET) (B/E AEROSPACE STOCK CODE PF-50254)		1
-190	11094-1		• FILTER OUTLET (SUPSD BY ITEM 190A)		1
190A	62-0020-5		• FILTER ASSEMBLY (SUPSDS ITEM 190)		1
-195	E0540-802-016		DELETED		DEL
200	SS400-2-2		• ELBOW (V02570) (TPN SS-400-2-2) (SUPSD BY ITEM 200A)		1
-200A	SS400-2-2		• ELBOW (V02570) (TPN SS-400-2-2) (B/E AEROSPACE STOCK CODE PF-50256) (SUPSDS ITEM 200)		1
205	SS402-1		• • NUT (V02570) (TPN SS-402-1) (B/E AEROSPACE STOCK CODE PF-50262)		1
210	SS400SET		• • FERRULE SET (V02570) (TPN SS-400-2-2) (B/E AEROSPACE STOCK CODE PF-50254)		1
215	11051-9		• TANK SUBASSEMBLY		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
4					
-220	11051-13		• • TUBE (4.00 OD X 0.065 WALL)		1
-225	11051-5		• • END CAP, UPPER		1
-230	11051-7		• • END CAP, LOWER		1
-235	11052-1		• • MOUNTING, BRACKET ASSEMBLY		1
240	11053-1		• • ATTACH MOD, BRACKET TANK		2

- ITEM NOT ILLUSTRATED

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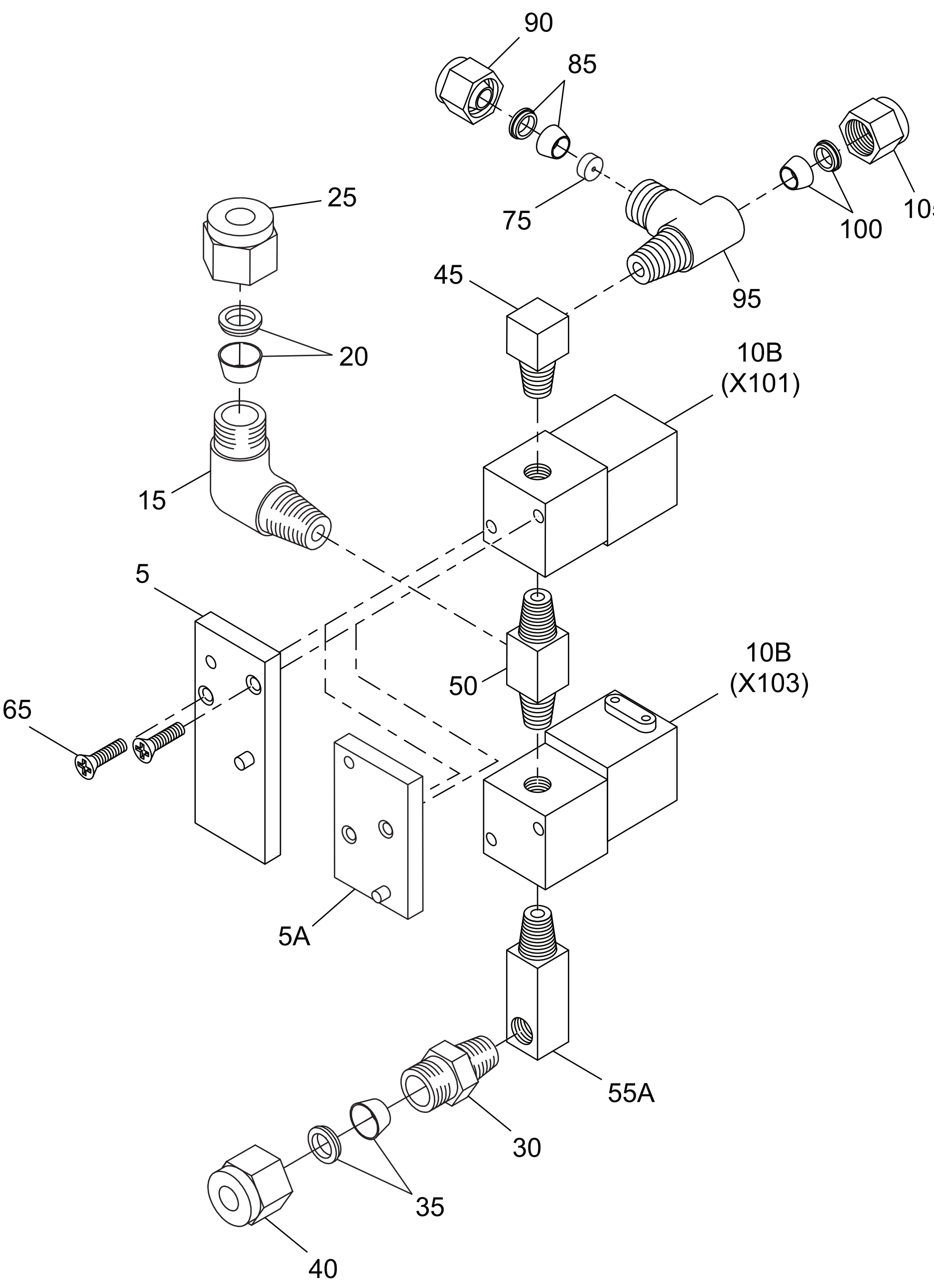

GRAPHIC 25-30-03-99B-020-A01

IPL Figure 5
Flow Control Assembly

TABLE 25-30-03-950-808-A01

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
5					
-1	11122-3		ASSEMBLY, FLOW CONTROL (SEE IPL FIGURE 1 FOR NHA) (SUPSD BY ITEM 1A)		RF
-1A	3520-1054-03		ASSEMBLY, FLOW CONTROL (SEE IPL FIGURE 1 FOR NHA) (SUPSDS ITEM 1)		RF
5	11107-1		• PLATE ASSEMBLY, SOLENOID MOUNTING (SUPSD BY ITEM 5A)		1
5A	11107-5		• PLATE, SOLENOID MOUNTING ASSEMBLY (SUPSDS ITEM 5) (SUPSD BY ITEM 5B)		1
-5B	3520-1038-01		• ADAPTER PLATE ASSEMBLY (SUPSDS ITEM 5A)		1
-10	U8225B6V		• VALVE, SOLENOID (V04845) (101, 103) (B/E AEROSPACE STOCK CODE VA-50104) (SUPSD BY ITEM 10A)		2
-10A	UX8225B6V		• VALVE, SOLENOID (V04845) (X101, X103) (B/E AEROSPACE STOCK CODE VA-50104) (SUPSDS ITEM 10) (SUPSD BY ITEM 10B)		2
10B	UX8225B006V		• VALVE, SOLENOID (V04845) (X101, X103) (B/E AEROSPACE STOCK CODE VA-50104) (SUPSDS ITEM 10A) (SUPSD BY ITEM 10C) (PRE SIL 3520-1071 & 1072-SIL-001)		2
-10C	3520-1071-01		• VALVE ASSEMBLY, 2 WAY (X101, X103) (SUPSDS ITEM 10B) (POST SIL 3520-1071 & 1072-SIL-001)		2
15	SS400-2-2		• ELBOW, MALE, 90° (0.25 OD X 0.125) (V02570) (TPN SS-400-2-2) (B/E AEROSPACE STOCK CODE PF-50256)		1
20	SS400SET		• • FERRULE SET (V02570) (TPN SS-402-SET) (B/E AEROSPACE STOCK CODE PF-50254)		1
25	SS402-1		• • NUT (V02570) (TPN SS-402-1) (B/E AEROSPACE STOCK CODE PF-50262)		1
30	SS400-1-2		• CONNECTOR, MALE (0.25 OD X 0.125) (V02570) (TPN SS-400-1-2) (B/E AEROSPACE STOCK CODE PF-50261)		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
5					
35	SS400SET		• • FERRULE SET (V02570) (TPN SS-402-SET) (B/E AEROSPACE STOCK CODE PF-50254)		1
40	SS402-1		• • NUT (V02570) (TPN SS-402-1) (B/E AEROSPACE STOCK CODE PF-50262)		1
45	11176-1		• ELBOW, STREET		1
50	11102-1		• TEE		1
-55	11111-1		• FITTING, WATER (SUPSD BY ITEM 55A)		1
55A	11111-3		• FITTING, WATER (SUPSDS ITEM 55)		1
-60	4443K721		• PLUG (V3A054) (USE WITH ITEM 55)		1
65	MS24693C50		• SCREW, FLAT HD (8-32 X 0.500) (SUPSD BY ITEM 65A)		2
-65A	SC50277		• SCREW, FLAT HD (M3 X 0.50 X 12 L) (TPN SC-50277) (SUPSDS ITEM 65)		2
-70	R4105SBT		• TERMINAL, RING (V14726) (B/E AEROSPACE STOCK CODE TE-097)		4
75	11172-1		• FLOW RESTRICTOR (SEE IPL FIGURE 1, ITEM 307)		RF
-80	12406-1		• HOSE ASSEMBLY, PRESSURE RELIEF (SEE IPL FIGURE 4, ITEM 140)		RF
85	SS400SET		• • FERRULE SET (V02570) (TPN SS-400-SET) (B/E AEROSPACE STOCK CODE PF-50254) (SEE IPL FIGURE 4, ITEM 150)		RF
90	SS402-1		• • NUT (V02570) (TPN SS-402-1) (B/E AEROSPACE STOCK CODE PF-50262) (SEE IPL FIGURE 4, ITEM 145)		RF
95	SS400-3TMT		• TEE, MALE RUN (V02570) (TPN SS-400-3TMT) (B/E AEROSPACE STOCK CODE PF-50258)		1
100	SS400SET		• • FERRULE SET (V02570) (TPN SS-400-SET) (B/E AEROSPACE STOCK CODE PF-50254) (SEE IPL FIGURE 4, ITEM 150)		1
105	SS402-1		• • NUT (V02570) (TPN SS-402-1) (B/E AEROSPACE STOCK CODE PF-50262) (SEE IPL FIGURE 4, ITEM 145)		1

- ITEM NOT ILLUSTRATED

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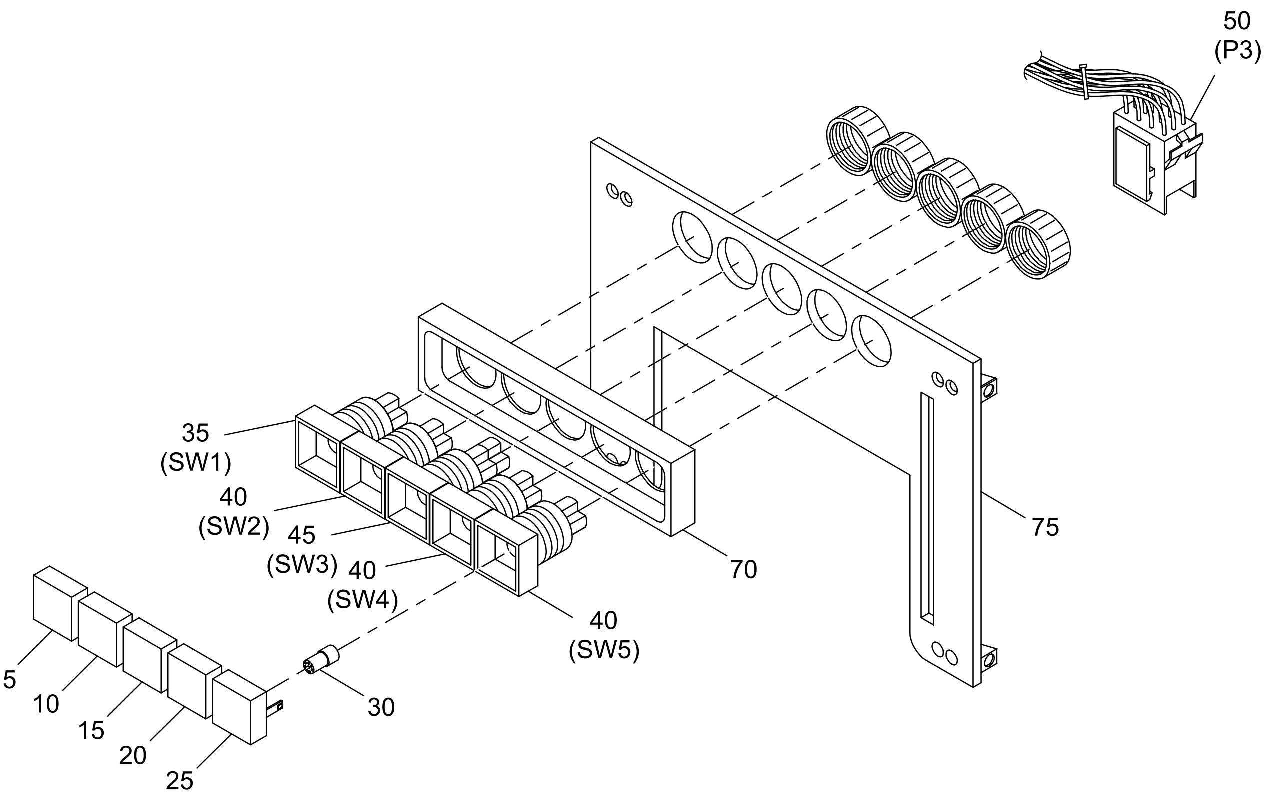

GRAPHIC 25-30-03-99B-021-A01

IPL Figure 6
Switch Assembly

TABLE 25-30-03-950-809-A01

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
6					
-1	11081-3		ASSEMBLY, SWITCH MOUNT (SEE IPL FIGURE 1 FOR NHA)		RF
-1A	11081-7		DELETED		RF
5	11240-1		• LENS, ON/OFF		1
10	11240-2		• LENS, BREW		1
15	11240-3		• LENS, HOT PLATE		1
-15A	31-9310		DELETED		1
20	11240-4		• LENS, HOT WATER		1
25	11240-6		• LENS ASSEMBLY, TEST-LOW WATER		1
30	6MGMRG28N		• LAMP, LED, WARM WHITE (PRE SB 25-30-03-7) (SUPSD BY ITEM 30A)		5
-30A	10-23161089		• LAMP, LED, WARM WHITE (V61706) (POST SB 25-30-03-7) (SUPSDS ITEM 30) (SUPSD BY ITEM 30B)		5
-30B	EUS1512151W3D		• LED, WHITE DIFFUSE (36V) (V61706) (TPN EUS-1512151W3D) (B/E AEROSPACE STOCK CODE LP-50099-1089) (SUPSDS ITEM 30A)		5
-30C	10-23161089		DELETED		4
35	31-281025		• SWITCH, 1PDT MAINTAIN (V61706) (SW1) (SUPSD BY ITEM 35A)		1
-35A	31-281-025		• SWITCH, 1PDT MAINTAIN (V61706) (SW1) (B/E AEROSPACE STOCK CODE SW-50212) (SUPSDS ITEM 35)		1
40	31-151025		• SWITCH, 1PDT MOMENTARY (V61706) (SW2-SW5) (SUPSD BY ITEM 40A)		3
-40A	31-151-025		• SWITCH, 1PDT MOMENTARY (V61706) (SW2, SW4, SW5) (B/E AEROSPACE STOCK CODE SW-50217) (SUPSDS ITEM 40)		3
-40B	31-151025		DELETED		4
45	31-282025		• PUSH BUTTON ILLUMINATED (V61706) (SW3) (SUPSD BY ITEM 45A)		1
-45A	31-2820252		• PUSH BUTTON ILLUMINATED (V61706) (SW3) (TPN 31-282.0252) (B/E AEROSPACE STOCK CODE SW-50279) (SUPSDS ITEM 45)		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
6					
50	1-640512-0		• HOUSING CAP, 12 POSITION (P3) (V00779) (B/E AEROSPACE STOCK CODE CO-50422)		1
-55	640545-2		• PIN (V61706) (B/E AEROSPACE STOCK CODE TE-50446) (SUPSD BY ITEM 55A)		12
-55A	350967-2		• PIN (V00779) (B/E AEROSPACE STOCK CODE TE-50446) (SUPSDS ITEM 55)		12
-60	1N4005		• DIODE (CR102-CR105) (SUPSD BY ITEM 60A)		4
-60A	1N4005		• DIODE (CR102-CR105) (V12060) (B/E AEROSPACE STOCK CODE RC-50105) (SUPSDS ITEM 60)		4
-60B	1N4005		DELETED		3
-65	31818		DELETED		2
70	11201-1		• SWITCH GUARD		1
75	11058-1		• PLATE, SWITCH MOUNT		1
-80	MS3367-4-9		• WIRE TIE, NYLON, 4 IN (101.60 MM)		4
-85	11204-1		• SWITCH WIRING DIAGRAM (REF) (SEE SCHEMATIC AND WIRING DIAGRAM SECTION) (SUPSD BY ITEM 85A)		RF
-85A	11274-1		• SWITCH WIRING DIAGRAM (REF) (SEE SCHEMATIC AND WIRING DIAGRAM SECTION) (SUPSDS ITEM 85)		RF
-90	12678-1		DELETED		RF

- ITEM NOT ILLUSTRATED

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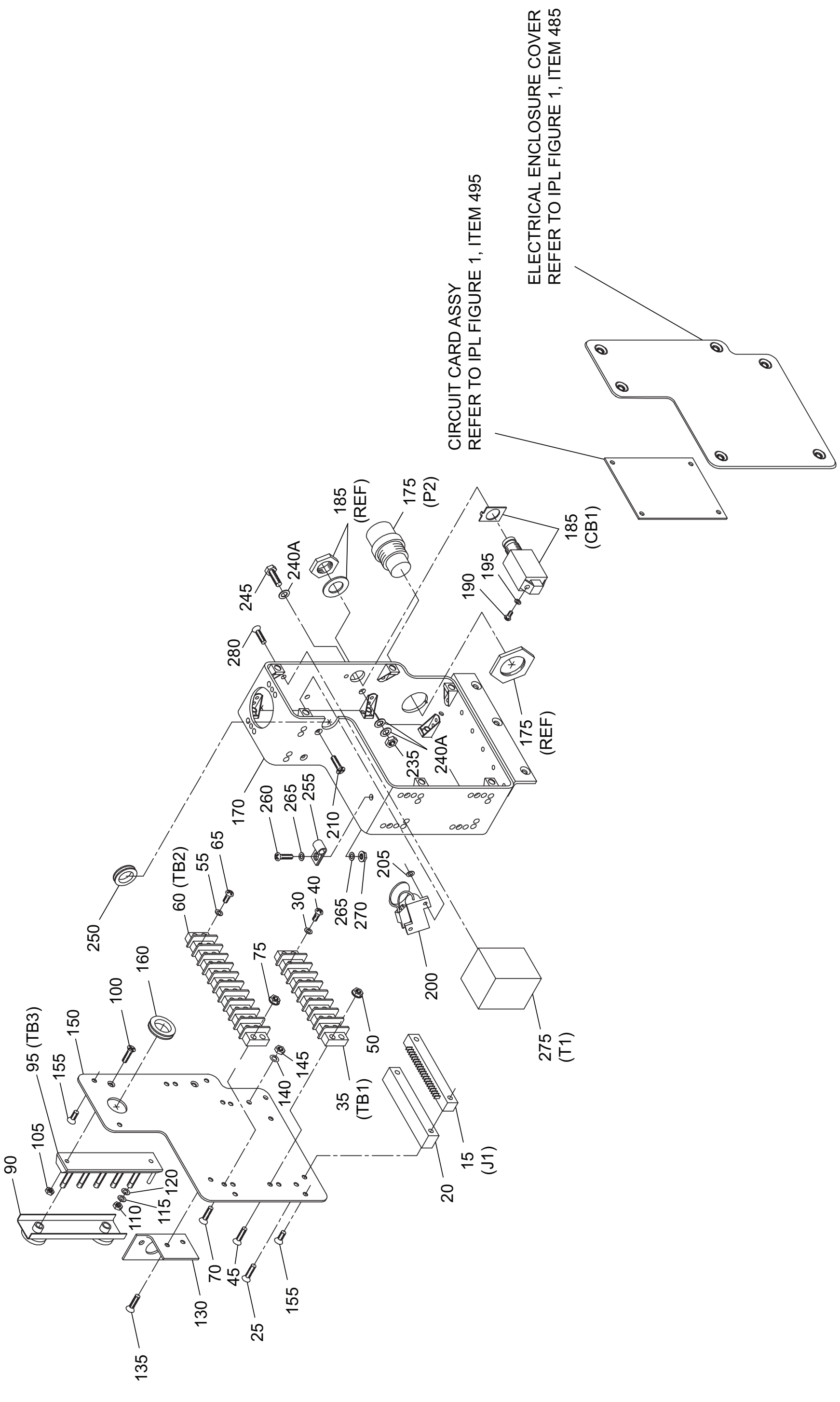

GRAPHIC 25-30-03-99B-022-A01

IPL Figure 7
Electrical Enclosure Assembly

TABLE 25-30-03-950-810-A01

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
7					
-1	11090-1		ELECTRICAL ENCLOSURE ASSEMBLY (SEE IPL FIGURE 1 FOR NHA)		RF
-5	11090-301		• COVER, SUB ASSEMBLY		1
-10	11093-1		• • CONNECTOR HARNESS ASSEMBLY		1
15	C403B021R		• • • CONNECTOR (V1NUQ8) (J1) (SUPSD BY ITEM 15A)		1
-15A	C403B021R		• • • CONNECTOR (V06928) (J1) (TPN C403B-021R) (B/E AEROSPACE STOCK CODE CO-50414) (SUPSDS ITEM 15)		1
20	11224-1		• • SUPPORT, CIRCUIT CARD		1
			ATTACHING PARTS		
25	MS24693C8		• • SCREW (4-40 X 0.75)		2
			* * *		
30	MS35338-136		• • WASHER, LOCKING (NO. 6)		16
35	600A8		• • TERMINAL BLOCK (V26405) (TB1) (SUPSD BY ITEM 35A)		1
-35A	600AGP8		• • TERMINAL BLOCK (V26405) (TB1) (B/E AEROSPACE STOCK CODE TE-50445-08) (SUPSDS ITEM 35)		1
40	KU63214		• • • SCREW, 6-32 X 0.25 IN NI PLATED BRASS (V26405) (SUPSD BY ITEM 40A)		16
-40A	9813504		• • • SCREW, 6-32 X 0.25 IN NI PLATED BRASS (PROVIDED WITH TB1) (V26405) (B/E AEROSPACE STOCK CODE SC-50269) (SUPSDS ITEM 40) (SUPSD BY ITEM 40B)		16
-40B	SC50269		• • • SCREW, 6-32 X 0.25 IN NI PLATED BRASS (PROVIDED WITH TB1) (TPN SC-50269) (SUPSDS ITEM 40A)		16
			ATTACHING PARTS		
45	MS24693C28		• • SCREW, FLAT HD (6-32 X 0.5)		4
50	MS21042-06		• • NUT, (6-32)		4
			* * *		
55	MS35338-136		• • WASHER, LOCKING (NO. 6)		20

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
7					
60	600A10		• • TERMINAL BLOCK (V26405) (TB2) (SUPSD BY ITEM 60A)		1
-60A	600AGP10		• • TERMINAL BLOCK (V26405) (TB2) (B/E AEROSPACE STOCK CODE TE-50445-10) (SUPSDS ITEM 60)		1
65	KU63214		• • • SCREW, 6-32 X 0.25 IN NI PLATED BRASS (V26405) (SUPSD BY ITEM 65A)		20
-65A	9813504		• • • SCREW, 6-32 X 0.25 IN NI PLATED BRASS (PROVIDED WITH TB1) (V26405) (B/E AEROSPACE STOCK CODE SC-50269) (SUPSDS ITEM 65) (SUPSD BY ITEM 65B)		20
-65B	SC50269		• • • SCREW, 6-32 X 0.25 IN NI PLATED BRASS (PROVIDED WITH TB1) (TPN SC-50269) (SUPSDS ITEM 65A)		20
			ATTACHING PARTS		
70	MS24693C28		• • SCREW, FLAT HD (6-32 X 0.5)		4
75	MS21042-06		• • NUT, (6-32)		4
			* * *		
-80	11246-1		• • STANDOFF ASSEMBLY		2
			ATTACHING PARTS		
-85	NDP24693C04		• • SCREW, FLAT HD (4-40 X 0.375 IN) (V4Y667) (B/E AEROSPACE STOCK CODE SC-50268-04)		2
			* * *		
90	MS18029-1S5		• • COVER, TERMINAL BLOCK (TPN MS18029-1S-5) (SEE IPL FIGURE 1, ITEM 520)		RF
95	MS27212-1-5		• • TERMINAL BLOCK (TB3)		1
			ATTACHING PARTS		
100	NDP24693C4		• • SCREW, FLAT HD (4-40 X 0.375 IN) (V4Y667) (B/E AEROSPACE STOCK CODE SC-50268-04) (SUPSD BY ITEM 100A)		2
-100A	MS24693C4		• • SCREW, FLAT HD (4-40 X 0.35 IN) (SUPSDS ITEM 100)		2

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
7					
105	MS35649-244		• • NUT (4-40)		2
110	NAS671C6		• • NUT (6-32)		5
115	MS35338-136		• • WASHER, LOCKING (NO. 6)		5
120	NAS1149CN616R		• • WASHER, FLAT (NO. 6)		5
-125	R4105SBT		• • TERMINAL, RING (V14726) (B/E AEROSPACE STOCK CODE TE-097)		10
			* * *		
130	11112-1		• • ANGLE		1
			ATTACHING PARTS		
135	MS24693C26		• • SCREW, FLAT HD (6-32 X 0.375)		2
140	AN960C6		• • WASHER, FLAT (SUPSD BY ITEM 140A)		1
-140A	NAS1149CN632R		• • WASHER, FLAT (NO. 6) (SUPSDS ITEM 140)		1
145	MS21042-06		• • NUT (6-32)		1
			* * *		
150	11098-1		• • COVER, BACK		1
			ATTACHING PARTS		
155	MS24693C24		• • SCREW, FLAT HD (6-32 X 0.25)		5
			* * *		
160	1126		• • GROMMET, 0.625 IN (15.875 MM) (V**4**) (SUPSD BY ITEM 160A)		1
-160A	GMT1056605		• • GROMMET, 0.625 IN (15.875 MM) (V**9**) (TPN GMT-1056605) (SUPSDS ITEM 160) (SUPSD BY ITEM 160B)		1
-160B	GMT105660S		• • GROMMET, 0.625 IN (15.875 MM) (V0AA66) (TPN GMT-105660S) (B/E AEROSPACE STOCK CODE GR-50057) (SUPSDS ITEM 160A)		1
-165	11090-303		• ENCLOSURE, SUBASSEMBLY		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
7					
170	11059-1		• • ENCLOSURE, ELECTRICAL		1
175	11327-1		• • CONNECTOR ASSEMBLY (P2)		1
-180	CDC06E16S1P		• • • CONNECTOR (V71771) (P2) (SUPSD BY ITEM 180A)		1
-180A	CO6505		• • • CONNECTOR POWER (TPN CO-6505) (SUPSDS ITEM 180)		1
185	7277-2-1		• • CIRCUIT BREAKER 1A (V82647) (CB1) (SUPSD BY ITEM 185A)		1
-185A	MS26574-1		• • CIRCUIT BREAKER 1A (CB1) (SUPSDS ITEM 185)		1
-185B	7274-2-1		• • CIRCUIT BREAKER 1A (V82647) (CB1) (OPTIONAL) (SUPSD BY ITEM 185C)		1
-185C	7274-2-1		• • CIRCUIT BREAKER 1A (CB1) (OPTIONAL) (SUPSDS ITEM 185B)		1
190	KU63214		• • • SCREW, 6-32 X 0.25 IN NI PLATED BRASS (V26405) (SUPSD BY ITEM 190A)		2
-190A	9813504		• • • SCREW, 6-32 X 0.25 IN NI PLATED BRASS (V26405) (B/E AEROSPACE STOCK CODE SC-50269) (SUPSDS ITEM 190) (SUPSD BY ITEM 190B)		2
-190B	SC50269		• • • SCREW, 6-32 X 0.25 IN NI PLATED BRASS (TPN SC-50269) (SUPSDS ITEM 190A)		2
195	MS35338-136		• • • WASHER, LOCKING (NO. 6)		2
200	11269-1		• • SCR ASSEMBLY		1
			ATTACHING PARTS		
205	MS21042-04		• • NUT, LOCKING (4-40)		2
210	MS24693C4		• • SCREW, FLAT HD (4-40 X 0.375)		2
			* * *		
-215	RH25-16		• • • RESISTOR, 25 WATT, 16 OHM (V18612) (R101) (B/E AEROSPACE STOCK CODE RE-50107)		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
7					
-220	MCR264-008		• • • SCR (V04713) (Q101) (SUPSD BY ITEM 220A)		1
-220A	TYN640		• • • SCR (V50088) (Q101) (B/E AEROSPACE STOCK CODE IC-50045) (SUPSDS ITEM 220)		1
-225	V250LA40A		• • • VARISTOR (V34371) (CR101) (SUPSD BY ITEM 225A)		1
-225A	V250LA40A		• • • VARISTOR (V75915) (CR101) (B/E AEROSPACE STOCK CODE RE-50108) (SUSPSDS ITEM 225)		1
-230	XR1903SNT		• • TERMINAL RING (V14726) (B/E AEROSPACE STOCK CODE TE-215)		2
			ATTACHING PARTS		
235	MS21042-3		• • NUT, SELF-LOCKING (10-32)		1
-240	AN960D10L		• • WASHER, FLAT (SUPSD BY ITEM 240A)		2
240A	NAS1149D0316H		• • WASHER, ALUMINIUM (NO. 10) (SUPSDS ITEM 240)		3
245	MS51958-62		• • SCREW (SUPSD BY ITEM 245A)		1
-245A	NAS1801-3-7		• • SCREW (SUPSDS ITEM 245) (SUPSD BY ITEM 245B)		1
-245B	NAS1801-3-8		• • SCREW, HEX HD (10-32 X 0.437) (SUPSDS ITEM 245A)		1
			* * *		
250	1125		• GROMMET, 0.50 IN (12.7 MM) (V**4**) (SUPSD BY ITEM 250A)		1
-250A	910SBR		• GROMMET 0.50 IN (12.7 MM) (V98893) (B/E AEROSPACE STOCK CODE GR-50054) (SUPSDS ITEM 250)		1
255	8876T13		• WIRE CLAMP 0.25 IN (6.35 MM) (V3A054) (SUPSD BY ITEM 255A)		RF
-255A	CCS25S8M0		• WIRE CLAMP 0.25 IN (6.35 MM) (V06383) (SUPSDS ITEM 255) (SUPSD BY ITEM 255B)		RF
-255B	F4NY250-0		• WIRE CLAMP 0.25 IN (6.35 MM) (V85480) (B/E AEROSPACE STOCK CODE CL-3201) (SUPSDS ITEM 255A)		RF
			ATTACHING PARTS		

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
7					
260	MS51957-45		• SCREW, PAN HD (8-32 X 0.50) (SUPSD BY ITEM 260A)		1
-260A	MS24693C50		• SCREW, FLAT HD (8-32 X 0.50) (SUPSDS ITEM 260)		1
265	AN960-08		• WASHER, FLAT (SUPSD BY ITEM 265A)		2
-265A	NAS1149FN832P		• WASHER, NO.8 (SUPSDS ITEM 265)		2
270	MS21042-08		• NUT, SELF-LOCKING (8-32)		1
			* * *		
275	200298		• TRANSFORMER (V17547) (T1) (B/E AEROSPACE STOCK CODE TFMR-50016)		1
			ATTACHING PARTS		
280	MS24693C25		• SCREW, FLAT HD (6-32 X 0.3125)		2
			* * *		

- ITEM NOT ILLUSTRATED

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GRAPHIC 25-30-03-99B-023-A01

IPL Figure 8
Circuit Card Assembly - Deleted

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TABLE 25-30-03-950-811-A01

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
8					
-1	11095-503		CIRCUIT CARD ASSEMBLY (SEE IPL FIGURE 1 FOR NHA)		RF
5	11206-1		DELETED		DEL
10	XAL50V330		DELETED		DEL
-10A	XAL50V330		DELETED		DEL
-15	11095-301		DELETED		DEL
20	11095E		DELETED		DEL
25	T85N11D114-12		DELETED		DEL
-25A	V23105A3003B201		DELETED		DEL
-25B	8-1343792-8		DELETED		DEL
-25C	V23105A50003A201		DELETED		DEL
30	L7812CV		DELETED		DEL
-30A	L7812CV		DELETED		DEL
-35	LM239AJ		DELETED		DEL
-35A	LM139AJ		DELETED		DEL
-35B	LM239N		DELETED		DEL
40	CD4049UBF		DELETED		DEL
-40A	CD4049UBF		DELETED		DEL
-40B	CD4049UBE		DELETED		DEL
45	CD4011BF		DELETED		DEL
-45A	CD4011BF		DELETED		DEL
-45B	CD4011BE		DELETED		DEL
50	CD4082BF		DELETED		DEL
-50A	CD4082BF		DELETED		DEL
-50B	CD4082BE		DELETED		DEL
55	CD4081BF		DELETED		DEL
-55A	CD4081BF		DELETED		DEL
-55B	CD4081BE		DELETED		DEL
60	W02G		DELETED		DEL
-60A	W02G		DELETED		DEL

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
8					
-60B	W02F		DELETED		DEL
-60C	W02F		DELETED		DEL
65	1.5KE39A		DELETED		DEL
70	1N755A		DELETED		DEL
75	1N914		DELETED		DEL
80	RFL1N08		DELETED		DEL
-80A	MPF960		DELETED		DEL
-80B	ZVN4306A		DELETED		DEL
85	2N7000		DELETED		DEL
-90	4610X101-103		DELETED		DEL
-90A	4610X101-103		DELETED		DEL
95	4606X102-223		DELETED		DEL
-95A	4606X102-223		DELETED		DEL
100	750K OHM		DELETED		DEL
-100A	750K OHM		DELETED		DEL
105	2M OHM		DELETED		DEL
-105A	2M OHM		DELETED		DEL
110	510K OHM		DELETED		DEL
-110A	510K OHM		DELETED		DEL
115	12K OHM		DELETED		DEL
-115A	12K OHM		DELETED		DEL
120	511 OHM		DELETED		DEL
-120A	511 OHM		DELETED		DEL
125	11222-1		DELETED		DEL
-125A	11222		DELETED		DEL
130	150 OHM		DELETED		DEL
-130A	150 OHM		DELETED		DEL
135	5.6K OHM		DELETED		DEL
-135A	5.6K OHM		DELETED		DEL
140	3.3K OHM		DELETED		DEL

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
8					
-140A	3.3K OHM		DELETED		DEL
145	1M OHM		DELETED		DEL
-145A	1M OHM		DELETED		DEL
150	47 OHM		DELETED		DEL
-150A	47 OHM		DELETED		DEL
155	10 OHM		DELETED		DEL
-155A	10 OHM		DELETED		DEL
160	8.45K OHM		DELETED		DEL
-160A	8.45K OHM		DELETED		DEL
165	3386P200		DELETED		DEL
-165A	3386P1-200		DELETED		DEL
-165B	3386P1-200		DELETED		DEL
170	SA115E334M		DELETED		DEL
-170A	SA115E334M		DELETED		DEL
175	SA115E224M		DELETED		DEL
-175A	SA115E224M		DELETED		DEL
180	SA305E105M		DELETED		DEL
-180A	SA305E105M		DELETED		DEL
185	.22M35		DELETED		DEL
-185A	.22M35		DELETED		DEL
190	.10M35		DELETED		DEL
-190A	.10M35		DELETED		DEL
195	SA105C103M		DELETED		DEL
-195A	SA105C103M		DELETED		DEL
-200	XRL16V470		DELETED		DEL
200A	XRL16V470		DELETED		DEL
205	SA115E104M		DELETED		DEL
-205A	SA105E104M		DELETED		DEL
-205B	SA105E104M		DELETED		DEL
-210	43-05-2		DELETED		DEL

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
8					
-210A	43-05-2		DELETED		DEL
-215	KU63214		DELETED		DEL
-215A	KU63214		DELETED		DEL
-215B	9813504		DELETED		DEL
-220	MS21042-06		DELETED		DEL
225	7-179BA		DELETED		DEL
-225A	7-179BA		DELETED		DEL
-230	1B31		DELETED		DEL
-230A	1B31		DELETED		DEL
-235	12356-1		DELETED		DEL

- ITEM NOT ILLUSTRATED

REMOVAL

TASK 25-30-03-010-801-A01

1. General

   A. This section contains the removal procedures for the Coffee Maker Assembly.

TASK 25-30-03-010-802-A01

2. Procedure

   WARNING: IF THE COFFEE MAKER WAS USED RECENTLY THE REMOVAL HANDLE AND THE CARRY HANDLE WILL BE WARM.

   CAUTION: DO NOT USE SERVER BASE TO REMOVE BREWER.

   A. Coffee Maker Removal

      (1) Before removing the Coffee Maker from the galley it is best (but not required) to bleed the water pressure from the tank. This can be done by turning off the galley water supply and drawing a cup of water from the faucet by pushing the HOT WATER button while power is on.

      (2) To remove the Coffee Maker from the galley compartment, unscrew the secondary restraint with a bladed screwdriver and raise the LOCK lever in the server cavity. Then pull straight out on the removal handle in the server cavity until the carry handle is exposed. Support the Coffer Maker weight with the carry handle and pull clear of galley compartment.

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INSTALLATION

TASK 25-30-03-410-801-A01

1. General

   A. This section contains the installation procedures for the Coffee Maker Assembly.

TASK 25-30-03-410-802-A01

2. Installation

   A. Mechanical

      (1) Verify that the valve stem assembly is closed (turn CW). Position the LOCK lever (in server cavity) up. Engage both Coffee Maker rail guides into the end of a standard Coffee Maker rail. Hold Coffee Maker level and slide onto rail until power and water connectors fully engage. Lower the LOCK lever until it is straight down, align secondary restraint screw and hand tighten, then turn the screw an additional 1/8 turn using a blade screwdriver.

      (2) Turn on galley water (if it is not already on). Tank will fill part way with water.

   B. Power

      (1) Galley power is obtained through the rail.

   C. Water

      (1) Galley potable water is obtained through the rail.

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STORAGE

TASK 25-30-03-550-801-A01

1. General

   A. This section contains the storage procedures for the Coffee Maker Assembly.

TASK 25-30-03-550-802-A01

2. Procedure

   A. Draining

      (1) To drain the water tank turn the valve stem assembly CCW until water flows out of the manifold.

      (2) To enable aircraft potable water drainage, reinsert Coffee Maker onto rail with the drain valve open.

      (3) Turn the valve stem assembly CW to close after the Coffee Maker has drained for storage off the aircraft.

   B. Storage

      (1) After the unit assembly is complete, put the unit in a clean, dust-free, moisture-proof container. A sealed, polyethylene plastic bag is permitted.

      (2) The unit can be kept in any normal, ambient temperature, pressure, and humidity environment.

      (3) Put the correct part identification (include the name of the unit, part number, unit condition, and date of repair) on the unit.

      (4) For long storage, put the full unit in a polyethylene plastic bag, and put it in a cardboard container.

      (5) The Coffee Maker should be stored in a dirt free environment.

      (6) Maintain storage temperature between -50° F to 150° F.

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